

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 2

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\410-E-31008.pvdb

Date of Analysis : May 21, 2026 5:55pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 2, 2025

Diameter Spec : 1200.000 x 2300.000 mm ID
 Vessel Design Length, Tangent to Tangent 1122.42 cm
 Specified Datum Line Distance 0.00 cm
 Shell Side Design Temperature 80 °C
 Channel Side Design Temperature 80 °C
 Shell Side Design Pressure 23.454 kgf/cm²
 Channel Side Design Pressure 163.158 kgf/cm²
 Shop Shell Side Test Pressure 29.878 kgf/cm²
 Shop Channel Side Test Pressure 220.427 kgf/cm²
 Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N+IT
Head	SA-516 70	...	...	K02700	+N+IT
Cone	SA-516 70	...	...	K02700	+N+IT
Flange	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-350 LF2	1	...	K03011	
Nozzle Flg	SA-350 LF1	1	...	K03009	+IT
Tubes	SA-334 1	...	...	K03008	+IT
Tubesheet	SA-350 LF2	1	...	K03011	+IT
Flg Bolting	SA-320 L43	...	<= 100	G43400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	+N+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (kgf/cm² & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
TS END COVER	163.158	0.00	171.000	1.5000	N/A	No
TS FLG1	163.235	0.00	164.000	1.5000	N/A	No
CH. SHELL	163.235	0.00	173.000	1.5000	N/A	No
SH.1	23.531	0.00	42.000	1.5000	N/A	No
SH.2	23.601	1.05	24.000	1.5000	N/A	No
SH.3	23.601	1.05	28.000	1.5000	N/A	No
DISH END	23.601	1.05	24.000	1.5000	No	No

Liquid Level: 230.00 cm Dens.: 639.707 kg/m³ Sp. Gr.: 0.640

Element Types and Properties:

Element Type	"To" Elev cm	Element Length cm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Body Flg	30.6	30.3	311.0	303.0	295.4	...	1.00	1.00
Body Flg	70.4	39.8	296.0	293.0	286.3	...	1.00	1.00
Cylinder	195.4	125.0	68.0	68.0	64.0	...	1.00	1.00
Cylinder	256.4	30.0	16.0	16.0	9.5	...	1.00	1.00
Conical	447.4	191.0	21.0	21.0	20.6	...	1.00	1.00
Cylinder	1117.4	670.0	21.0	21.0	18.1	...	1.00	1.00
Ellipse	1122.4	5.0	21.0	17.0	16.4	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
20	30	303	200.355	...	No Calc
30	40	293	286.283	...	No Calc
40	Tsht	68	No Calc	...	151.127
Tsht	50	68	No Calc	...	No Calc
50	60	16	No Calc	...	31.1031
60	70	21	8.9296	1.05461	10.2094
70	80	21	12.5282	1.05461	4.45564
80	90	17	6.91643	1.05461	8.32114

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
20	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	124.997	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	30.0031	No Calc	No Calc	No Calc
60	70	191	191	No Calc	No Calc
70	80	694.167	2710.68	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
NOZZLE N1	WRC-297	0.946	Shell	Passed
NOZZLE N2	WRC-297	0.946	Shell	Passed
Noz N4	WRC-297	0.846	Shell	Passed
Noz N5	WRC-297	0.846	Shell	Passed
Noz N3	WRC-297	0.847	Shell	Passed
Noz N6	WRC-297	0.851	Shell	Passed
Noz N7	WRC-297	0.851	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

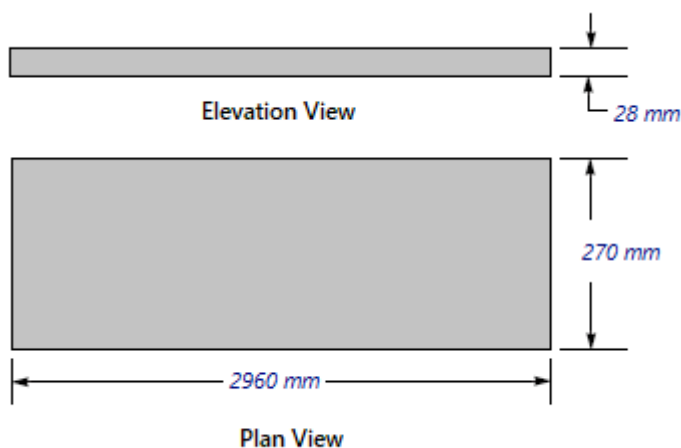
Saddle Width	240.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	1630.000 mm
Wear Pad Width	480.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	550.000 mm

Baseplate Length	2960.000 mm
Baseplate Thickness	28.000 mm
Baseplate Width	270.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	28.000 mm



Web Thickness	28.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	4

Baseplate Sketch



Maximum Tensile Bolt Load	1. kgf
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Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	71882.67 kgf
Maximum Transverse Saddle Shear Load	3158.99 kgf
Maximum Longitudinal Saddle Shear Load	6317.98 kgf

The Wind and Seismic loads were not load factored in this analysis.

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	70078.41 kgf
Maximum Transverse Saddle Shear Load	684.98 kgf
Maximum Longitudinal Saddle Shear Load	407.10 kgf

Weights:

Fabricated - Bare w/o removable internals	46108.1 kg
Shop Test - Fabricated + water (full)	81159.0 kg
Shipping - Fab. + rem. intls.+ shipping app.	46503.2 kg
Erected - Fab. + rem. intls.+ insul. (etc)	46503.2 kg
Empty - Fab. + intls. + details + wghts.	46503.2 kg
Operating - Empty + operating liquid (no ca)	70014.6 kg

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Nozzle Calculation Summary (kgf/cm² & mm):

Description	MAWP	Ext	MAPNC	Tbl 4.5.2	Weld	Pass/Fail
NOZZLE N1	173	...	...	OK	OK	Passed
NOZZLE N2	173	...	...	OK	OK	Passed
Noz N9	200	...	...	OK	OK	Passed
Noz N8	42	...	...	OK	OK	Passed
Noz N4	28	OK	...	OK	OK	Passed
Noz N5	28	OK	...	OK	OK	Passed
Noz N3	28	OK	...	OK	OK	Passed
Noz N6	28	OK	...	OK	OK	Passed
Noz N7	28	OK	...	OK	OK	Passed
Noz K2A	28	OK	...	OK	OK	Passed
Noz K2B	28	OK	...	OK	OK	Passed
Noz K3	28	OK	...	OK	OK	Passed
Noz K4	28	OK	...	OK	OK	Passed
Noz K5	28	OK	...	OK	OK	Passed
Noz K1A	25	OK	...	OK	OK	Passed
Noz K1B	25	OK	...	OK	OK	Passed
MH1	25	OK	...	OK	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 25 Nozzle : Noz K1B [Shellside]
 Minimum MAWP Nozzles : 173 Nozzle : NOZZLE N2 [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

VIII-2 Nozzle to Gross Structural Discontinuity Check:

No Nozzles were too close to any gross structural discontinuities per 4.5.13.1

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Inside Radius mm
40	NOZZLE N1	1329.175	90.000	195.175
40	NOZZLE N2	1329.175	-90.000	195.175
40	Noz N9	0.000	0.000	22.925
50	Noz N8	2414.175	-90.000	22.925
70	Noz N4	5074.175	90.000	196.675
70	Noz N5	7674.175	90.000	196.675
70	Noz N3	6574.174	30.823	196.675
70	Noz N6	10854.175	-90.000	94.275
70	Noz N7	6574.174	-58.646	28.400
70	Noz K2A	4609.174	30.823	26.900
70	Noz K2B	4609.174	-30.823	26.900
70	Noz K3	7674.175	30.823	20.550
70	Noz K4	5974.175	90.000	26.900
70	Noz K5	6774.175	90.000	26.900
80	Noz K1A	0.000	90.000	26.900
80	Noz K1B	0.000	270.000	26.900
80	MH1	0.000	0.000	275.363

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

PV Elite 28 Licensee: SL User
FileName : 410-E-31008
Nozzle Summary:

Step: 41 5:55pm May 21,2026

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Noz K3	1.500 in	Actual	LWN	2.760	16.000	...	...	246.85	300
Noz N9	2.000 in	160	WNF	2.375	8.738	...	...	237.29	1500
Noz N8	2.000 in	160	WNF	2.375	8.738	160.32	12.00	166.76	300
Noz N7	2.000 in	Actual	LWN	3.299	16.500	...	...	392.53	300
Noz K2A	2.000 in	Actual	LWN	3.299	16.500	...	...	251.42	300
Noz K2B	2.000 in	Actual	LWN	3.299	16.500	...	...	251.42	300
Noz K4	2.000 in	Actual	LWN	3.299	16.500	...	...	221.76	300
Noz K5	2.000 in	Actual	LWN	3.299	16.500	...	...	221.76	300
Noz K1A	2.000 in	Actual	LWN	3.299	16.500	...	...	232.14	300
Noz K1B	2.000 in	Actual	LWN	3.299	16.500	...	...	232.14	300
Noz N6	8.000 in	120	WNF	8.625	18.263	319.08	12.00	296.23	300
NOZZLE N1	18.000 in	120	None	18.000	34.925	...	...	724.01	...
NOZZLE N2	18.000 in	120	None	18.000	34.925	...	...	874.01	...
Noz N4	18.000 in	120	WNF	18.000	34.925	657.20	12.00	313.95	300
Noz N5	18.000 in	120	WNF	18.000	34.925	657.20	12.00	313.95	300
Noz N3	18.000 in	120	WNF	18.000	34.925	657.20	12.00	478.16	300
MH1	24.000 in	80	WNF	24.000	30.937	809.60	12.00	339.41	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Noz K3	SA-106 B	...	10.000	...	...	...
Noz N9	SA-333 6	8.738	10.000	...	...	...
Noz N8	SA-333 6	8.738	10.000	12.000	12.000	...
Noz N7	SA-333 6	...	10.000	...	...	...
Noz K2A	SA-106 B	...	10.000	...	...	...
Noz K2B	SA-106 B	...	10.000	...	...	...
Noz K4	SA-106 B	...	10.000	...	...	...
Noz K5	SA-106 B	...	10.000	...	...	...
Noz K1A	SA-106 B	...	10.000	...	...	...
Noz K1B	SA-106 B	...	10.000	...	...	...
Noz N6	SA-333 6	12.700	10.000	12.000	12.000	...
NOZZLE N1	SA-350 LF2	66.000	10.000	...	...	...
NOZZLE N2	SA-350 LF2	66.000	10.000	...	...	...
Noz N4	SA-333 6	14.275	10.000	12.000	12.000	...
Noz N5	SA-333 6	12.000	12.000	12.000	12.000	...
Noz N3	SA-333 6	14.275	10.000	12.000	12.000	...
MH1	SA-333 6	21.000	10.000	12.000	12.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum cm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Noz K3	767.417	30.8	200.00	0.00	SH.3
Noz N9	105.418	0.0	200.00	0.00	NOZZLE N2
Noz N8	241.418	-90.0	150.00	0.00	SH.1



Noz N7	657.417	-58.6	270.00	0.00	SH.3
Noz K2A	460.917	30.8	200.00	0.00	SH.3
Noz K2B	460.917	-30.8	200.00	0.00	SH.3
Noz K4	597.417	90.0	200.00	0.00	SH.3
Noz K5	677.417	90.0	200.00	0.00	SH.3
Noz K1A	...	90.0	200.00	0.00	DISH END
Noz K1B	...	270.0	200.00	0.00	DISH END
Noz N6	1085.417	-90.0	270.00	0.00	SH.3
NOZZLE N1	132.917	90.0	582.00	0.00	CH. SHELL
NOZZLE N2	132.917	-90.0	732.00	0.00	CH. SHELL
Noz N4	507.417	90.0	270.00	0.00	SH.3
Noz N5	767.417	90.0	270.00	0.00	SH.3
Noz N3	657.417	30.8	270.00	0.00	SH.3
MH1	...	0.0	300.00	0.00	DISH END

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 311.0mm THK X 1750.0mm OD	SA-350 LF2
1	WELD NECK FLANGE: 296.0mm THK X 1750.0mm OD	SA-350 LF2
1	CYLINDER: 16.0mm THK X 1200.0mm ID X 30.0cm	SA-516 70
1	ECCENTRIC CONE: 21.0mm THK X 2300.0mm X 1200.0cm ID X 191.0 cm	SA-516 70
1	CYLINDER: 21.0mm THK X 2300.0mm ID X 670.0cm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 21.0mm THK X 2300.0mm ID X 5.0cm	SA-516 70
1	INSULATION: 303mm X 115mm THK	
1	INSULATION: 397mm X 115mm THK	
1	INSULATION: 1250mm X 115mm THK	
1	TUBESHEET: 1336.0mm X 310.0mm THK	SA-350 LF2
1066	TUBES: 7997.0mm X 25.0mm DIA X 2.5mm THK	SA-334 1
1	INSULATION: 300mm X 115mm THK	
1	INSULATION: 1910mm X 115mm THK	
1	SADDLE: 240.0mm X 120 DEG	SA-516 70
1	SADDLE: 270.0mm X 120 DEG	SA-516 70
1	INSULATION: 6700mm X 115mm THK	
1	INSULATION: 625mm X 115mm THK	
1	CLASS 1500 GR 1.1, 2.0" BLIND FLANGE(S)	SA-105
1	CLASS 300 GR 1.1, 2.0" BLIND FLANGE(S)	SA-350 LF2
1	CLASS 300 GR 1.1, 24.0" BLIND FLANGE(S)	SA-350 LF2
1	GASKET: 1229.0mm OD X 1203.0mm ID	...
24	BODY FLANGE BOLTS: 88.9mm DIA	SA-320 L43
48	NUTS FOR BODY FLANGE BOLTS: 88.9mm DIA	...
1	CYLINDRICAL SEGMENT 68.0mm THK X 3047.3mm X 125.0cm	SA-516 70
1	CYLINDRICAL SEGMENT 68.0mm THK X 722.6mm X 125.0cm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	PWHT	Thickness ratio	Gov Thk mm	E*
Notes							
SH.1	[8]	!	-29	No			
SH.2	[8]	!	-29	No			
SH.3	[8]	!	-29	No			
DISH END	[10]	!	-29	No			
DISH END	[7]	!	-29	No			
Noz N8	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-85	No		
Noz N4	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-85	No		
Noz N5	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-85	No		
Noz N3	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-85	No		
Noz N6	[1]	!	-46	No			
Nozzle Flg	[5]	!	-29	-68	No		
Noz N7	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-46	No		
Noz K2A	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K2B	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K3	[1]	B	-104	-104	No	0.020	16.000
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K4	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K5	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K1A	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
Noz K1B	[1]	B	-104	-104	No	0.026	16.500
Nozzle Flg	[5]	B	-46	-48	No		1.00
MH1	[1]	!	-46	No			
Nozzle Flg	[5]	!	-46	-85	No		
Tubesheet: SS	[13]	!	-46	No			

Warmest MDMT: -29 -46

TS END COVER	[11]	!	-46	No			
TS FLG1	[11]	!	-46	Yes			
CH. SHELL	[8]	!	-29	Yes			
NOZZLE N1	[1]	!	-46	Yes			
NOZZLE N2	[1]	!	-46	Yes			
Noz N9	[1]	!	-46	Yes			
Nozzle Flg	[5]	A	-18	-38	Yes		
Tubesheet: CS	[14]	!	-46	Yes			
Bolting	[21]		-101				

Warmest MDMT: -18 -38

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-28.9	Pass
Channel/Tube	-29.0	-28.9	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.1.
- [5] - ASME Flange MDMT Calcs; Thickness ratio per equation 3.3.
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.



- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per 3.11.2.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.

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Units used in this Analysis (Kgfc^{m2}):

Name	System Unit	Constant	User Unit
Length	Feet	30.4800	cm
Force	Pounds	0.4536	kgf
Mass	Pounds	0.4536	kg
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	kgf-m.
Stress	lbs./sq.in.	0.0703	kgf/cm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0703	kgf/cm ²
Elast. Modulus	lbs./sq.in.	0.0703	kgf/cm ²
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	16.0183	kg/m ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kgf/cm ²
Inertia	in.4	41.6231	cm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8824	kgf/m ²
Elevation	Feet	0.3048	m
Volume	in.3	16.3871	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	23.454	kgf/cm ²
Channel Side Design Pressure	163.16	kgf/cm ²
Shell Side Design Temperature	80.0	°C
Channel Side Design Temperature	80.0	°C
Radiography, Shell Side	None	
Radiography, Channel Side	None	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	kgf/cm ²
User defined MAWP, Channel Side	0	kgf/cm ²
User defined MAPnc, Shell Side	0	kgf/cm ²
User defined MAPnc, Channel Side	0	kgf/cm ²
User defined Test Pres., Shell Side	0	kgf/cm ²
User defined Test Pres., Channel Side	0	kgf/cm ²

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	N	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175



0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test	Hydrostatic
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.1	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
-------------	------



0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration Yes

Design Pressure + Static Head Y

Consider MAP New and Cold in Noz. Design N

Consider External Loads for Nozzle Des. Y

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g_0 is taken as $\min(g_0, \text{attached shell thk})$ Yes

Using Metric Material Databases, ASME II D Yes

Using ASME VIII-1 Appendix 46 No

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	20
Element To Node	30
Element Type	Flange
Description	TS END COVER
Distance "FROM" to "TO"	30.3 cm
Flange Outside Diameter	1750 mm
Element Thickness	303 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	311 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Allowable Stress, Ambient	1682.4 kgf/cm ²
Allowable Stress, Operating	1568.8 kgf/cm ²
Allowable Stress, Hydrotest	2421.7 kgf/cm ²
Material Density	0.00775 kg/cm ³
P Number Thickness	0 mm
Yield Stress, Operating	2349.6 kgf/cm ²
MDMT Curve	Impact Tested
External Pressure Chart Name	CS-2
UNS Number	K03011



Class / Thickness / Grade	1::
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Insulation
Detail ID	Ins: 20
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	30.3 cm
Thickness of Insulation	115 mm
Density	0.40000E-04 kg/cm ³

Element#	2/7
Element From Node	30
Element To Node	40
Element Type	Flange
Description	TS FLG1
Distance "FROM" to "TO"	39.8 cm
Flange Inside Diameter	1200 mm
Element Thickness	293 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	296 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-350 LF2 +IT
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	120 cm
Liquid Density	639.71 kg/m ³

Element From Node	30
Detail Type	Insulation
Detail ID	Ins: 30
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	39.8 cm
Thickness of Insulation	115 mm
Density	0.40000E-04 kg/cm ³

Element#	3/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	CH. SHELL
Distance "FROM" to "TO"	125 cm
Inside Diameter	1200 mm
Element Thickness	68 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	163.16 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Allowable Stress, Ambient	1784.4 kgf/cm ²



Allowable Stress, Operating	1650.4	kgf/cm ²
Allowable Stress, Hydrotest	2518.6	kgf/cm ²
Material Density	0.00775	kg/cm ³
P Number Thickness	0	mm
Yield Stress, Operating	2477.8	kgf/cm ²
MDMT Curve	Impact Tested	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	120	cm
Liquid Density	639.71	kg/m ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	125	cm
Thickness of Insulation	115	mm
Density	0.40000E-04	kg/cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	NOZZLE N1	
Dist. from "FROM" Node / Offset dist	62.5	cm
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	None	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	381.44	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	NOZZLE N2	
Dist. from "FROM" Node / Offset dist	62.5	cm
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	None	
Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	435.33	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Noz N9	
Dist. from "FROM" Node / Offset dist	35	cm
Nozzle Diameter	2	in.
Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	26.471	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	



Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	SH.1
Distance "FROM" to "TO"	30 cm
Inside Diameter	1200 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	23.454 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	0 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	120 cm
Liquid Density	639.71 kg/m ³

Element From Node	50
Detail Type	Insulation
Detail ID	Ins: 50
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	30 cm
Thickness of Insulation	115 mm
Density	0.40000E-04 kg/cm ³

Element From Node	50
Detail Type	Nozzle
Detail ID	Noz N8
Dist. from "FROM" Node / Offset dist	15 cm
Nozzle Diameter	2 in.
Nozzle Schedule	160
Nozzle Class	300
Layout Angle	-90.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	10.901 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-333 6 +IT

Element#	5/7
Element From Node	60
Element To Node	70
Element Type	Conical
Description	SH.2
Distance "FROM" to "TO"	191 cm
Inside Diameter	1200 mm
Element Thickness	21 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	21 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	23.454 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	2300 mm



Design Length of Cone 1910 mm
 Half Apex Angle of Cone 29.938337 degrees
 Toriconical (Y/N) N
 Weld is pre-Heated No

Element From Node 60
 Detail Type Liquid
 Detail ID Liquid: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Liquid 230 cm
 Liquid Density 639.71 kg/m³

Element From Node 60
 Detail Type Insulation
 Detail ID Ins: 60
 Dist. from "FROM" Node / Offset dist 0 cm
 Height/Length of Insulation 191 cm
 Thickness of Insulation 115 mm
 Density 0.40000E-04 kg/cm³

Element# 6/7
 Element From Node 70
 Element To Node 80
 Element Type Cylinder
 Description SH.3
 Distance "FROM" to "TO" 670 cm
 Inside Diameter 2300 mm
 Element Thickness 21 mm
 Internal Corrosion Allowance 1.5 mm
 Nominal Thickness 21 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 23.454 kgf/cm²
 Design Temperature Internal Pressure 80 °C
 Design External Pressure 1.0546 kgf/cm²
 Design Temperature External Pressure 37 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70 +N+IT
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 70
 Detail Type Saddle
 Detail ID Left Saddle
 Dist. from "FROM" Node / Offset dist 55 cm
 Width of Saddle 240 mm
 Height of Saddle at Bottom 1630 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 480 mm
 Thickness of Wear Plate 12 mm
 Contact Angle, Wear Plate (degrees) 132.0
 Friction coefficient 0.0
 Moment Factor 3.0
 Dimension E at base (optional) 0 mm
 Circumferential Eff. over Saddle 1.0
 Circumferential Eff. at Midspan 1.0
 Tangent to Tangent dist. (optional) 0 cm

Element From Node 70
 Detail Type Saddle
 Detail ID RIGHT SADDLE
 Dist. from "FROM" Node / Offset dist 480 cm
 Width of Saddle 270 mm
 Height of Saddle at Bottom 1630 mm
 Saddle Contact Angle 120.0
 Height of Composite Ring Stiffener 0 mm
 Width of Wear Plate 480 mm



Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	cm
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Liquid	230	cm
Liquid Density	639.71	kg/m ³
Element From Node	70	
Detail Type	Insulation	
Detail ID	Ins: 70	
Dist. from "FROM" Node / Offset dist	0	cm
Height/Length of Insulation	670	cm
Thickness of Insulation	115	mm
Density	0.40000E-04	kg/cm ³
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N4	
Dist. from "FROM" Node / Offset dist	60	cm
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	264.2	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N5	
Dist. from "FROM" Node / Offset dist	320	cm
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	310.14	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N3	
Dist. from "FROM" Node / Offset dist	210	cm
Nozzle Diameter	18	in.
Nozzle Schedule	120	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	264.2	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N6	
Dist. from "FROM" Node / Offset dist	638	cm
Nozzle Diameter	8	in.
Nozzle Schedule	120	
Nozzle Class	300	



Layout Angle	-90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	61.858	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz N7	
Dist. from "FROM" Node / Offset dist	210	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	-58.646301	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	11.579	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K2A	
Dist. from "FROM" Node / Offset dist	13.5	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6863	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K2B	
Dist. from "FROM" Node / Offset dist	13.5	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	-30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	9.6863	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K3	
Dist. from "FROM" Node / Offset dist	320	cm
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	30.822701	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	7.7689	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	70	
Detail Type	Nozzle	
Detail ID	Noz K4	
Dist. from "FROM" Node / Offset dist	150	cm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	70
Detail Type	Nozzle
Detail ID	Noz K5
Dist. from "FROM" Node / Offset dist	230 cm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	7/7
Element From Node	80
Element To Node	90
Element Type	Elliptical
Description	DISH END
Distance "FROM" to "TO"	5 cm
Inside Diameter	2300 mm
Element Thickness	17 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	21 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	23.454 kgf/cm ²
Design Temperature Internal Pressure	80 °C
Design External Pressure	1.0546 kgf/cm ²
Design Temperature External Pressure	37 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N+IT
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Liquid	230 cm
Liquid Density	639.71 kg/m ³

Element From Node	80
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 cm
Height/Length of Insulation	62.5 cm
Thickness of Insulation	115 mm
Density	0.40000E-04 kg/cm ³

Element From Node	80
Detail Type	Nozzle
Detail ID	Noz K1A
Dist. from "FROM" Node / Offset dist	650 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	80
Detail Type	Nozzle
Detail ID	Noz K1B
Dist. from "FROM" Node / Offset dist	650 mm



Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	80	
Detail Type	Nozzle	
Detail ID	MH1	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	24	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-333 6 +IT	



XY Coordinate Calculations:

From	To	X (Horiz.) cm	Y (Vert.) cm	DX (Horiz.) cm	DY (Vert.) cm
TS END COVER		30.6175	...	30.3	...
TS FLG1		70.4175	...	39.8	...
CH. SHELL		195.417	...	125	...
SH.1		256.418	...	30	...
SH.2		447.418	...	191	...
SH.3		1117.42	...	670	...
DISH END		1122.42	...	5	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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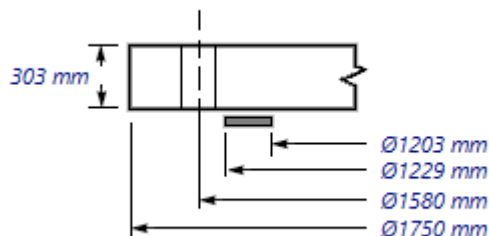


Flange Input Data Values

Description: FLANGE :

TS END COVER

Flange Type		Blind	
Design Pressure	P	163.16	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Outside Diameter	A	1750.000	mm
Flange Thickness	t	303.0000	mm
Flange Material		SA-350 LF2 +IT	
Flange Material UNS number		K03011	
Flange Allowable Stress At Temperature	Sfo	1568.83	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material		SA-320 L43	
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Perimeter along the Center of the Bolts	L	4963.717	mm
Diameter of Bolt Circle	C	1580.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		24	
Flange Face Outside Diameter	Fod	1235.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1229.000	mm
Gasket Inside Diameter	Gi	1203.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1203.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²



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Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$b_o = N/2$	6.500 mm
Effective Gasket Width,	$b = C_b * \text{sqrt}(b_o)$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 * b$	1216.151 mm

Basic Flange and Bolt Loads:



Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1216.1509^2 \cdot 163.158$$

$$= 1895281.375 \text{ kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 1216.1509 \cdot 3.75 \cdot 163.16$$

$$+ 2 \cdot 1203.0 \cdot 5.0 \cdot 3.75 \cdot 163.1584$$

$$= 373970.688 \text{ kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(1895281 + 373971 + 0, 0)$$

$$= 2269252.000 \text{ kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 534.33 \cdot 6.4246 \cdot \pi \cdot 1216.151 + 534.33 \cdot 5.0 \cdot 1203.0$$

$$= 163297.297 \text{ kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((2269252 + 0 + 4 \cdot 0/1216.15)/1754, 163297/1754)$$

$$= 1293.893 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1293.893	1354.681	
Radial Distance between Bolts and Edge:	82.550	85.000	
Circ. Spacing between the Bolts:	180.975	206.231	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 1354.681 \cdot 1753.82 / (534.33 \cdot \pi (1229.0 + 1203.0))$$

$$= 58.196 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 1354.681$$

$$= 2375862.75 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 2269252.00 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1580.0 - 1216.1509) / 2$$

$$= 181.9246 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 2269252 \cdot 181.9246 \cdot 1.0 / (303.0^2 \cdot 1216.1509) + 0.3 \cdot 1.0 \cdot 163.16 \cdot 1216.1509^2 / 303.0^2$$

$$= 1491.05 \text{ kgf/cm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 2375863 \cdot 181.9246 \cdot 1.0 / (303.0^2 \cdot 1216.1509)$$

$$= 735.52 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$



= 2269252/1354.681
 = 1675.12 kgf/cm²

Bolt Stress, Seating [BSa]:

= (Wm2 / Ab)
 = (163297/1354.681)
 = 120.54 kgf/cm²

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1491.05	1568.83	735.52 1682.44
Bolting	1675.12	1753.82	120.54 1753.82

Reqd. Blind Flange Thickness at Center 295.393 mm
 Reqd. Blind Flange Thickness at Gasket 202.760 mm

Estimated M.A.W.P. (Operating) 171 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 5648.5 kg
 Estimated Unfinished Weight of Forging at given Thk 5648.5 kg

Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Tangential Flange	1491.05	1682.44	735.52 1682.44
Bolting	1675.12	1753.82	120.54 1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 285.267 mm

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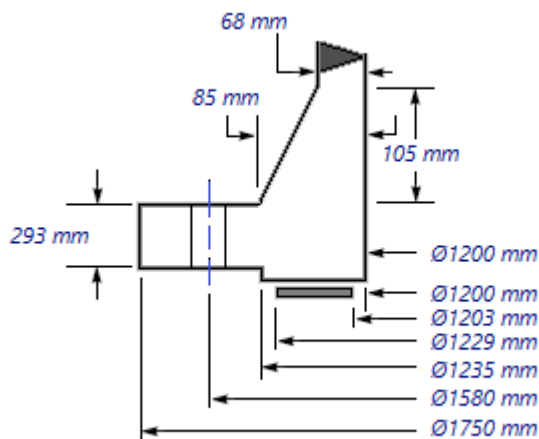


Flange Input Data Values

Description: FLANGE :

TS FLG1

Flange Type	Integral Weld Neck		
Design Pressure	P	163.24	kgf/cm ²
Design Temperature		80	°C
Internal Corrosion Allowance	ci	1.5000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		No	
Flange Inside Diameter	B	1200.000	mm
Flange Outside Diameter	A	1750.000	mm
Flange Thickness	t	293.0000	mm
Thickness of Hub at Small End	go	68.0000	mm
Thickness of Hub at Large End	gl	85.0000	mm
Length of Hub	h	105.0000	mm
Flange Material	SA-350 LF2 +IT		
Flange Material UNS number	K03011		
Flange Allowable Stress At Temperature	Sfo	1568.83	kgf/cm ²
Flange Allowable Stress At Ambient	Sfa	1682.44	kgf/cm ²
Bolt Material	SA-320 L43		
Bolt Allowable Stress At Temperature	Sb	1753.82	kgf/cm ²
Bolt Allowable Stress At Ambient	Sa	1753.82	kgf/cm ²
Diameter of Bolt Circle	C	1580.000	mm
Nominal Bolt Diameter	a	88.9000	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	24		
Flange Face Outside Diameter	Fod	1235.000	mm
Flange Face Inside Diameter	Fid	1200.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1229.000	mm
Gasket Inside Diameter	Gi	1203.000	mm
Gasket Factor	m	3.7500	
Gasket Design Seating Stress	y	534.33	kgf/cm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	1203.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.7500	
Partition Gasket Design Seating Stress	yPart	534.33	kgf/cm ²





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Hub Small End Required Thickness due to Internal Pressure:

$$\begin{aligned} &= (P(R + Ca)/(S - 0.5 \cdot P)) + (Ca + Ca_{ext}) \\ &= (163.235(1200.0 + 1.5)/(1568.835 - 0.5 \cdot 163.235)) + (1.5) \\ &= 67.5199 \text{ mm} \end{aligned}$$

Small End Hub MAWP:

$$\begin{aligned} &= S \cdot \log((2 \cdot T + R)/R) \\ &= 1568.835 \cdot \log((2 \cdot 66.5 + 601.5)/601.5) \\ &= 164.511 \text{ kgf/cm}^2 \end{aligned}$$

Small End Hub MAPnc:

$$\begin{aligned} &= S_a \cdot \log((2.0 \cdot T + R)/R) \\ &= 1682.443 \cdot \log((2.0 \cdot 68.0 + 600.0)/600.0) \\ &= 180.625 \text{ kgf/cm}^2 \end{aligned}$$

Corroded Flange ID,	$B_{cor} = B + 2 \cdot c_i$	1203.000 mm
Corroded Large Hub,	$g1_{cor} = g1 - c_i$	83.500 mm
Corroded Small Hub,	$g0_{cor} = g0 - c_i$	66.500 mm
Code R Dimension,	$R = ((C - B_{cor})/2) - g1_{cor}$	105.000 mm

Gasket Contact Width,	$N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width,	$bo = N/2$	6.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter,	$G = G_o - 2 \cdot b$	1216.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$\begin{aligned} &= \pi/4 \cdot G^2 \cdot P \\ &= \pi/4 \cdot 1216.1509^2 \cdot 163.235 \\ &= 1896173.000 \text{ kgf} \end{aligned}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$\begin{aligned} &= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P \\ &= 2 \cdot 6.4246 \cdot \pi \cdot 1216.1509 \cdot 3.75 \cdot 163.24 \\ &\quad + 2 \cdot 1203.0 \cdot 5.0 \cdot 3.75 \cdot 163.2352 \\ &= 374146.625 \text{ kgf} \end{aligned}$$

Hydrostatic End Load at Flange ID [Hd]:

$$\begin{aligned} &= \pi/4 \cdot B_{cor}^2 \cdot P \\ &= \pi/4 \cdot 1203.0^2 \cdot 163.2352 \\ &= 1855386.000 \text{ kgf} \end{aligned}$$

Pressure Force on Flange Face [Ht]:

$$\begin{aligned} &= H - H_d \\ &= 1896173 - 1855386 \\ &= 40786.992 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Operating Condition [Wo]:

$$\begin{aligned} &= \max(W_o, 1 + W_o, 2 + H'p, 0) \\ &= \max(1896173 + 374147 + 0, 0) \\ &= 2270319.500 \text{ kgf} \end{aligned}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$\begin{aligned} &= y \cdot b \cdot \pi \cdot G + y_{part} \cdot b_{part} \cdot l_p \\ &= 534.33 \cdot 6.4246 \cdot \pi \cdot 1216.151 + 534.33 \cdot 5.0 \cdot 1203.0 \\ &= 163297.297 \text{ kgf} \end{aligned}$$

Required Bolt Area [Am]:

$$\begin{aligned} &= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a) \\ &= \max((2270320 + 0 + 4 \cdot 0/1216.15)/1754, 163297/1754) \\ &= 1294.501 \text{ cm}^2 \end{aligned}$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$\begin{aligned} &= 2a + 6t/(m + 0.5) \\ &= 2 \cdot 88.9 + 6 \cdot 293.0/(3.75 + 0.5) \\ &= 591.447 \text{ mm} \end{aligned}$$



Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 1580.0 \cdot \sin(\pi / 24)$$

$$= 206.231 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{206.231 / (2 \cdot 88.9 + 293.0)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1294.501	1354.681	
Radial Distance between Hub and Bolts:	104.775	105.000	
Radial Distance between Bolts and Edge:	82.550	85.000	
Circ. Spacing between the Bolts:	180.975	206.231	591.447

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 1354.681 \cdot 1753.82 / (534.33 \cdot \pi(1229.0 + 1203.0))$$

$$= 58.196 \text{ mm [Note: Exceeds actual gasket width, 13.000]}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 1753.82 \cdot 1354.681$$

$$= 2375864.00 \text{ kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 2270320 - 1896173$$

$$= 374146.62 \text{ kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1580.0 - 1216.1509) / 2$$

$$= 181.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((1580.0 - 1203.0) / 2 + 181.9246)$$

$$= 185.2123 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (1580.0 - 1203.0 - 83.5) / 2$$

$$= 146.7500 \text{ mm}$$

Summary of Moments for Internal Pressure: (kgf-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1855386.	146.7500	1.0000	272278.
Face Pressure, Mt	40787.	185.2123	1.0000	7554.
Gasket Load, Mg	374147.	181.9246	1.0000	68067.
Gasket Seating, Matm	2375864.	181.9246	1.0000	432229.

Total Moment for Operation, Mo 347899. kgf-m.
 Total Moment for Gasket seating, Mg 432229. kgf-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 282.842 mm
 Hub Ratio, h/h0 = HL / H0 0.371



Thickness Ratio, $g1/g0 = (g1Cor/goCor)$ 1.256

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.883
 Factor V per 4.16.5 0.417
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1750.0/1203.0
 = 1.455

Stress factor from Table 4.16.4 [Y]:

= $(0.66845 + (5.7169(K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 = $(0.66845 + (5.7169(1.455^2 \cdot \log_{10}(1.455))) / (1.455^2 - 1)) / (1.455 - 1)$
 = 5.350

Stress factor from Table 4.16.4 [T]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 = $(1.455^2(1 + 8.55246 \cdot \log_{10}(1.455)) - 1) / ((1.0472 + 1.9448 \cdot 1.455^2) \cdot (1.455 - 1))$
 = 1.730

Stress factor from Table 4.16.4 [U]:

= $(K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$
 = $(1.455^2(1 + 8.55246 \cdot \log_{10}(1.455)) - 1) / (1.36136(1.455^2 - 1) \cdot (1.455 - 1))$
 = 5.879

Stress factor from Table 4.16.4 [Z]:

= $(K^2 + 1) / (K^2 - 1)$
 = $(1.455^2 + 1) / (1.455^2 - 1)$
 = 2.792

Stress factor from Table 4.16.4 for integral flanges [d]:

= $U \cdot g0^2 \cdot h0 / V$
 = $5.879 \cdot 66.5^2 \cdot 282.842 / 0.417$
 = 17636692.000 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.883/282.842
 = 0.003 1/mm

Stress factor from Table 4.16.4 [L]:

= $(t \cdot e + 1) / T + t^3/d$
 = $(293.0 \cdot 0.003 + 1) / 1.73 + 293.0^3/17636692$
 = 2.533

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= $(f \cdot Mop / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 347899/1203.0) / (2.533 \cdot 83.5^2)$
 = 1637.49 kgf/cm²

Longitudinal Hub Stress, Seating [SHa]:

= $(f \cdot Matm / Bcor) / (L \cdot g1^2)$
 = $(1.0 \cdot 432229/1203.0) / (2.533 \cdot 83.5^2)$
 = 2034.41 kgf/cm²

Radial Flange Stress, Operating [SRo]:

= $(\beta \cdot Mop / Bcor) / (L \cdot t^2)$
 = $(2.2203 \cdot 347899/1203.0) / (2.533 \cdot 293.0^2)$
 = 295.27 kgf/cm²

Radial Flange Stress, Seating [SRa]:

= $(\beta \cdot Matm / Bcor) / (L \cdot t^2)$
 = $(2.2203 \cdot 432229/1203.0) / (2.533 \cdot 293.0^2)$
 = 366.84 kgf/cm²

Tangential Flange Stress, Operating [STo]:



$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRO$$

$$= (5.3502 \cdot 347899 / (293.0^2 \cdot 1203.0)) - 2.7919 \cdot 295$$

$$= 977.93 \text{ kgf/cm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRA$$

$$= (5.3502 \cdot 432229 / (293.0^2 \cdot 1203.0)) - 2.7919 \cdot 367$$

$$= 1214.97 \text{ kgf/cm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (1637 + \max(295, 978)) / 2$$

$$= 1307.71 \text{ kgf/cm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHA + \max(SRa, STa)) / 2$$

$$= (2034 + \max(367, 1215)) / 2$$

$$= 1624.69 \text{ kgf/cm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 2270320 / 1354.681$$

$$= 1675.91 \text{ kgf/cm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (163297 / 1354.681)$$

$$= 120.54 \text{ kgf/cm}^2$$

Flange Stress Analysis Results: kgf/cm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	1637.49	2353.25	2034.41	2523.66
Radial Flange	295.27	1568.83	366.84	1682.44
Tangential Flange	977.93	1568.83	1214.97	1682.44
Maximum Average	1307.71	1568.83	1624.69	1682.44
Bolting	1675.91	1753.82	120.54	1753.82

Minimum Required Flange Thickness 286.283 mm

Estimated M.A.W.P. (Operating) 165 kgf/cm²
 Estimated Finished Weight of Flange at given Thk. 3143.4 kg
 Estimated Unfinished Weight of Forging at given Thk 3930.8 kg

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 432228.7 / (1.0 \cdot 100000.023 \cdot 0.417 / (2.425 \cdot 2052965 \cdot 66.5^2 \cdot 282.842 \cdot 0.3))$$

$$= 0.503 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347899.1 / (1.0 \cdot 100000.023 \cdot 0.417 / (2.425 \cdot 2018573 \cdot 66.5^2 \cdot 282.842 \cdot 0.3))$$

$$= 0.411 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 432228.7 / (1.0 \cdot 100000.023 \cdot 0.417 / (2.533 \cdot 2052965 \cdot 66.5^2 \cdot 282.842 \cdot 0.3))$$

$$= 0.481 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 347899.1 / (1.0 \cdot 100000.023 \cdot 0.417 / (2.533 \cdot 2018573 \cdot 66.5^2 \cdot 282.842 \cdot 0.3))$$

$$= 0.394 \text{ (should be } \leq 1)$$



Minimum Design Metal Temperature Results:

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: kgf/cm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	1631.90	2523.66	2017.16	2523.66
Radial Flange	303.45	1682.44	375.09	1682.44
Tangential Flange	962.38	1682.44	1189.58	1682.44
Maximum Average	1297.14	1682.44	1603.37	1682.44
Bolting	1675.91	1753.82	120.54	1753.82

Minimum Req'd Thickness, ambient, no Corrosion: 283.566 mm

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kgf/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) kgf/cm ²	Element#
TS END COVER		163.16	311	1.5	1750	1568.8	1
TS FLG1		163.24	296	1.5	1200	1568.8	2
CH. SHELL		163.24	68	1.5	1200	1650.4	3
SH.1		23.531	16	1.5	1200	1784.4	4
SH.2		23.601	21	1.5	2300	1650.9	5
SH.3		23.601	21	1.5	2300	1650.9	6
DISH END		23.601	21	1.5	2300	1650.9	7

Element Required Thickness and MAWP :

From	To	Design Pressure kgf/cm ²	M.A.W.P. Corroded kgf/cm ²	M.A.P. New & Cold kgf/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
TS END COVER		163.158	170.824	170.824	303	295.393	1
TS FLG1		163.158	164.434	170.824	293	286.283	2
CH. SHELL		163.158	172.987	191.572	68	64.0336	3
SH.1		23.454	42.4285	46.9608	16	9.48443	4
SH.2		23.454	23.9028	28.0163	21	20.6335	5
SH.3		23.454	27.5751	32.2915	21	18.0804	6
DISH END		23.454	24.4659	29.1898	17	16.3605	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

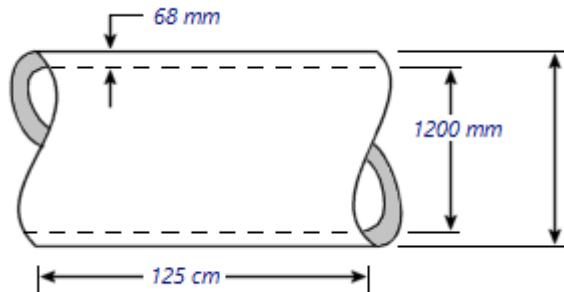
Shell Side MAWP = 24 kgf/cm²
 Shell Side MAPnc = 28 kgf/cm²
 Channel Side MAWP = 164 kgf/cm²
 Channel Side MAPnc = 171 kgf/cm²

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: CH. SHELL

Cylindrical Shell Design Information:					
Design Pressure	P	163.235 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		125.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope) S	1650.4 kgf/cm²	
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb) Sa	1784.4 kgf/cm²	
Inside Diameter	Di	1200.000 mm	Actual Stress	1476.5 kgf/cm²	
Spec. Min. Thk t _e , or sa		68.0000 mm	Spec. Nominal Thk.	68.000 mm	
Surface Area		52464.6 cm²	SG of Contents	0.640	
Element Volume		1413719.9 cm³	Weight of Contents	904.4 kgf	
Empty Weight		2624.3 kgf	Operating Weight	3528.7 kgf	
			Comp. All. Ope. Stress	1251.5 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1203.0 \cdot (\exp(163.235/(1650.405 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 62.5336 + 1.5 + 0.0 = 64.0336 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1650.405 \cdot 1.0 \cdot \ln((2 \cdot 66.5 + 1203.0) / 1203.0) - 0.077$$

$$= 173.064 - 0.077 = 172.9873 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784.411 \cdot 1.0 \cdot \ln((2 \cdot 68.0 + 1200.0) / 1200.0)$$

$$= 191.5717 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 163.235 \cdot 1203.0 / (1.0 (1336.0 - 1203.0))$$

$$= 1476.4813 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 163.235 / (1.0 \cdot \ln((2 \cdot 66.5 + 1203.0) / 1203.0))$$
$$= 1556.6729 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.363 %

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Note:

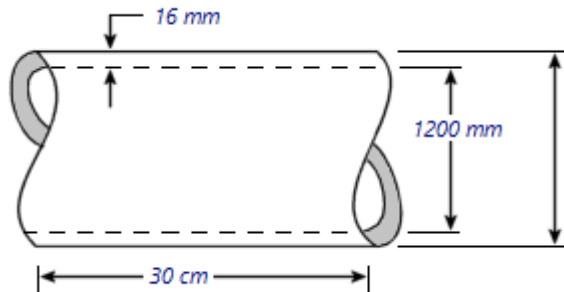
Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.1

Cylindrical Shell Design Information:					
Design Pressure	P	23.531 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		30.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1784.4 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	1200.000 mm	Actual Stress	976.1 kgf/cm²	
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.	16.000 mm	
Surface Area		11611.3 cm²	SG of Contents	0.640	
Element Volume		182311.0 cm³	Weight of Contents	116.6 kgf	
Empty Weight		142.1 kgf	Operating Weight	258.7 kgf	
			Comp. All. Ope. Stress	1137.6 kgf/cm²	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 1203.0 \cdot (\exp(23.531/(1784.411 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 7.9844 + 1.5 + 0.0 = 9.4844 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1784.411 \cdot 1.0 \cdot \ln((2 \cdot 14.5 + 1203.0) / 1203.0) - 0.077$$

$$= 42.505 - 0.077 = 42.4285 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784.411 \cdot 1.0 \cdot \ln((2 \cdot 16.0 + 1200.0) / 1200.0)$$

$$= 46.9608 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 23.531 \cdot 1203.0 / (1.0 (1232.0 - 1203.0))$$

$$= 976.1268 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 23.531 / (1.0 \cdot \ln((2 \cdot 14.5 + 1203.0) / 1203.0))$$
$$= 987.8453 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

Cylinders formed from Plate $(50 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 1.316 %

Minimum Design Metal Temperature Results:

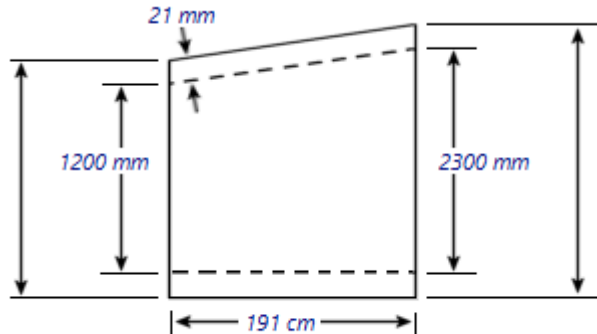
Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.2

Conical Head/Section Design Information:					
Design Pressure	P	23.601 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		191.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.9 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress	1608.3 kgf/cm²	
Spec. Min. Thk t _e , or sa		21.0000 mm	Spec. Nominal Thk.	21.000 mm	
Surface Area		112301.1 cm²	SG of Contents	0.640	
Element Volume		3745909.5 cm³	Weight of Contents	2396.3 kgf	
Empty Weight		1999.5 kgf	Operating Weight	4395.8 kgf	
			Comp. All. Ope. Stress	1204.4 kgf/cm²	
			To End Diameter	1200.000 mm	

Radiography Information:			
...			...
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Thickness due to Internal Pressure: Equation 4.3.4.1 [tr]:

$$= 0.5 \cdot D_c / \cos(\alpha) \cdot (\exp(P / (S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2303.0 / \cos(29.938) \cdot (\exp(23.601 / (1650.863 \cdot 1.0)) - 1) + c_i + c_o$$

$$= 19.1335 + 1.5 + 0.0 = 20.6335 \text{ mm}$$

Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor}$$

$$= 1650.863 \cdot 1.0 \cdot \ln(2 \cdot 19.5 \cdot \cos(29.938) + 2303.0) / 2303.0$$

$$= 24.05 \text{ kgf/cm}^2$$

Maximum Allowable Pressure New and Cold [MAPnc]:

$$= S_a \cdot E \cdot \ln(2 \cdot t \cdot \cos(\alpha) + D) / D$$

$$= 1784.442 \cdot 1.0 \cdot \ln(2 \cdot 21.0 \cdot \cos(29.938) + 2300.0) / 2300.0$$

$$= 28.0163 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D) \cos(\alpha))$$

$$= 23.601 \cdot 2303.0 / (1.0 (2342.0 - 2303.0) \cos(29.938))$$

$$= 1608.292 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln(2 \cdot t_{cor} \cdot \cos(\alpha) + D_{cor}) / D_{cor})$$



$$= 23.601 / (1.0 \cdot \ln((2 \cdot 19.5 \cdot \cos(29.938) + 2303.0) / 2303.0))$$
$$= 1620.058 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cones formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 1.715 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: SH.2

Cone From 60 to 70 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 2204.11 / \sqrt{(1355.047 \cdot 19.5)}$$
$$= 13.559$$

Compute the Factor [Ch]:

$$\text{Since } M_x > 13 \text{ and } M_x < 2 \cdot (D_o/t)_{0.94}, \text{ Ch} = 1.12 \cdot M_x - 1.058$$
$$= 1.12 \cdot 13.559 - 1.058$$
$$= 0.0710$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot \text{Ch} \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.071 \cdot 2057748 (19.5 / 2710.094)$$
$$= 1682.177 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= F_{he} / F_y$$
$$= 1682.2 / 2671.6$$
$$= 0.630$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 2671.6 / 4921.5$$
$$= 0.543$$

Ratio [AE]:

$$= F_{he} / E$$
$$= 1682.2 / 2057748$$
$$= 0.000817$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / (\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.543) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 2671.6 (1 + 0.002) / \ln(1 + 0.002)^{0.133118}$$
$$= 6123.286 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.543)$$
$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (4921.5 \cdot \exp(0.274)) / (0.274^{0.274})$$

= 9232.343 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.5431.5 - 0.5 \cdot 0.5432.5 - 0.5433.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.904E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.176E-03
3	1845	-3.966	0.249E-06	0.248E-06	0.279E-05	-0.278E-05	0.101E+07	0.101E-02
4	1537	-4.698	0.759E-07	0.759E-07	0.172E-05	-0.172E-05	0.157E+07	0.165E-03
5	1384	-5.064	0.382E-07	0.382E-07	0.130E-05	-0.130E-05	0.178E+07	0.386E-04
6	1461	-4.881	0.543E-07	0.543E-07	0.150E-05	-0.150E-05	0.168E+07	0.521E-04
7	1422	-4.973	0.457E-07	0.457E-07	0.140E-05	-0.140E-05	0.173E+07	0.437E-05
8	1403	-5.018	0.418E-07	0.418E-07	0.135E-05	-0.135E-05	0.175E+07	0.177E-04
9	1413	-4.995	0.437E-07	0.437E-07	0.138E-05	-0.138E-05	0.174E+07	0.680E-05
10	1417	-4.984	0.447E-07	0.447E-07	0.139E-05	-0.139E-05	0.174E+07	0.125E-05
11	1420	-4.978	0.452E-07	0.452E-07	0.139E-05	-0.139E-05	0.173E+07	0.155E-05
12	1419	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.147E-06
13	1418	-4.983	0.448E-07	0.448E-07	0.139E-05	-0.139E-05	0.174E+07	0.552E-06
14	1418	-4.982	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.203E-06
15	1418	-4.982	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.277E-07
16	1418	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.598E-07
17	1418	-4.981	0.449E-07	0.449E-07	0.139E-05	-0.139E-05	0.174E+07	0.161E-07

Factor of Safety [FS]:

Since Fic [1418.9] < 0.55 * 2671.64, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic}/\text{FS}$$

$$= 1418.901/2.0$$

$$= 709.451 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

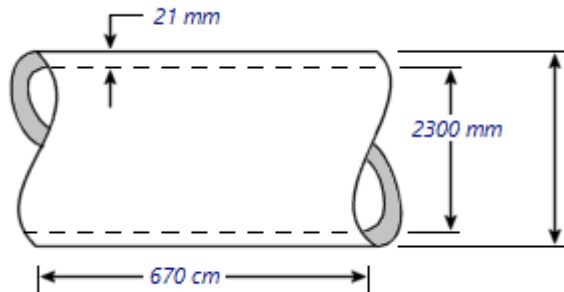
$$= 2 \cdot \text{Fha} (t / D_o)$$

$$= 2 \cdot 709.451 (19.5/2710.094)$$

$$= 10.209 \text{ kgf/cm}^2$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: SH.3

Cylindrical Shell Design Information:					
Design Pressure	P	23.601 kgf/cm²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		670.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.9 kgf/cm²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm²
Inside Diameter	Di	2300.000 mm	Actual Stress	1393.7 kgf/cm²	
Spec. Min. Thk t _e , or s _a		21.0000 mm	Spec. Nominal Thk.	21.000 mm	
Surface Area		492960.0 cm²	SG of Contents	0.640	
Element Volume		24471222.0 cm³	Weight of Contents	15654.5 kgf	
Empty Weight		7950.9 kgf	Operating Weight	23605.4 kgf	
			Comp. All. Ope. Stress	1067.1 kgf/cm²	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N+IT

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 2303.0 \cdot (\exp(23.601/(1650.863 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 16.5804 + 1.5 + 0.0 = 18.0804 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1650.863 \cdot 1.0 \cdot \ln((2 \cdot 19.5 + 2303.0) / 2303.0) - 0.147$$

$$= 27.722 - 0.147 = 27.5751 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New & Cold [MAPnc]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D)$$

$$= 1784.442 \cdot 1.0 \cdot \ln((2 \cdot 21.0 + 2300.0) / 2300.0)$$

$$= 32.2915 \text{ kgf/cm}^2$$

Computed Stress Intensity at Design Pressure [Si]:

$$= P \cdot D / (E (D_o - D))$$

$$= 23.601 \cdot 2303.0 / (1.0 (2342.0 - 2303.0))$$

$$= 1393.6862 \text{ kgf/cm}^2$$

Computed Membrane Stress at Design Pressure [Sm]:

$$= P / (E \cdot \ln((2 \cdot t + D) / D))$$



$$= 23.601 / (1.0 \cdot \ln((2 \cdot 19.5 + 2303.0) / 2303.0))$$
$$= 1405.4550 \text{ kgf/cm}^2$$

Percent Fiber Elongation per Table 6.1

$$\text{Cylinders formed from Plate} \quad (50 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) \quad 0.905 \%$$

Minimum Design Metal Temperature Results:

Note: This Element/Detail was specified as being Impact Tested.

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: SH.3

Cylindrical Shell From 70 to 80 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Determine Parameter [Mx]:

$$= L / \sqrt{(R_o \cdot t)}$$
$$= 6941.667 / \sqrt{(1171.0 \cdot 19.5)}$$
$$= 45.938$$

Compute the Factor [Ch]:

Since $M_x > 13$ and $M_x < 2 \cdot (D_o/t)0.94$, $Ch = 1.12 \cdot M_x - 1.058$

$$= 1.12 \cdot 45.938 - 1.058$$
$$= 0.0195$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$= 1.6 \cdot Ch \cdot E_y (t / D_o)$$
$$= 1.6 \cdot 0.02 \cdot 2057748 (19.5 / 2342.0)$$
$$= 535.309 \text{ kgf/cm}^2$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$= Fhe / F_y$$
$$= 535.3 / 2671.6$$
$$= 0.200$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$= \sigma_{ys} / \sigma_{uts}$$
$$= 2671.6 / 4921.5$$
$$= 0.543$$

Ratio [AE]:

$$= Fhe / E$$
$$= 535.3 / 2057748$$
$$= 0.000260$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$
$$= (\ln(0.543) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$$
$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$$
$$= 2671.6 (1 + 0.002) / \ln(1 + 0.002)^{0.133118}$$
$$= 6123.286 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m2f(m2f2 - R)$$
$$= 0.6(1.0 - 0.543)$$
$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$$
$$= (4921.5 \cdot \exp(0.274)) / (0.274^{0.274})$$



= 9232.343 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$$

$$= 1.5 \cdot 0.5431.5 - 0.5 \cdot 0.5432.5 - 0.5433.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.959E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.382E-03
3	615	-6.894	0.194E-09	0.194E-09	0.152E-06	-0.152E-06	0.206E+07	0.391E-04
4	307	-7.626	0.213E-11	0.213E-11	0.244E-07	-0.244E-07	0.206E+07	0.111E-03
5	461	-7.260	0.299E-10	0.299E-10	0.712E-07	-0.712E-07	0.206E+07	0.359E-04
6	538	-7.077	0.815E-10	0.815E-10	0.107E-06	-0.107E-06	0.206E+07	0.154E-05
7	499	-7.169	0.503E-10	0.503E-10	0.880E-07	-0.880E-07	0.206E+07	0.172E-04
8	519	-7.123	0.643E-10	0.643E-10	0.973E-07	-0.973E-07	0.206E+07	0.783E-05
9	528	-7.100	0.725E-10	0.725E-10	0.102E-06	-0.102E-06	0.206E+07	0.315E-05
10	533	-7.089	0.769E-10	0.769E-10	0.105E-06	-0.105E-06	0.206E+07	0.806E-06
11	535	-7.083	0.792E-10	0.792E-10	0.106E-06	-0.106E-06	0.206E+07	0.365E-06
12	534	-7.086	0.780E-10	0.780E-10	0.105E-06	-0.105E-06	0.206E+07	0.220E-06
13	535	-7.084	0.786E-10	0.786E-10	0.106E-06	-0.106E-06	0.206E+07	0.724E-07
14	534	-7.085	0.783E-10	0.783E-10	0.105E-06	-0.105E-06	0.206E+07	0.739E-07
15	535	-7.085	0.784E-10	0.784E-10	0.105E-06	-0.105E-06	0.206E+07	0.775E-09

Factor of Safety [FS]:

Since Fic [535.13] < 0.55 * 2671.64, FS = 2

Allowable Circumferential Compressive Stress [Fha]:

$$= \text{Fic}/\text{FS}$$

$$= 535.134/2.0$$

$$= 267.567 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

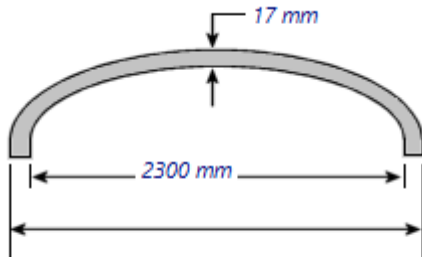
$$= 2 \cdot \text{Fha}(t / \text{Do})$$

$$= 2 \cdot 267.567(19.5/2342.0)$$

$$= 4.456 \text{ kgf/cm}^2$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 2, 2025



Internal Pressure Results for: DISH END

Elliptical Head Design Information:					
Design Pressure	P	23.601 kgf/cm ²	Int. Design Temperature	80 °C	
Material		SA-516 70	Ext. Design Temperature	37 °C	
Length		5.000 cm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	1.5000 mm	Allowable Stress (ope)	S	1650.9 kgf/cm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	1784.4 kgf/cm ²
Inside Diameter	Di	2300.000 mm	Actual Stress	1583.0 kgf/cm ²	
Spec. Min. Thk t _e , or s _a		17.0000 mm	Spec. Nominal Thk.	21.000 mm	
Surface Area		63135.0 cm ²	SG of Contents	0.640	
Element Volume		1800398.8 cm ³	Weight of Contents	1151.7 kgf	
Empty Weight		1108.7 kgf	Operating Weight	2260.4 kgf	
			Comp. All. Ope. Stress	1067.1 kgf/cm ²	
			Aspect Ratio	2.00	

Radiography Information:			
...		...	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N+IT

Head Calculation - Section 4.3.6 (and Section 4.3.7 for Ellipsoidal Head)

Computed Minimum Required Thickness [t]:

tr = 14.8605 mm

$$\begin{aligned}
 t &= tr + ci + co \\
 &= 14.8605 + 1.5 + 0.0 \\
 &= 16.3605 \text{ mm - see below for the derivation}
 \end{aligned}$$

For Elliptical heads: Compute [h, k, r,] per 4.3.7.1

$$\begin{aligned}
 h &= D/(2 \cdot \text{Aspect Ratio}) \\
 &= 2300.0/(2 \cdot 2.0) \\
 &= 575.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 k &= D/(2 \cdot h) \\
 &= 2300.0/(2 \cdot 575.0) \\
 &= 2.000
 \end{aligned}$$

$$r = D(0.5/k - 0.08) + ca$$



$$= 2300.0(0.5/2.0 - 0.08) + 1.5$$

$$= 392.500 \text{ mm}$$

$$L = D(0.44 \cdot k + 0.02) + ca$$

$$= 2300.0(0.44 \cdot 2.0 + 0.02) + 1.5$$

$$= 2071.500 \text{ mm}$$

Step 3: Geometric Constants [βTh , ϕTh , Rth]:

Equations 4.3.8 to 4.3.11 as appropriate

t (corroded) and P for the calc.: $t = 14.8605 \text{ mm}$, $P = 23.6013 \text{ kgf/cm}^2$

Angle [βTh]:

$$= \arccos[(0.5 \cdot D - r)/(L - r)]$$

$$= \arccos[(0.5 \cdot 2303.0 - 392.5)/(2071.5 - 392.5)]$$

$$= 1.1017 \text{ radians - Equation 4.3.8}$$

$$\phi Th = \sqrt{(L \cdot t) / r}$$

$$= \sqrt{(2071.5 \cdot 14.861) / 392.5}$$

$$= 0.447 \text{ radians - Equation 4.3.9}$$

$\phi Th < \beta Th$, Thus:

$$Rth = (0.5 \cdot D - r) / \cos(\beta Th - \phi Th) + r$$

$$= (0.5 \cdot 2303.0 - 392.5) / \cos(1.102 - 0.447) + 392.5$$

$$= 1349.3585 \text{ mm - Equation 4.3.10}$$

Step 4: Coefficients: Equations 4.3.12 to 4.3.15 [$C1$ and $C2$]:

$r/D > 0.08$, Thus:

$$C1 = 0.692(r/D) + 0.605$$

$$= 0.692(392.5/2303.0) + 0.605$$

$$= 0.7229 \text{ - Equation 4.3.13}$$

$$C2 = 1.46 - 2.6(r/D) = 1.46 - 2.6(392.5/2303.0)$$

$$= 1.0169, \text{ Equation 4.3.15}$$

Step 5: Pressure to produce knuckle elastic buckling [$Peth$]:

$$= C1 \cdot Et \cdot t^2 / (C2 \cdot Rth(0.5 \cdot Rth - r))$$

$$= 0.723 \cdot 2032634 \cdot 14.861^2 / (1.017 \cdot 1349.359(0.5 \cdot 1349.359 - 392.5))$$

$$= 838.12 \text{ kgf/cm}^2, \text{ Equation 4.3.16}$$

Step 6: Internal Pressure for max stress in knuckle equals yield [Py]:

$C3 = Sy = 2477.825 \text{ kgf/cm}^2$, Not in the creep range - 4.3.6.1(f)

$$Py = C3 \cdot t / (C2 \cdot Rth(0.5 \cdot Rth / r - 1))$$

$$= 2477.825 \cdot 14.861 / (1.017 \cdot 1349.359(0.5 \cdot 1349.359 / 392.5 - 1))$$

$$= 37.327 \text{ kgf/cm}^2, \text{ Equation 4.3.17}$$

Step 7: Value of Internal Press. resulting in buckling of knuckle [G and Pck]:

$$G = Peth/Py = 838.116/37.327 = 22.453$$

$G > 1$, Thus:

Buckling Pressure [Pck]:

$$= ((0.77508 \cdot G - 0.20354 \cdot G^2 + 0.019274 \cdot G^3) /$$

$$(1 + 0.19014 \cdot G - 0.089534 \cdot G^2 + 0.0093965 \cdot G^3)) Py$$

$$= ((0.77508 \cdot 22.453 - 0.20354 \cdot 22.453^2 + 0.019274 \cdot 22.453^3) /$$

$$(1 + 0.19014 \cdot 22.453 - 0.089534 \cdot 22.453^2 + 0.0093965 \cdot 22.453^3)) \cdot 37.327$$

$$= 74.64 \text{ kgf/cm}^2, \text{ Equation 4.3.19}$$

Step 8: Compute Allowable Pressure base on buckling of knuckle [Pak]:

$$= Pck/1.5 = 74.64/1.5 = 49.76 \text{ kgf/cm}^2 \text{ - Equation 4.3.21}$$

Step 9: Compute allowable pressure based on rupture of crown [Pac]:

$$= 2 \cdot S \cdot E / (L/t + 0.5)$$

$$= 2 \cdot 1650.863 \cdot 1.0 / (2071.5/14.861 + 0.5)$$

$$= 23.6 \text{ kgf/cm}^2, \text{ Equation 4.3.22}$$

Step 10: Compute final pressure [Pa]:

$$= \min(Pak, Pac) = \min(49.759, 23.601)$$

$$= 23.6 \text{ kgf/cm}^2, \text{ Equation 4.3.23: (same as original design pressure)}$$



Allowable pressure in Crown at given thickness [Pac]:

$$\begin{aligned} &= 2 \cdot S \cdot E / (L / t + 0.5) \\ &= 2 \cdot 1650.863 \cdot 1.0 / (2071.5 / 15.5 + 0.5) \\ &= 24.613 \text{ kgf/cm}^2 \end{aligned}$$

Computed Maximum Allowable Pressure [MAWP]:

$$= \text{MaxPressure} - \text{Pliq} = 24.6131 - 0.147 = 24.4659 \text{ kgf/cm}^2$$

Computed Maximum Allowable Pressure New and Cold [MAPnc]:

$$= 29.1898 \text{ kgf/cm}^2$$

Computed Membrane Stress Intensity at Design Pressure [Sm]:

$$= 1583.0 \text{ kgf/cm}^2$$

Required Thickness of Straight Flange = 18.080 mm

Percent Fiber Elongation per Table 6.1

$$\begin{aligned} \text{For Heads formed from Segments } (75 \cdot t_{\text{nom}} / R_f) \cdot (1 - R_f / R_o) & \quad 3.923 \% \\ \text{For one Piece doubly Curved Heads } 100 \cdot \ln(D_b / (D_f - 2t)) & \quad 20.922 \% \end{aligned}$$

Where D_b is the blank diameter for one piece doubly curved surfaces [D_b]:

$$\begin{aligned} &= \text{square root}(4 \cdot \text{Head Surface Area} / 3.14159) \\ &= \text{square root}(4 \cdot 63135.0 / 3.14159) \\ &= 2835.242 \text{ mm} \end{aligned}$$

MDMT Calculations in the Knuckle Portion:

Note: This Element/Detail was specified as being Impact Tested.

MDMT Calculations in the Head Straight Flange:

Note: This Element/Detail was specified as being Impact Tested.

Tolerance for Formed Heads per 4.3.2.2:

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 28.750 mm
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 14.375 mm

Material treatment options: +N+IT

EXTERNAL PRESSURE RESULTS for: DISH END

Elliptical Head From 80 to 90 Ext. Chart: CS-2 at 37 °C

Maximum External Pressure per ASME Division 2 Section 4.4:

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} &= 0.075 \cdot E_y(t / R_o) \\ &= 0.075 \cdot 2057748(15.5 / 2071.95) \\ &= 1154.532 \text{ kgf/cm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} &= F_{he} / S_y \\ &= 1154.5 / 2671.6 \\ &= 0.432 \end{aligned}$$

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} &= \sigma_y / \sigma_{uts} \\ &= 2671.6 / 4921.5 \\ &= 0.543 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} &= F_{he} / E \\ &= 1154.5 / 2057748 \\ &= 0.000561 \end{aligned}$$

Factor [m1]:

$$= (\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$$

$$= (\ln(0.543) + (0.00002 - 0.002)) / (\ln(1 + 0.00002) / \ln(1 + 0.002))$$

$$= 0.133$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$= \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys}) m_1$$

$$= 2671.6 (1 + 0.002) / \ln(1 + 0.002) 0.133118$$

$$= 6123.286 \text{ kgf/cm}^2$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$= m_2 f(m_2^2 - R)$$

$$= 0.6 (1.0 - 0.543)$$

$$= 0.274$$

Curve Fitting Constant [A2]:

$$= (\sigma_{uts} \cdot \exp(m_2)) / (m_2 m_2)$$

$$= (4921.5 \cdot \exp(0.274)) / (0.274 0.274)$$

$$= 9232.343 \text{ kgf/cm}^2$$

Material Parameter for Stress-Strain Curve Model [K]:

$$= 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5}$$

$$= 1.5 \cdot 0.5431.5 - 0.5 \cdot 0.5432.5 - 0.5433.5$$

$$= 0.374$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	2460	-2.502	0.162E-05	0.156E-05	0.597E-05	-0.564E-05	0.250E+06	0.929E-02
2	1230	-5.430	0.177E-07	0.177E-07	0.954E-06	-0.954E-06	0.192E+07	0.806E-04
3	615	-6.894	0.194E-09	0.194E-09	0.152E-06	-0.152E-06	0.206E+07	0.262E-03
4	922	-6.162	0.273E-08	0.273E-08	0.446E-06	-0.446E-06	0.203E+07	0.108E-03
5	1076	-5.796	0.744E-08	0.744E-08	0.670E-06	-0.670E-06	0.200E+07	0.218E-04
6	1153	-5.613	0.117E-07	0.117E-07	0.804E-06	-0.804E-06	0.196E+07	0.264E-04
7	1115	-5.705	0.935E-08	0.935E-08	0.735E-06	-0.735E-06	0.198E+07	0.169E-05
8	1095	-5.750	0.835E-08	0.835E-08	0.702E-06	-0.702E-06	0.199E+07	0.102E-04
9	1105	-5.727	0.884E-08	0.884E-08	0.719E-06	-0.719E-06	0.199E+07	0.430E-05
10	1110	-5.716	0.909E-08	0.909E-08	0.727E-06	-0.727E-06	0.198E+07	0.131E-05
11	1112	-5.710	0.922E-08	0.922E-08	0.731E-06	-0.731E-06	0.198E+07	0.188E-06
12	1111	-5.713	0.915E-08	0.915E-08	0.729E-06	-0.729E-06	0.198E+07	0.562E-06
13	1112	-5.712	0.919E-08	0.918E-08	0.730E-06	-0.730E-06	0.198E+07	0.187E-06
14	1112	-5.711	0.920E-08	0.920E-08	0.731E-06	-0.731E-06	0.198E+07	0.238E-09

Factor of Safety [FS]:

$$\text{Since } F_{ic} [1112.32] < 0.55 \cdot 2671.64, FS = 2$$

Allowable Circumferential Compressive Stress [Fha]:

$$= F_{ic} / FS$$

$$= 1112.322 / 2.0$$

$$= 556.161 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot F_{ha} [t/R_o]$$

$$= 2 \cdot 556.161 [15.5/2071.95]$$

$$= 8.321 \text{ kgf/cm}^2$$

Hydrostatic Test Pressure Results:

Hydrotest Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{ShellSide MAWP(St/S)}$$

$$= 1.25 \cdot 23.903(1.0)$$

$$= 29.878 \text{ kgf/cm}^2$$

Hydrotest Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Channel Side MAWP(St/S)}$$

$$= 1.25 \cdot 164.434(1.072)$$

$$= 220.427 \text{ kgf/cm}^2$$

Horizontal Test performed per: per VIII-2, Section 8.2

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (kgf/cm²):

From To	Stress	Allowable	Ratio	Pressure
CH. SHELL	2103.2	2518.6	0.835	220.55
SH.1	1259.4	2518.6	0.500	30.00
SH.2	2066.7	2518.6	0.821	30.11
SH.3	1793.0	2518.6	0.712	30.11
DISH END	2019.4	2518.6	0.802	30.11

Stress ratios for Nozzle and Pad Materials (kgf/cm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
NOZZLE N1	Nozzle	1682.44	1568.83	1.072
NOZZLE N2	Nozzle	1682.44	1568.83	1.072
Noz N9	Nozzle	1641.66	1522.22	1.078
Noz N8	Nozzle	1641.66	1522.22	1.078
Noz N8	Pad	1784.41	1650.40	1.081
Noz N4	Nozzle	1641.66	1522.22	1.078
Noz N4	Pad	1784.41	1650.40	1.081
Noz N5	Nozzle	1641.66	1522.22	1.078
Noz N5	Pad	1784.41	1650.40	1.081
Noz N3	Nozzle	1641.66	1522.22	1.078
Noz N3	Pad	1784.41	1650.40	1.081
Noz N6	Nozzle	1641.66	1522.22	1.078
Noz N6	Pad	1784.41	1650.40	1.081
Noz N7	Nozzle	1641.66	1522.22	1.078
Noz K2A	Nozzle	1641.66	1522.22	1.078
Noz K2B	Nozzle	1641.66	1522.22	1.078
Noz K3	Nozzle	1641.66	1522.22	1.078
Noz K4	Nozzle	1641.66	1522.22	1.078
Noz K5	Nozzle	1641.66	1522.22	1.078
Noz K1A	Nozzle	1641.66	1522.22	1.078
Noz K1B	Nozzle	1641.66	1522.22	1.078
MH1	Nozzle	1641.66	1522.22	1.078
MH1	Pad	1784.41	1650.40	1.081

Minimum

1.072

Stress ratios for Pressurized Vessel Elements (kgf/cm²):

Description	Ambient	Operating	Ratio
TS END COVER	1682.44	1568.83	1.072
TS FLG1	1682.44	1568.83	1.072
CH. SHELL	1784.41	1650.40	1.081
SH.1	1784.41	1784.41	1.000
SH.2	1784.44	1650.86	1.081
SH.3	1784.44	1650.86	1.081



DISH END	1784.44	1650.86	1.081
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Minimum 1.000

Stress ratios for Exchanger Materials (kgf/cm²):

Description	Ambient	Operating	Ratio
Tube Material	1407.13	1308.09	1.076
Tubesheet Material	1682.44	1568.83	1.072

Minimum 1.072

Hoop Stress in Nozzle Wall during Pressure Test (kgf/cm²):

Description	Ambient	Operating	Ratio
NOZZLE N1	1395.18	2421.70	0.576
NOZZLE N2	1395.18	2421.70	0.576
Noz N9	803.81	2324.83	0.346
Noz N8	109.33	2324.83	0.047
Noz N4	200.16	2324.83	0.086
Noz N5	200.16	2324.83	0.086
Noz N3	200.16	2324.83	0.086
Noz N6	200.65	2324.83	0.086
Noz N7	77.42	2324.83	0.033
Noz K2A	67.94	2324.83	0.029
Noz K2B	67.94	2324.83	0.029
Noz K3	56.39	2324.83	0.024
Noz K4	67.94	2324.83	0.029
Noz K5	67.94	2324.83	0.029
Noz K1A	67.94	2324.83	0.029
Noz K1B	67.94	2324.83	0.029
MH1	296.44	2324.83	0.128

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: U-TUBE SHEET

Shell Data:

Main Shell Description: SH.1

Shell Test Pressure		29.88 kgf/cm ²
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1200.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	1784.41 kgf/cm ²
Shell Allowable Stress at Ambient		1784.41 kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder
Channel Test Pressure		220.43 kgf/cm ²
Channel Thickness	tc	68.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1200.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	1784.41 kgf/cm ²
Channel Allowable Stress at Ambient		1784.41 kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1066
Tube Wall Thickness	et	2.5000 mm
Tube Outside Diameter	D	25.0000 mm
Total Straight Tube Length	Lt	7997.00 mm
Straight Tube Length (bet. inner tubsht faces) L		7687.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-334 1
Tube Material UNS Number		K03008
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		1407.13 kgf/cm ²
Tube Allowable Stress At Ambient		1407.13 kgf/cm ²
Tube Yield Stress At design Temperature	Syt	2090.31 kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.5000 mm
Groove Weld Leg	ag	2.5000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		4.21.2
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	584.500 mm
Largest Center-to-Center Tube Distance	Ul	47.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	8.0000 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-350 LF2

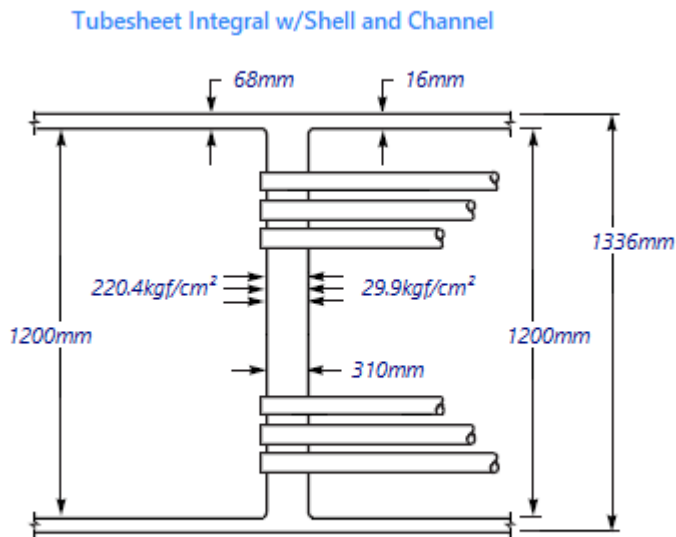
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number		K03011
Tubesheet Allowable Stress at Temperature	S	1682.44 kgf/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1682.44 kgf/cm ²
Thickness of Tubesheet	h	310.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	1336.000 mm

Area of the Untubed Lanes	AL	549.4 cm ²
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Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Tubesheet Integral with	Both
Tubesheet Extended as Flange	No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20670E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20670E+07 kgf/cm ²
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Elastic Mod. at Ambient Temperature 21.1 °C 0.20670E+07 kgf/cm²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

$$\begin{aligned} & \text{Computed Minimum Required Thickness [t]:} \\ & = 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i \quad \text{Equation 4.3.1} \\ & = 0.5 \cdot 20.0 \cdot (\exp(220.427/(1407.1 \cdot 1.0)) - 1) + 0 \\ & = 1.6959 + 0 \\ & = 1.6959 \text{ mm} \end{aligned}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

$$\begin{aligned} & = L / \sqrt{(R_o \cdot t)} \\ & = 8271.5 / \sqrt{(12.5 \cdot 2.5)} \\ & = 1479.651 \end{aligned}$$

Compute the Factor [Ch]:

$$\begin{aligned} & \text{Since } M_x > 2 \cdot (D_o/t)^{0.94}, \text{ Ch} = 0.55 / (D_o/t) \\ & = 0.55 / (25.0/2.5) \\ & = 0.0550 \end{aligned}$$

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

$$\begin{aligned} & = 1.6 \cdot \text{Ch} \cdot E_y(t / D_o) \\ & = 1.6 \cdot 0.055 \cdot 2067025 (2.5/25.0) \\ & = 18189.824 \text{ kgf/cm}^2 \end{aligned}$$

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

$$\begin{aligned} & = F_{he} / F_y \\ & = 18189.8 / 2090.3 \\ & = 8.702 \end{aligned}$$

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

$$\begin{aligned} & = \sigma_{ys} / \sigma_{uts} \\ & = 2090.3 / 3866.9 \\ & = 0.541 \end{aligned}$$

Ratio [AE]:

$$\begin{aligned} & = F_{he} / E \\ & = 18189.8 / 2067025 \\ & = 0.008800 \end{aligned}$$

Factor [m1]:

$$\begin{aligned} & = (\ln(r) + (\epsilon_{ps} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{ps}) / \ln(1 + \epsilon_{sys})) \\ & = (\ln(0.541) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002)) \\ & = 0.134 \end{aligned}$$

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

$$\begin{aligned} & = \sigma_{ys} (1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1} \\ & = 2090.3 (1 + 0.002) / \ln(1 + 0.002)^{0.134034} \\ & = 4818.264 \text{ kgf/cm}^2 \end{aligned}$$

Curve Fitting Exponent for, Ferritic Steel [m2]:

$$\begin{aligned} & = m2f / (m2f^2 - R) \\ & = 0.6 / (1.0 - 0.541) \\ & = 0.276 \end{aligned}$$

Curve Fitting Constant [A2]:

$$\begin{aligned} & = (\sigma_{uts} \cdot \exp(m2)) / (m2^{m2}) \\ & = (3866.9 \cdot \exp(0.276)) / (0.276^{0.276}) \\ & = 7266.839 \text{ kgf/cm}^2 \end{aligned}$$

Material Parameter for Stress-Strain Curve Model [K]:

$$\begin{aligned} & = 1.5 \cdot R_{1.5} - 0.5 \cdot R_{2.5} - R_{3.5} \\ & = 1.5 \cdot 0.5411.5 - 0.5 \cdot 0.5412.5 - 0.5413.5 \\ & = 0.373 \end{aligned}$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1933	-2.474	0.212E-05	0.205E-05	0.770E-05	-0.724E-05	0.196E+06	0.107E-02
2	966	-5.395	0.241E-07	0.241E-07	0.125E-05	-0.125E-05	0.188E+07	0.829E-02
3	1450	-3.934	0.331E-06	0.330E-06	0.361E-05	-0.360E-05	0.866E+06	0.713E-02
4	1691	-3.204	0.896E-06	0.889E-06	0.542E-05	-0.535E-05	0.428E+06	0.485E-02
5	1812	-2.839	0.140E-05	0.138E-05	0.650E-05	-0.632E-05	0.291E+06	0.257E-02
6	1873	-2.656	0.173E-05	0.169E-05	0.708E-05	-0.680E-05	0.239E+06	0.964E-03
7	1903	-2.565	0.192E-05	0.186E-05	0.738E-05	-0.702E-05	0.216E+06	0.525E-05
8	1918	-2.520	0.202E-05	0.195E-05	0.754E-05	-0.713E-05	0.206E+06	0.519E-03
9	1910	-2.542	0.197E-05	0.191E-05	0.746E-05	-0.708E-05	0.211E+06	0.253E-03
10	1907	-2.554	0.194E-05	0.188E-05	0.742E-05	-0.705E-05	0.214E+06	0.123E-03
11	1905	-2.560	0.193E-05	0.187E-05	0.740E-05	-0.704E-05	0.215E+06	0.585E-04
12	1904	-2.562	0.192E-05	0.187E-05	0.739E-05	-0.703E-05	0.216E+06	0.266E-04
13	1903	-2.564	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.107E-04
14	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.270E-05
15	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.128E-05
16	1903	-2.565	0.192E-05	0.186E-05	0.739E-05	-0.703E-05	0.216E+06	0.708E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$= 2.407 - 0.741 \cdot Fic / F_y$

$= 2.407 - 0.741 \cdot 1903.41/2090.31$

$= 1.732$

Allowable Circumferential Compressive Stress [Fha]:

$= Fic/FS$

$= 1903.41/1.732$

$= 1098.805 \text{ kgf/cm}^2$

Allowable External Pressure at the given Thickness [Pa]:

$= 2 \cdot Fha(t / Do)$

$= 2 \cdot 1098.805 (2.5/25.0)$

$= 219.761 \text{ kgf/cm}^2$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.6959 mm
Allowable Internal Pressure at Corroded thickness	313.99 kgf/cm ²
Required Internal Design Pressure	220.43 kgf/cm ²
Allowable External Pressure at Corroded thickness	219.76 kgf/cm ²
Required External Design Pressure	29.88 kgf/cm ²
Required Thickness due to Shell Side pressure	0.6804 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$

$= 2 \cdot 584.5 + 25.0$

$= 1194.000 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$

$= (32.0 - 25.0) / 32.0$

$= 0.219$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Ds / Do$ (Configurations a, b, c)



$$= 1200.0/1194.0$$

$$= 1.0050$$

Ratio [pc]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1200.0/1194.0$$

$$= 1.0050$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:

$$= l_{tx} / h$$

$$= 0/310.0$$

$$= 0.000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$= \max(dt - 2tt(E_t/E) \cdot (StT/S) \cdot (\rho); dt - 2tt)$$

$$= \max(25.0 - 2 \cdot 2.5 (2067025/2067025) \cdot$$

$$(1407/1682) \cdot (0); 25.0 - 2 \cdot 2.5)$$

$$= 25.0000 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$$

$$= 32.0 / \sqrt{ (1 - 4 \cdot \min(549.43 ; 4 \cdot 1194.0 \cdot 32.0) / (\pi \cdot 1194.0^2)) }$$

$$= 32.8152 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.8152 - 25.0) / 32.8152$$

$$= 0.2382$$

$$h/p = 9.687500 ; \mu^* = 0.238159 ; E^* = 416977 \text{ kgf/cm}^2$$

$$E^*/E = 0.201728 ; \nu^* = 0.409245$$

Note: As h/p (9.688) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{ ((D_s + t_s) t_s) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1200.0 + 16.0) 16.0) }$$

$$= 0.0130 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.013 \cdot 2067025 \cdot 16.0^3 / (6(1 - 0.3^2))$$

$$= 20208870.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1200.0 + 16.0) \cdot 20208870 / 310.0^3 \cdot (1 + 310.0 \cdot 0.013 + (310.0 \cdot 0.013)^2 / 2)$$

$$= 65336.6 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1200.0^2 / (4 \cdot 2067025 \cdot 16.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 16.0)))$$

$$= 0.92739\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.005 \cdot 20208870 \cdot 0.013 \cdot 0.009274 (1 + 310.0 \cdot 0.013)$$

$$= 12372.2441 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((D_c + t_c) t_c) }$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((1200.0 + 68.0) 68.0) }$$

$$= 0.0062 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2))$$



$$= 0.006 \cdot 2067025 \cdot 68.0^3 / (6(1 - 0.3^2))$$

$$= 736922176.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1200.0 + 68.0 / \cos(0)) \cdot 0.73692E+09 / 310.0^3 \cdot (1.0 + 310.0 \cdot 0.006 \cdot (1.0 + 1.0) / 2 + (310.0 \cdot 0.006)^2 / 2 \cdot 1.0)$$

$$= 895927.6 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1200.0^2 / (4 \cdot 2067025 \cdot 68.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 68.0)))$$

$$= 0.21976E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.005 \cdot 0.73692E+09 [0.002198 (1.0 + 0.006 \cdot 1.0 \cdot 310.0) - 0 (1.0 / 310.0 + 1.0 \cdot 0.006 / 2)]$$

$$= 29413.7246 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1336.0 / 1194.0$$

$$= 1.1189$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 416977 \cdot (65336.55 + 895927 + 2067025 \cdot \ln(1.12))$$

$$= 1.6909$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$$

$$= 1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 29.878 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 220.427]$$

$$= -171492.3438 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -171492.34 + 29413.725 \cdot 220.427 - 12372.244 \cdot 29.878$$

$$= 5942417.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2 / 32 \cdot F (P_s - P_t) / (1 + F)$$

$$= (5942417) - 1194.0^2 / 32 \cdot 1.691 (29.88 - 220.43) / (1 + 1.69)$$

$$= 7542738.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2 / 64 \cdot (3 + v^*) \cdot (P_s - P_t)$$

$$= 7542738 + 1194.0^2 / 64 \cdot (3 + 0.409) \cdot (29.88 - 220.43)$$

$$= -6928045.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|7542738|; |-6928045.0|)$$

$$= 7542738.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$



ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 7542738 / (0.2382 (310.0 - 8.0)^2) \\ &= 2083.5286 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot \beta T \cdot S_y \\ &= 2 \cdot 0.95 \cdot 2549.2 \\ &= 4843.40 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 7586449 / (0.2382 (206.9446 - 8.0)^2) \\ &= 4829.0166 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [H_{reqB}]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 206.9446 + 0 + 0 \\ &= 206.9446 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [H_{req}]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(206.9446; 134.212) \\ &= 206.9446 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |29.88 - 220.43| \\ &= 190.548 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot 0.8 \cdot \beta T \cdot S_y \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot 0.8 \cdot 0.95 \cdot 2549.156 \\ &= 440.124 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / (0.8 \cdot \beta T \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1937.359) \cdot |29.88 - 220.43| + 0 \\ &= 134.2120 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 16.000mm

Min. Shell len. adjacent to the tubesheet is: 249.415 mm

Note: For a given Channel thickness of: 68.000 mm

Min. Channel len. adjacent to the tubesheet is: 514.183mm

Tube Metal Area [A_t]:

$$\begin{aligned} &= \pi (d_o - t) t \\ &= \pi (25.0 - 2.5) 2.5 \\ &= 1.767 \text{ cm}^2 \end{aligned}$$

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [F_t]:

$$\begin{aligned} &= \pi \cdot t (d_o - t) S \\ &= \pi \cdot 2.5 (25.0 - 2.5) 1985.793 \\ &= 3509.180 \text{ kgf} \end{aligned}$$

Groove Weld Strength [F_g]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$



$$= 0.85 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1985.79$$

$$= 3509.1802 \text{ kgf}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 1985.793/1985.793$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1985.79$$

$$= 2288.1804 \text{ kgf}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.0)^2 + 1.07 \cdot 2.5 \cdot (25.0 - 2.5) 1.0 \cdot 1.0)} - 0.75 \cdot 25.0 \right)$$

$$= 3.083 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm

Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 3509.18 is $\geq$ tube strength 3509.18 kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2083.5 $\leq$	4843.4 kgf/cm ²	Ok
Tubesheet shear stress	838.8 $\leq$	1937.4 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	206.945	310.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.500	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.500	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	219.957 ...	2350	4843 970 1937	Ps+Pt D1	Ok
D2uc	89.433 ...	361	4843 132 1937	Ps+Pt D2	Ok
D3uc	206.945 ...	2084	4843 839 1937	Ps+Pt D3	Ok
Max:	219.9567 ... mm	0.485	0.501 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures



[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.							
Case#	Shell	Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	...	3778	...	3778	Ok
D2uc	...	...	...	3778	...	3778	Ok
D3uc	...	...	...	3778	...	3778	Ok
Max	...	...	...	...	...	...	...

Note:
 Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	219.957	310.000	Ok
Tube thickness :	1.696	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.500	Ok
Tube-tubesheet groove weld leg :	1.542	2.500	Ok

Given Shell Thickness: 16.0000 mm

Given Channel Thickness: 68.0000 mm

Min Shell length of thk, (16.000) adj. to tubesheet: 249.415 mm

Min Channel length of thk, (68.000) adj. to tubesheet: 514.183 mm

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 2, 2025

External Pressure Calculations:

From	To	Section Length cm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B kgf/cm ²
20	30	No Calc	...	301.5	No Calc	No Calc
30	40	No Calc	...	291.5	No Calc	No Calc
40	Tsht	124.997	1336	66.5	No Calc	No Calc
Tsht	50	No Calc	...	66.5	No Calc	No Calc
50	60	30.0031	1232	14.5	No Calc	No Calc
60	70	191	2348.47	19.5	No Calc	No Calc
70	80	694.167	2342	19.5	No Calc	No Calc
80	90	No Calc	2334	15.5	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure kgf/cm ²	External M.A.W.P. kgf/cm ²
20	30	303	200.355	...	No Calc
30	40	293	286.283	...	No Calc
40	Tsht	68	No Calc	...	151.127
Tsht	50	68	No Calc	...	No Calc
50	60	16	No Calc	...	31.1031
60	70	21	8.9296	1.05461	10.2094
70	80	21	12.5282	1.05461	4.45564
80	90	17	6.91643	1.05461	8.32114

Minimum

4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners cm	Allowable Length Bet. Stiffeners cm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
20	30	No Calc	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	Tsht	124.997	No Calc	No Calc	No Calc
Tsht	50	No Calc	No Calc	No Calc	No Calc
50	60	30.0031	No Calc	No Calc	No Calc
60	70	191	191	No Calc	No Calc
70	80	694.167	2710.68	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume cm ³	Corroded Metal Wgt. kg	Corroded ID Volume cm ³	Extra due Misc % kg
20	30	5797.61	...	5797.61	...	579.76
30	40	3173.06	455197	3155.46	455792	317.306
40	50	2624.28	1413719	2569.43	1420797	262.428
50	60	142.117	182311	128.953	184010	14.2117
60	70	1999.46	3745909	1858.45	3764104	199.946
70	80	7950.93	24471222	7387.8	24543884	795.093
80	90	1108.66	1800398	1029.47	1807181	110.866
Total		22796	32068758.00	21927	32175768.00	2279



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. cm	Y Offset, Dtl. Cent. cm	Z Offset Dtl. Cent. cm	Description
20	Ins1	15.0528	15.15	...	...	Ins: 20
30	Liqd	291.195	19.9	...	...	Liquid: 30
30	Ins1	9.01284	19.9	...	...	Ins: 30
40	Liqd	904.373	62.5	...	...	Liquid: 40
40	Ins1	26.2413	62.5	...	...	Ins: 40
40	Noz1	419.579	62.5	79.3675	...	NOZZLE N1
40	Noz1	478.862	62.5	-79.3675	...	NOZZLE N2
40	Noz1	29.1176	35	...	62.1425	Noz N9
50	Liqd	116.626	15	...	...	Liquid: 50
50	Ins1	5.84652	15	...	...	Ins: 50
50	Noz1	11.9912	15	-62.1425	...	Noz N8
60	Liqd	2396.29	75.3861	-27.5	...	Liquid: 60
60	Ins1	59.8939	95.5	...	...	Ins: 60
70	Sad1	927.866	55	137.35	...	Left Saddle
70	Sad1	815.247	480	137.35	...	RIGHT SADDLE
70	Liqd	15654.5	335	...	...	Liquid: 70
70	Ins1	238.171	335	...	...	Ins: 70
70	Noz1	290.62	60	134.368	...	Noz N4
70	Noz1	341.159	320	134.368	...	Noz N5
70	Noz1	290.62	210	68.8476	115.389	Noz N3
70	Noz1	68.0443	638	-124.127	...	Noz N6
70	Noz1	12.7368	210	-100.376	61.1584	Noz N7
70	Noz1	10.6549	13.5	60.2255	100.938	Noz K2A
70	Noz1	10.6549	13.5	-60.2255	100.938	Noz K2B
70	Noz1	8.54578	320	59.9002	100.393	Noz K3
70	Noz1	10.6549	150	117.54	...	Noz K4
70	Noz1	10.6549	230	117.54	...	Noz K5
80	Liqd	1151.74	24.1667	...	...	Liquid: 80
80	Ins1	40.961	31.25	...	...	Ins: 80
80	Noz1	10.6549	52.4342	65	...	Noz K1A
80	Noz1	10.6549	52.4342	-65	...	Noz K1B
80	Noz1	844.121	62.5	...	...	MH1
40	FTsh	2321.98	140.5	...	...	U-TUBE SHEET
40	Tube	14107.9	524.85	...	...	Tubes

Total Weight of Each Detail Type:

Saddles	1743.1
Liquid	20514.7
Insulation	395.2
Nozzles	2859.3



Exchanger Components	16429.9
Liquid in Tubes	2996.6
Sum of the Detail Weights	44938.9 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	25075.7	25075.7	25075.7	25075.7	25075.7	25075.7
Saddles	1743.1	1743.1	1743.1	1743.1	1743.1	1743.1
Nozzles	2859.3	2859.3	2859.3	2859.3	2859.3	2859.3
Exchanger	16429.9	16429.9	16429.9	16429.9	16429.9	16429.9
Insulation	...	...	395.2	395.2	395.2	395.2
Ope. Liquid	...	...	...	...	...	20514.7
Tube Ope Lqd	...	...	...	...	...	2996.6
Test Liquid	...	32054.3	...	...	...	...
Tube Tst Lqd	...	2996.6	...	...	...	...
Totals	46108.1	81159.0	46503.2	46503.2	46503.2	70014.6

Field Installation Options:**Miscellaneous Weight Percent: 10.0 %**

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	46108.1 kg
Shop Test Wt.	- Fabricated weight + water (full)	81159.0 kg
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	46503.2 kg
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	46503.2 kg
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	46503.2 kg
Operating Wt.	- Empty weight + operating liq. uncorroded	70014.6 kg
Oper. Wt. + CA	- Corr wt. + operating liquid	69058.8 kg

Exchanger Tube Data

Volume of Exchanger tubes :	2997990.8 cm ³
Weight of Ope Liq in tubes :	2996.6 kg
Weight of Water in tubes :	2996.6 kg

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
20	30	41150.9	...
30	40	33535.7	15117.3
40	50	52464.6	47123.9
50	60	11611.3	11309.7
60	70	112301	109275
70	80	492960	484120
80	90	63135	60955.7

Total	807158.6	727900.8 cm ²
Total	868.8	783.5 ft ²

Nozzle Flange MAWP Results: (kgf/cm² & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
Noz N9	244.39	260.14	80	1500	GR 1.1	...	...	...	...
Noz N8	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz N4	48.89	52.01	80	300	GR 1.1	0.664	48.887	48.887	48.887
Noz N5	48.89	52.01	80	300	GR 1.1	0.643	48.887	48.887	48.887
Noz N3	48.89	52.01	80	300	GR 1.1	0.664	48.887	48.887	48.887
Noz N6	48.89	52.01	80	300	GR 1.1	1.162	48.887	48.887	48.887
Noz N7	48.89	52.01	80	300	GR 1.1	2.840	48.887	48.887	48.887
Noz K2A	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K2B	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K3	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K4	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K5	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K1A	48.89	52.01	80	300	GR 1.1	...	...	...	...
Noz K1B	48.89	52.01	80	300	GR 1.1	...	...	...	...
MH1	48.89	52.01	80	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 48.887 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 52.007 kgf/cm²

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 244.394 kgf/cm²
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 260.136 kgf/cm²

Selected Method for Derating ASME B16.5 Flange MAWP: ASME 4.16.12

The 4.16.12 Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the 4.16.12 method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME kgf-m.	Axial Force FE kgf	MAWP PD kgf/cm ²	Maximum Pressure Pmax kgf/cm ²	Pass Fail
Noz N4	500.888	0.50	4	1275	23.903	48.887	Pass
Noz N5	500.888	0.50	4	1234	23.903	48.887	Pass
Noz N3	500.888	0.50	4	1275	23.903	48.887	Pass
Noz N6	248.539	0.50	1	551	23.903	48.887	Pass
Noz N7	77.851	0.50	0	133	23.903	48.887	Pass

Evaluating Flange on Nozzle [Noz N4]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4) + 4 \cdot \text{abs}(1275)500.888 \leq \pi \cdot 500.888^3 \cdot ((48.887 - 23.903) + 500 \cdot 48.887)$$

$$= 0.26200E+04 \leq 0.19514E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle Noz N4 [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.887, 0.500 \cdot 48.887 + 48.887 - 2620.0 / (\pi \cdot 500.888^3))$$

$$= \min(48.887, 72.667)$$

$$= 48.887 \text{ kgf/cm}^2$$

Evaluating Flange on Nozzle [Noz N5]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(4) + 4 \cdot \text{abs}(1234)500.888 \leq \pi \cdot 500.888^3 \cdot ((48.887 - 23.903) + 500 \cdot 48.887)$$

$$= 0.25383E+04 \leq 0.19514E+06 \text{ Check Passes}$$



Maximum Pressure for Flange on Nozzle Noz N5 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(48.887, 0.500 \cdot 48.887 + 48.887 - 2538.3 / (\pi \cdot 500.888^3)) \\ &= \min(48.887, 72.688) \\ &= 48.887 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [Noz N3]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4) + 4 \cdot \text{abs}(1275) 500.888 \leq \pi \cdot 500.888^3 [(48.887 - 23.903) + .500 \cdot 48.887] \\ &= 0.26200E+04 \leq 0.19514E+06 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle Noz N3 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(48.887, 0.500 \cdot 48.887 + 48.887 - 2620.0 / (\pi \cdot 500.888^3)) \\ &= \min(48.887, 72.667) \\ &= 48.887 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [Noz N6]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(1) + 4 \cdot \text{abs}(551) 248.539 \leq \pi \cdot 248.539^3 [(48.887 - 23.903) + .500 \cdot 48.887] \\ &= 0.56050E+03 \leq 0.23840E+05 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle Noz N6 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(48.887, 0.500 \cdot 48.887 + 48.887 - 560.5 / (\pi \cdot 248.539^3)) \\ &= \min(48.887, 72.168) \\ &= 48.887 \text{ kgf/cm}^2 \end{aligned}$$

Evaluating Flange on Nozzle [Noz N7]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(0) + 4 \cdot \text{abs}(133) 77.851 \leq \pi \cdot 77.851^3 [(48.887 - 23.903) + .500 \cdot 48.887] \\ &= 0.42098E+02 \leq 0.73268E+03 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle Noz N7 [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(48.887, 0.500 \cdot 48.887 + 48.887 - 42.1 / (\pi \cdot 77.851^3)) \\ &= \min(48.887, 70.491) \\ &= 48.887 \text{ kgf/cm}^2 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	km/hr
Wind Zone Number	2	
Base Elevation	30	cm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.1
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 5.403

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.1

= 54.997 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1814.775, 1.0 • 0.89 • 1.0 • 0.6 • 55.0²)

= 1616.3 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.7 • 0.808 • 1616.335

= 940.3 N

From	Height m	k1	k2	k3	Vz m/sec	Cf
20	1.93	1.0000	1.0000	1.0000	55.00	0.8081
30	1.93	1.0000	1.0000	1.0000	55.00	0.8081
40	1.93	1.0000	1.0000	1.0000	55.00	0.8081
50	1.93	1.0000	1.0000	1.0000	55.00	0.8081
60	1.93	1.0000	1.0000	1.0000	55.00	0.8081
70	1.93	1.0000	1.0000	1.0000	55.00	0.8081
80	1.93	1.0000	1.0000	1.0000	55.00	0.8081

Wind Load Calculation:

From	To	Wind Height cm	Wind Diameter cm	Wind Area cm²	Wind Pressure kgf/m²	Element Wind Load kgf
20	30	193	237.6	7199.28	164.888	95.887
30	40	193	171.6	6829.68	164.888	90.9643
40	50	193	187.92	23490	164.888	312.863
50	60	193	175.44	5263.2	164.888	70.1004
60	70	193	242.64	46344.2	164.888	617.258
70	80	193	308.64	206789	164.888	2754.21
80	90	193	307.68	15775.7	164.888	210.116



Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 7479.52 • 0.0939 • 1.0
= 702.0 kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 67316 • 1.0
= 4212.0 kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height cm	Weight kgf	Ope Load kgf
20	30	87.5	7479.52	701.998
30	40	60	7479.52	701.998
40	50	60	7479.52	701.998
50	60	60	7479.52	701.998
60	70	60	7479.52	701.998
70	Sad1	115	7479.52	701.998
Sad1	80	115	7479.52	701.998
70	80	115	7479.52	701.998
80	90	115	7479.52	701.998

The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	701.188 cm
Center of Gravity of the Liquid	707.849 cm
Center of Gravity of the Insulation	658.081 cm
Center of Gravity of the Nozzles	650.095 cm
Center of Gravity of the Tubesheet(s)	210.917 cm
Center of Gravity of the Tubes	595.268 cm
Center of Gravity of Bare Shell New and Cold	388.373 cm
Center of Gravity of Bare Shell Corroded	376.668 cm
Vessel CG in the Operating Condition	544.742 cm
Vessel CG in the Fabricated (Shop/Empty) Condition	472.389 cm
Vessel CG in the Test Condition	571.033 cm

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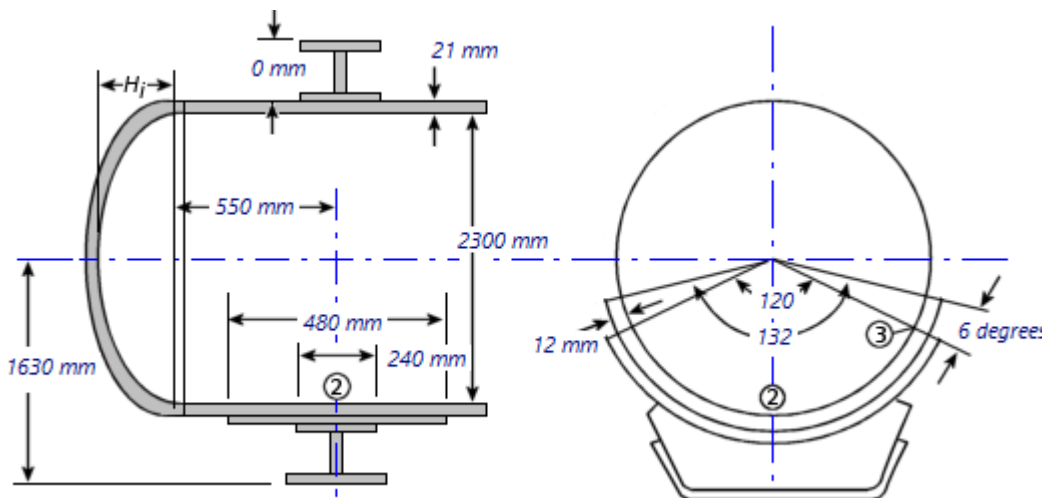


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.41	kgf/cm ²
Shell Allowable Stress used in Calculation		1650.86	kgf/cm ²
Head Allowable Stress used in Calculation		1568.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		70954.81	kgf
Pressure used in Analysis	P	23.528	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		548.93	1650.86
Long. Stress at Bottom of Midspan		852.17	1650.86
Long. Stress at Top of Saddles		696.99	1650.86
Long. Stress at Bottom of Saddles		704.11	1650.86
Tangential Shear in Shell		275.71	1320.68
Tangential Shear in Head		18.44	1252.34
Circ. Stress at Horn of Saddle		265.61	2063.58
Circ. Compressive Stress in Shell		36.08	1650.86

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:

$$\begin{aligned} &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3.0(4151.4/2 + 0) \cdot 1630.0/2049.0164 \\ &= 4953.7 \text{ kgf} \end{aligned}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$\begin{aligned} &= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls} \\ &= \max(1233.62, 0) \cdot 1630.0/4250.0 \\ &= 473.1 \text{ kgf} \end{aligned}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$\begin{aligned} &= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls} \\ &= \max(6317.98, 0) \cdot 1630.0/4250.0 \\ &= 2423.1 \text{ kgf} \end{aligned}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$\begin{aligned} &= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E \\ &= 3.0(6318/2 + 0) \cdot 1630.0/2049.0164 \\ &= 7539.0 \text{ kgf} \end{aligned}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$\begin{aligned} &= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst}) \\ &= 63416 + \max(473, 4954, 2423, 7539) \\ &= 70954.8 \text{ kgf} \end{aligned}$$

Longitudinal Wind Force [Fl]:

$$\begin{aligned} &= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2)) \\ &= 1.0 \cdot 1616.989(0.0 + \pi/4(2.572 \cdot 1.2)^2) \\ &= 12097.753 \text{ N} \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	71882.67	kgf
Transverse Shear Load	3158.99	kgf
Longitudinal Shear Load	6317.98	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$\begin{aligned} &= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm}) \\ &= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25) \\ &= 2049.016 \text{ mm} \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$\begin{aligned} &= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})] \\ &= -70955 \cdot 55.0 [1 - (1 - 55.0/896.0 + (116.125^2 - 0.0^2) / (2 \cdot 55.0 \cdot 896.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 896.0))] \\ &= 2943.9 \text{ kgf-m.} \end{aligned}$$

Moment per Equation 4.15.2 [M2]:

$$\begin{aligned} &= Q \cdot \text{L} / 4 ((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - 4a/\text{L}) \\ &= 70955 \cdot 896 / 4 ((1 + 2(116^2 - 0^2)/(896^2)) / (1 + (4 \cdot 0)/(3 \cdot 896)) - 4 \cdot 55/896) \\ &= 125253.2 \text{ kgf-m.} \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned} &= P \cdot \text{Rm} / (2t) - \text{M2} / (\pi \cdot \text{Rm}^2 t) \\ &= 23.528 \cdot 1161.25 / (2 \cdot 19.5) - 125253.2 / (\pi \cdot 1161.2^2 \cdot 19.5) \\ &= 548.93 \text{ kgf/cm}^2 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) + 125253.2/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 852.17 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) - 2943.9/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 696.99 \text{ kgf/cm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) + 2943.9/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 704.11 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 70955(896.0 - 2 \cdot 55.0)/(896.0 + (4 \cdot 0.0/3))$$

$$= 62243.8 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 70955/(1161.25 \cdot 19.5)$$

$$= 275.71 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_{3*}]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 70955/(1161.25 \cdot 291.5)$$

$$= 18.44 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 55.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(1650.408/1650.863, 1)$$

$$= 0.9997$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [$\sigma_{6,r}$]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 70955 \cdot 0.1/(474.75(19.5 + 1 \cdot 12.0))$$

$$= -36.08 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [$\sigma_{7,r*}$]:

$$= -Q/(4(t+\eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm/(L(t+\eta \cdot tr)^2)$$

$$= -70955/(4(19.5 + 1 \cdot 12.0)474.75) -$$

$$12 \cdot 0.013 \cdot 70955 \cdot 1161.25/(896.0(19.5 + 1 \cdot 12.0)^2)$$

$$= -265.61 \text{ kgf/cm}^2$$

Distance between Saddle Supports [Ls]:

$$= 4250.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 4250.0(80.0 - 21.1)$$

$$= 2.996 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm



Web Location	Webloc	Center	
Saddle Design Temperature		80.0	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)^3) / 12$
 $= 44.104 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$
 $= 44.104 + 139.184 (169.914 - 9.75)^2$
 $= 35748.230 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.357E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.120E+05
Web	28.0	419.0	241.0	117.3	2827412.5	0.231E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.657E+05
Totals	...	...	...	389.7	6621617.0	0.137E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 6621617 / 389.704$
 $= 169.914 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle} / 2$
 $= 180 - 120.0 / 2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 70954.812$
 $= 14440.8418 \text{ kgf}$

Tension Stress, St = (Fh/As) = 57.6435 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1486.6764 kgf/cm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$
 $= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$
 $= 677.717 \text{ mm}$

Bending Moment, M = Fh * d = 9786.8174 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 121.8113 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1651.8627 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$= \sqrt[3]{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt[3]{(3(70955 + 928)270.0 / (4 \cdot 2960.0 \cdot 1651.863))}$
 $= 17.254 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$



= 2049.016 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2049.0161/(4 - 1)
 = 683.005 mm

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 683.0054 • 270.0/2
 = 922.057 cm²

Bearing Pressure [Bp]:
 = (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 71882.672/(2960.0 • 270.0)
 = 8.994 kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 922.1 • 8.99
 = 8293.286 kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 59.36 + 95.621
 = 154.981 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 8293.3/154.9808
 = 53.512 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	240.0	...	67.2	...	0.323E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 240.0 - 28.0
 = 212.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/154.981
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 240.0/2
 = 120.000 mm

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{3225.6/154.981}$
 = 45.621 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1630.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 28.0
 = 1010.500 mm

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2032632/2477.8}$
 = 127.251



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1010.5/45.621$$

$$= 22.150$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 6318.0 / (2 \cdot 2960.0) \cdot 683.005 \cdot 1010.5/2$$

$$= 368.288 \text{ kgf-m.}$$

Compressive Allowable, KL/r < Cc (22.1498 < 127.2511) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.15)^2 / (2 \cdot 127.25^2)) 2478 /$$

$$(5/3 + 3 \cdot (22.15) / (8 \cdot 127.25) - (22.15^3) / (8 \cdot 127.25^3))$$

$$= 1410 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 53.51/1409.51 + (368.29 \cdot 120.0/32256004) / 1651.86$$

$$= 0.121$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	212.0	0.0	59.4	0.0	0.322E+04
Web	683.0	28.0	0.0	191.2	0.0	125.
Totals	...	...	...	250.6	...	0.335E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/250.602$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1630.0 - \sqrt{ (1183.0 + 12.0)^2 - (683.005/2)^2 } - 28.0$$

$$= 469.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3345.4/250.602 }$$

$$= 36.537 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 469.364/36.537$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6317.981 / (2 \cdot 2960.0)$$

$$= 1.067 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.067 \cdot 683.005 \cdot 469.364$$

$$= 342.130 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 342.13/278785.188$$

$$= 122.722 \text{ kgf/cm}^2$$



Compressive Allowable, $KL/r < C_c$ (12.8462 < 127.2511) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.85)^2 / (2 \cdot 127.25^2)) 2478 / (5/3 + 3 \cdot (12.85) / (8 \cdot 127.25) - (12.85^3) / (8 \cdot 127.25^3))$$

$$= 1446 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 65.85 / 1446.36 + (342.13 \cdot 120.0 / 3345.419) / 1651.86$$

$$= 0.120$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1650.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	64343.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2423.1 kgf
Maximum Transverse Force	Ft	3159.0 kgf
Maximum Longitudinal Force	Fl	6318.0 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6318.0 / (2.929 \cdot 4)$$

$$= 539.3 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 3159.0 / (2.929 \cdot 4)$$

$$= 269.6 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 6317.98 / (1101.26 \cdot 4.0)$$

$$= 1.4343 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1630.0 \cdot 3158.99 + 0.0$$

$$= 5149.16 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 5149.16 / 64343.71$$

$$= 80.03 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 1.4343, 0.0]$$

$$= 1.4343 \text{ cm}^2$$



Note: Saddle and wear plate material are both specified as impact tested.

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (480.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(270.0 + 1.56 \cdot \sqrt{1161.25 \cdot 19.5}, 2 \cdot 1900.0)$$

$$= 504.7495 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	270.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	58.43	cm
Shell Allowable Stress used in Calculation		1650.86	kgf/cm ²
Head Allowable Stress used in Calculation		1650.86	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		15650.78	kgf
Pressure used in Analysis	P	23.528	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	696.52	1650.86
Long. Stress at Bottom of Midspan	704.58	1650.86
Long. Stress at Top of Saddles	784.24	1650.86
Long. Stress at Bottom of Saddles	654.16	1650.86
Tangential Shear in Shell	42.87	1320.68
Circ. Stress at Horn of Saddle	378.07	2063.58
Circ. Compressive Stress in Shell	12.09	1650.86

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (4151.4/2 + 0) \cdot 1630.0/2049.0164$$

$$= 4953.7 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(1233.62, 0) \cdot 1630.0/4250.0$$

$$= 473.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(6317.98, 0) \cdot 1630.0/4250.0$$

$$= 2423.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:



$$= \text{Ftr}(\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot \text{B} / \text{E}$$

$$= 3.0(6318/2 + 0) \cdot 1630.0/2049.0164$$

$$= 7539.0 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 8112 + \max(473, 4954, 2423, 7539)$$

$$= 15650.8 \text{ kgf}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1616.989(0.0 + \pi/4(2.572 \cdot 1.2)^2)$$

$$= 12097.753 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	16466.02	kgf
Transverse Shear Load	3158.99	kgf
Longitudinal Shear Load	6317.98	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$$

$$= 2049.016 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm})/3\text{L})]$$

$$= -15651 \cdot 190.0 [1 - (1 - 190.0/896.0 + (116.125^2 - 58.425^2)/$$

$$(2 \cdot 190.0 \cdot 896.0)) / (1 + (4 \cdot 58.43)/(3 \cdot 896.0))]$$

$$= -7370.6 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(\text{L}^2)) / (1 + (4\text{hm}^2)/(3\text{L})) - a/L)$$

$$= 15651 \cdot 896/4((1 + 2(116^2 - 58^2)/(896^2)) / (1 + (4 \cdot 58)/$$

$$(3 \cdot 896))) - 4 \cdot 190/896)$$

$$= 3326.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) - 3326.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 696.52 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot \text{Rm}/(2t) + \text{M2}/(\pi \cdot \text{Rm}^2 \cdot t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) + 3326.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 704.58 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot \text{Rm}/(2t) - \text{M1}/(\text{K1} \cdot \pi \cdot \text{Rm}^2 t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) - 7370.6/(0.1066 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 784.24 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot \text{Rm}/(2t) + \text{M1}/(\text{K1} \cdot \pi \cdot \text{Rm}^2 \cdot t)$$

$$= 23.528 \cdot 1161.25/(2 \cdot 19.5) + 7370.6/(0.1923 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 654.16 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(\text{L} - 2a)/(\text{L} + (4 \cdot \text{hm}/3))$$

$$= 15651(896.0 - 2 \cdot 190.0)/(896.0 + (4 \cdot 58.43/3))$$

$$= 8292.2 \text{ kgf}$$



Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 8292.23 / (1161.25 \cdot 19.5)$$

$$= 42.87 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 15651 \cdot 0.1 / (19.5(270.0 + 117.37 + 117.37))$$

$$= -12.09 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -15651 / (4 \cdot 19.5(270.0 + 117.375 + 117.375)) -$$

$$12 \cdot 0.053 \cdot 15651 \cdot 1161.25 / (896.0 \cdot 19.5^2)$$

$$= -378.07 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(270.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 190.0)$$

$$= 504.75 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		80.0	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 44.104 + 139.184(165.012 - 9.75)^2$$

$$= 33596.133 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.336E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.112E+05
Web	22.0	419.0	241.0	92.2	2221538.2	0.188E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.679E+05
Totals	...	...	...	364.6	6015743.0	0.131E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6015743 / 364.564$$

$$= 165.012 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$



Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 15650.775 \\ &= 3185.2722 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh / As) = 14.1329 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1486.6764 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472 \\ &= 677.717 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2158.7161 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 27.0927 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1651.8627 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\ &= \sqrt{(3(15651 + 815) \cdot 270.0) / (4 \cdot 2960.0 \cdot 1651.863)} \\ &= 8.258 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0) \\ &= 2049.016 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 2049.0161 / (4 - 1) \\ &= 683.005 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 683.0054 \cdot 270.0/2 \\ &= 922.057 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 16466.021 / (2960.0 \cdot 270.0) \\ &= 2.060 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 922.1 \cdot 2.06 \\ &= 1899.727 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 54.56 + 75.131 \\ &= 129.691 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / Ar \\ &= 1899.7 / 129.6906 \\ &= 14.648 \text{ kgf/cm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction



	B	D	Y	A	AY	Io
Rib+Web	22.0	270.0	...	59.4	...	0.361E+04

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 270.0 - 22.0
 = 248.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/129.691
 = 0.000 mm

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 270.0/2
 = 135.000 mm

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{3608.5/129.691}$
 = 52.749 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1630.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 28.0
 = 1010.500 mm

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 2032632/2477.8)}$
 = 127.251

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 1010.5/52.749
 = 19.157

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 6318.0/(2 • 2960.0) • 683.005 • 1010.5/2
 = 368.288 kgf-m.

Compressive Allowable, KL/r < Cc (19.1569 < 127.2511) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (19.16)^2 / (2 \cdot 127.25^2)) 2478 / (5/3 + 3 \cdot (19.16) / (8 \cdot 127.25) - (19.16^3) / (8 \cdot 127.25^3))$
 = 1422 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I)/Sba
 = 14.65/1422.03 + (368.29 • 135.0/36085492)/1651.86
 = 0.094

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	248.0	0.0	54.6	0.0	0.361E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	204.8	...	0.367E+04

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0.0/204.821
 = 0.000 mm

Distance to Centroid [C1]:



$$= \text{Saddle Width} / 2$$

$$= 270.0 / 2$$

$$= 135.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1630.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 28.0$$

$$= 469.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3667.2/204.821)}$$

$$= 42.314 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 469.364 / 42.314$$

$$= 11.093$$

Unit Force [Force,u]:

$$= \text{FI} / (2 \cdot \text{Baseplate Length})$$

$$= 6317.981 / (2 \cdot 2960.0)$$

$$= 1.067 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.067 \cdot 683.005 \cdot 469.364$$

$$= 342.130 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 342.13 / 271644.625$$

$$= 125.948 \text{ kgf/cm}^2$$

Compressive Allowable, $\text{KL}/r < C_c$ (11.0925 < 127.2511) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot C_c^2)) \cdot F_y / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot C_c) - (\text{KL}/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.09)^2 / (2 \cdot 127.25^2)) \cdot 2478 / (5/3 + 3 \cdot (11.09) / (8 \cdot 127.25) - (11.09^3) / (8 \cdot 127.25^3))$$

$$= 1453 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Mbase,c} \cdot \text{C1}/I) / \text{Sba}$$

$$= 19.07 / 1452.61 + (342.13 \cdot 135.0 / 3667.199) / 1651.86$$

$$= 0.089$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1396.96 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1650.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load Q0 (Weight)	Q0	8927.1 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	2423.1 kgf
Maximum Transverse Force	Ft	3159.0 kgf
Maximum Longitudinal Force	F1	6318.0 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= \text{FI} / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6318.0 / (2.929 \cdot 4)$$

$$= 539.3 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:



$$\begin{aligned} &= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 3159.0 / (2.929 \cdot 4) \\ &= 269.6 \text{ kgf/cm}^2. \text{ Must be less than } 1101.3 \text{ kgf/cm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

$$\begin{aligned} &\text{Bolt Area due to Shear Load [Bltarears]:} \\ &= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts}) \\ &= 6317.98 / (1101.26 \cdot 4.0) \\ &= 1.4343 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

$$\begin{aligned} &\text{Moment on Baseplate Due to Transverse Load [Rmom]:} \\ &= B \cdot F_t + \text{Sum of X Moments} \\ &= 1630.0 \cdot 3158.99 + 0.0 \\ &= 5149.16 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Eccentricity (e):} \\ &= R_{mom} / QO \\ &= 5149.16 / 8927.06 \\ &= 576.80 \text{ mm} > B_{plen} / 6 \text{ --> Uplift in Transverse direction} \end{aligned}$$

$$\begin{aligned} f &= B_{plen} / 2 - \text{Edgedis} \\ &= 2960.0 / 2 - 50.8 \\ &= 1429.20 \text{ mm} \end{aligned}$$

$$\begin{aligned} K1 &= 3 (e - 0.5 \cdot B_{plen}) \\ &= 3 (576.8 - 0.5 \cdot 2960.0) \\ &= -2709.59 \text{ mm} \end{aligned}$$

$$\begin{aligned} K2 &= 6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e) \\ &= 6 \cdot 1.0 \cdot 5.86 / 270.0 \cdot (1429.2 + 576.8) \\ &= 26114.21 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} K3 &= -K2 \cdot (0.5 \cdot B_{plen} + f) \\ &= -26114.21 \cdot (0.5 \cdot 2960.0 + 1429.2) \\ &= -75971459.31 \text{ mm}^3 \end{aligned}$$

$$\begin{aligned} &\text{Iteratively Solving for the Effective Bearing Length:} \\ &Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0 \\ &Y^3 + -2709.59 \cdot Y^2 + 26114.21 \cdot Y + -0.8E+08 = 0 \\ &Y = 2710.30 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Num} &= (B_{plen} / 2 - Y / 3 - e) \\ &= (2960.0 / 2 - 2710.3 / 3 - 576.8) \\ &= -0.24 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (B_{plen} / 2 - Y / 3 + f) \\ &= (2960.0 / 2 - 2710.3 / 3 + 1429.2) \\ &= 2005.77 \end{aligned}$$

$$\begin{aligned} &\text{Total Bolt Tension Force [Tforce]:} \\ &= - QO \cdot \text{Num} / \text{Denom} \\ &= - 8927.06 \cdot -0.24 / 2005.77 \\ &= 1.05 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= T_{force} / (Stba \cdot Nbt) \\ &= 1.05 / (1396.96 \cdot 2.0) \\ &= 0.0004 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 1.4343, 0.0004] \\ &= 1.4343 \text{ cm}^2 \end{aligned}$$



Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} &\text{Bearing Pressure (fc)} \\ &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(8927.06 + 1.05) / (2710.3 \cdot 270.0) \\ &= 2.44 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (Bplen - Weblength) / 2 \\ &= (2960.0 - 2909.2) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= Tforce \cdot Adist \\ &= 1.05 \cdot 2.54 \\ &= 0.03 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Equivalent Bearing Pressure (f1):} \\ &= fc \cdot (Y - Adist) / Y \\ &= 2.44 \cdot (2710.3 - 25.4) / 2710.3 \\ &= 2.42 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 270.0 / 6) \cdot (2.42 + 2 \cdot 2.44) \\ &= 2.12 \text{ kgf-m.} \end{aligned}$$

$$\begin{aligned} &\text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(0.03, 2.12) / (270.0 \cdot 2475.61))} \\ &= 1.3790 \text{ mm} \end{aligned}$$

Note: Saddle and wear plate material are both specified as impact tested.

Review notes about nozzle loadings and supports in the Warnings report.

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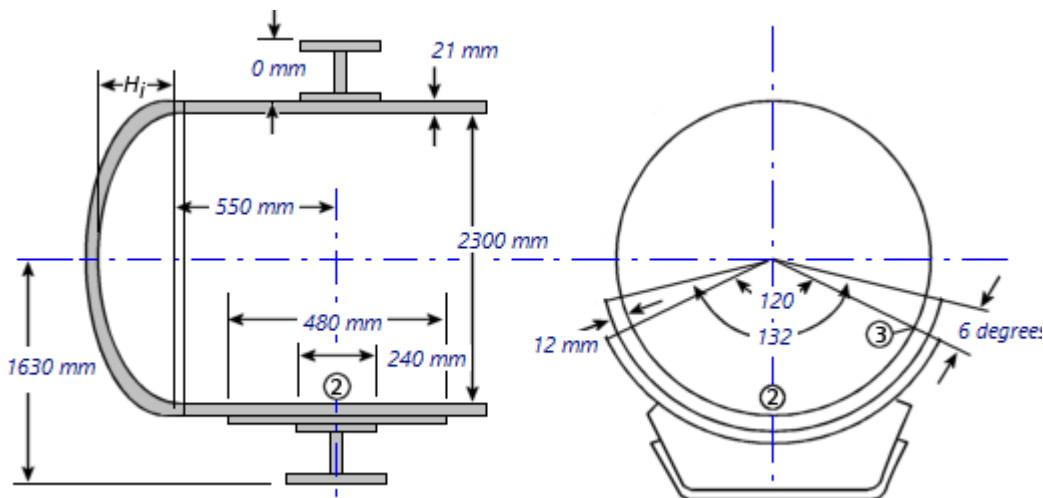


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.41	kgf/cm ²
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	P	69150.54	kgf
Pressure used in Analysis		29.995	kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		745.34	2518.57
Long. Stress at Bottom of Midspan		1040.87	2518.57
Long. Stress at Top of Saddles		889.63	2518.57
Long. Stress at Bottom of Saddles		896.58	2518.57
Tangential Shear in Shell		268.70	1413.05
Tangential Shear in Head		17.97	1252.34
Circ. Stress at Horn of Saddle		322.86	3777.85
Circ. Compressive Stress in Shell		40.47	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:



Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1370.0/2 + 0) \cdot 1630.0/2049.0164$
 $= 1634.7 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(407.1, 0) \cdot 1630.0/4250.0$
 $= 156.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 67516 + \max(156, 1635, 0, 0)$
 $= 69150.5 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 1616.989 (0.0 + \pi/4 (2.572 \cdot 1.2)^2)$
 $= 12097.753 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	70078.41	kgf
Transverse Shear Load	684.98	kgf
Longitudinal Shear Load	407.10	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2300.0/2 + 21.0 + 12.0) \sin(120.0/2), 2 \cdot 1161.25)$
 $= 2049.016 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
 $= -69151 \cdot 55.0 [1 - (1 - 55.0/896.0 + (116.125^2 - 0.0^2)/(2 \cdot 55.0 \cdot 896.0))/(1 + (4 \cdot 0.0)/(3 \cdot 896.0))]$
 $= 2869.1 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 69151 \cdot 896/4 ((1 + 2(116^2 - 0^2)/(896^2))/(1 + (4 \cdot 0)/(3 \cdot 896)) - 4 \cdot 55/896)$
 $= 122068.2 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 29.995 \cdot 1161.25/(2 \cdot 19.5) - 122068.2/(\pi \cdot 1161.2^2 \cdot 19.5)$
 $= 745.34 \text{ kgf/cm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 29.995 \cdot 1161.25/(2 \cdot 19.5) + 122068.2/(\pi \cdot 1161.2^2 \cdot 19.5)$
 $= 1040.87 \text{ kgf/cm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:
 $= P \cdot R_m/(2t) - M_1/(\pi \cdot R_m^2 \cdot t)$
 $= 29.995 \cdot 1161.25/(2 \cdot 19.5) - 2869.1/(\pi \cdot 1161.2^2 \cdot 19.5)$
 $= 889.63 \text{ kgf/cm}^2$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:



$$= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 t)$$

$$= 29.995 \cdot 1161.25 / (2 \cdot 19.5) + 2869.1 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 896.58 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 69151(896.0 - 2 \cdot 55.0) / (896.0 + (4 \cdot 0.0/3))$$

$$= 60661.1 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ3]:

$$= K3 \cdot Q / (Rm \cdot t)$$

$$= 0.8799 \cdot 69151 / (1161.25 \cdot 19.5)$$

$$= 268.70 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ3*]:

$$= K3 \cdot Q / (Rm \cdot th)$$

$$= 0.8799 \cdot 69151 / (1161.25 \cdot 291.5)$$

$$= 17.97 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 55.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(1650.408/2518.566, 1)$$

$$= 0.6553$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 69151 \cdot 0.1 / (474.75(19.5 + 0.655 \cdot 12.0))$$

$$= -40.47 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -69151 / (4(19.5 + 0.655 \cdot 12.0)474.75) -$$

$$12 \cdot 0.013 \cdot 69151 \cdot 1161.25 / (896.0(19.5 + 0.655 \cdot 12.0)^2)$$

$$= -322.86 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)19.5^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 44.104 + 139.184(169.914 - 9.75)^2$$

$$= 35748.230 \text{ cm}^4$$

B

D

Y

A

AY

I + AD²(2)



	mm	mm	mm	cm ²	mm ³	cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.357E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.120E+05
Web	28.0	419.0	241.0	117.3	2827412.5	0.231E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.657E+05
Totals	...	...	...	389.7	6621617.0	0.137E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6621617/389.704$$

$$= 169.914 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 69150.539$$

$$= 14073.6348 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 56.1778 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1590.6736 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 677.717 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9537.9541 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 118.7138 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1767.4150 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(69151 + 928)270.0 / (4 \cdot 2960.0 \cdot 1767.415))}$$

$$= 16.470 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2049.0161 / (4 - 1)$$

$$= 683.005 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 683.0054 \cdot 270.0/2$$

$$= 922.057 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 70078.406 / (2960.0 \cdot 270.0)$$

$$= 8.769 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$



$$= 922.1 \cdot 8.77$$

$$= 8085.124 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 59.36 + 95.621$$

$$= 154.981 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 8085.1/154.9808$$

$$= 52.169 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	240.0	...	67.2	...	0.323E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 240.0 - 28.0$$

$$= 212.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/154.981$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3225.6/154.981)}$$

$$= 45.621 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1630.0 - \cos(120.0/2) (1150.0 + 21.0 + 12.0) - 28.0$$

$$= 1010.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 2032632/2651.1)}$$

$$= 123.021$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1010.5/45.621$$

$$= 22.150$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 407.1 / (2 \cdot 2960.0) \cdot 683.005 \cdot 1010.5/2$$

$$= 23.730 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ ($22.1498 < 123.021$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (22.15)^2 / (2 \cdot 123.02^2)) 2651 /$$

$$(5/3 + 3 \cdot (22.15) / (8 \cdot 123.02) - (22.15^3) / (8 \cdot 123.02^3))$$

$$= 1505 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 52.17/1504.6 + (23.73 \cdot 120.0/32256004) / 1767.42$$



= 0.040

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	212.0	0.0	59.4	0.0	0.322E+04
Web	683.0	28.0	0.0	191.2	0.0	125.
Totals	...	...	...	250.6	...	0.335E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/250.602$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0/2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1630.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 28.0$$

$$= 469.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(3345.4/250.602)}$$

$$= 36.537 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 469.364/36.537$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 407.096/(2 \cdot 2960.0)$$

$$= 0.069 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.069 \cdot 683.005 \cdot 469.364$$

$$= 22.045 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 22.045/278785.188$$

$$= 7.908 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (12.8462 < 123.021) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.85)^2 / (2 \cdot 123.02^2)) 2651 / (5/3 + 3 \cdot (12.85) / (8 \cdot 123.02) - (12.85^3) / (8 \cdot 123.02^3))$$

$$= 1546 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 64.2/1545.81 + (22.04 \cdot 120.0/3345.419) / 1767.42$$

$$= 0.046$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm



Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	68443.7 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	156.1 kgf
Maximum Transverse Force	Ft	685.0 kgf
Maximum Longitudinal Force	Fl	407.1 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:

$$\begin{aligned} &= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 407.1 / (2.929 \cdot 4) \\ &= 34.7 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:

$$\begin{aligned} &= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts}) \\ &= 685.0 / (2.929 \cdot 4) \\ &= 58.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2. \end{aligned}$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$\begin{aligned} &= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolls}) \\ &= 407.1 / (1052.29 \cdot 4.0) \\ &= 0.0967 \text{ cm}^2 \end{aligned}$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 1630.0 \cdot 684.98 + 0.0 \\ &= 1116.52 \text{ kgf-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 1116.52 / 68443.69 \\ &= 16.31 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction} \end{aligned}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 0.0967, 0.0] \\ &= 0.0967 \text{ cm}^2 \end{aligned}$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (480.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\ &= \min(270.0 + 1.56 \cdot \sqrt{1161.25 \cdot 19.5}, 2 \cdot 1900.0) \\ &= 504.7495 \text{ mm} \end{aligned}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	270.00	mm



Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	58.43	cm
Shell Allowable Stress used in Calculation		2518.57	kgf/cm ²
Head Allowable Stress used in Calculation		2518.57	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		13929.95	kgf
Pressure used in Analysis	P	29.995	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	889.52	2518.57
Long. Stress at Bottom of Midspan	896.69	2518.57
Long. Stress at Top of Saddles	967.59	2518.57
Long. Stress at Bottom of Saddles	851.82	2518.57
Tangential Shear in Shell	38.16	1413.05
Circ. Stress at Horn of Saddle	336.50	3777.85
Circ. Compressive Stress in Shell	10.76	2518.57

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1370.0/2 + 0) \cdot 1630.0/2049.0164$
 $= 1634.7 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(407.1, 0) \cdot 1630.0/4250.0$
 $= 156.1 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 12295 + \max(156, 1635, 0, 0)$
 $= 13930.0 \text{ kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 1616.989 (0.0 + \pi/4 (2.572 \cdot 1.2)^2)$
 $= 12097.753 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	14745.20	kgf
Transverse Shear Load	684.98	kgf
Longitudinal Shear Load	407.10	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(2300.0/2 + 21.0 + 12.0) \sin(120.0/2), 2 \cdot 1161.25)$
 $= 2049.016 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.



Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -13930 \cdot 190.0 \left[1 - \left(1 - \frac{190.0}{896.0} + \frac{(116.125^2 - 58.425^2)}{(2 \cdot 190.0 \cdot 896.0)} \right) / \left(1 + \frac{(4 \cdot 58.43)}{3 \cdot 896.0} \right) \right]$$

$$= -6560.2 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - \frac{4a}{L} \right)$$

$$= 13930 \cdot 896 / 4 \left(\frac{(1 + 2(116.125^2 - 58.425^2)/(896^2))}{(1 + (4 \cdot 58.43)/(3 \cdot 896))} - \frac{4 \cdot 190}{896} \right)$$

$$= 2960.6 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 29.995 \cdot 1161.25 / (2 \cdot 19.5) - 2960.6 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 889.52 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 29.995 \cdot 1161.25 / (2 \cdot 19.5) + 2960.6 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 896.69 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 29.995 \cdot 1161.25 / (2 \cdot 19.5) - 6560.2 / (0.1066 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 967.59 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 29.995 \cdot 1161.25 / (2 \cdot 19.5) + 6560.2 / (0.1923 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 851.82 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 13930(896.0 - 2 \cdot 190.0) / (896.0 + (4 \cdot 58.43 / 3))$$

$$= 7380.5 \text{ kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 7380.49 / (1161.25 \cdot 19.5)$$

$$= 38.16 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 13930 \cdot 0.1 / (19.5(270.0 + 117.37 + 117.37))$$

$$= -10.76 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.23) [σ^*7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot Rm / (L \cdot t^2)$$

$$= -13930 / (4 \cdot 19.5(270.0 + 117.375 + 117.375)) - 12 \cdot 0.053 \cdot 13930 \cdot 1161.25 / (896.0 \cdot 19.5^2)$$

$$= -336.50 \text{ kgf/cm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(270.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 190.0)$$

$$= 504.75 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	



Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	2651.1	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5) + 480.0}) 19.5^3) / 12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 44.104 + 139.184 (165.012 - 9.75)^2$$

$$= 33596.133 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.336E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.112E+05
Web	22.0	419.0	241.0	92.2	2221538.2	0.188E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.679E+05
Totals	...	...	...	364.6	6015743.0	0.131E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6015743 / 364.564$$

$$= 165.012 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 13929.952$$

$$= 2835.0474 \text{ kgf}$$

Tension Stress, St = (Fh/As) = 12.5790 kgf/cm²
 Allowed Stress, Sa = 0.6 * Yield Str = 1590.6736 kgf/cm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 677.717 \text{ mm}$$

Bending Moment, M = Fh * d = 1921.3623 kgf-m.

Bending Stress, Sb = (M * C1 / I) = 24.1138 kgf/cm²
 Allowed Stress, Sa = 2/3 * Yield Str = 1767.4150 kgf/cm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(13930 + 815) 270.0 / (4 \cdot 2960.0 \cdot 1767.415)}$$

$$= 7.555 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$



$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$
$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$
$$= 2049.016 / (4 - 1)$$
$$= 683.005 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$
$$= 683.0054 \cdot 270.0 / 2$$
$$= 922.057 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$
$$= 14745.198 / (2960.0 \cdot 270.0)$$
$$= 1.845 \text{ kgf/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$
$$= 922.1 \cdot 1.84$$
$$= 1701.191 \text{ kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$
$$= 54.56 + 75.131$$
$$= 129.691 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$
$$= 1701.2 / 129.6906$$
$$= 13.117 \text{ kgf/cm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	270.0	...	59.4	...	0.361E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$
$$= 270.0 - 22.0$$
$$= 248.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$
$$= 0.0 / 129.691$$
$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$
$$= 270.0 / 2$$
$$= 135.000 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$
$$= \sqrt{(3608.5 / 129.691)}$$
$$= 52.749 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$
$$= 1630.0 - \cos(120.0/2) (1150.0 + 21.0 + 12.0) - 28.0$$
$$= 1010.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$
$$= \sqrt{(2 \cdot \pi^2 \cdot 2032632 / 2651.1)}$$
$$= 123.021$$



Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1010.5/52.749$$

$$= 19.157$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 407.1 / (2 \cdot 2960.0) \cdot 683.005 \cdot 1010.5 / 2$$

$$= 23.730 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < Cc$ (19.1569 < 123.021) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (19.16)^2 / (2 \cdot 123.02^2)) 2651 / (5/3 + 3 \cdot (19.16) / (8 \cdot 123.02) - (19.16^3) / (8 \cdot 123.02^3))$$

$$= 1519 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 13.12/1518.61 + (23.73 \cdot 135.0/36085492) / 1767.42$$

$$= 0.014$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	248.0	0.0	54.6	0.0	0.361E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	204.8	...	0.367E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/204.821$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 270.0/2$$

$$= 135.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1630.0 - \sqrt{ (1183.0 + 12.0)^2 - (683.005/2)^2 } - 28.0$$

$$= 469.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3667.2/204.821 }$$

$$= 42.314 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 469.364/42.314$$

$$= 11.093$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 407.096 / (2 \cdot 2960.0)$$

$$= 0.069 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.069 \cdot 683.005 \cdot 469.364$$

$$= 22.045 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 22.045/271644.625$$

$$= 8.115 \text{ kgf/cm}^2$$



Compressive Allowable, $KL/r < C_c$ (11.0925 < 123.021) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.09)^2 / (2 \cdot 123.02^2)) 2651 / (5/3 + 3 \cdot (11.09) / (8 \cdot 123.02) - (11.09^3) / (8 \cdot 123.02^3))$$

$$= 1553 \text{ kgf/cm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 17.08 / 1552.79 + (22.04 \cdot 135.0 / 3667.199) / 1767.42$$

$$= 0.016$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.82 kgf/cm ²
Bolt Corrosion Allowance	Bca	3.0 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
Nominal Bolt Diameter	Bnd	30.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	1784.41 kgf/cm ²
Area Available in a Single Bolt	BltArea	2.9291 cm ²
Saddle Load QO (Weight)	QO	13110.5 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	156.1 kgf
Maximum Transverse Force	Ft	685.0 kgf
Maximum Longitudinal Force	Fl	407.1 kgf
Saddle Bolted to Steel Foundation		Yes

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$$= Fl / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 407.1 / (2.929 \cdot 4)$$

$$= 34.7 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 685.0 / (2.929 \cdot 4)$$

$$= 58.5 \text{ kgf/cm}^2. \text{ Must be less than } 1052.3 \text{ kgf/cm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= Fl / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 407.1 / (1052.29 \cdot 4.0)$$

$$= 0.0967 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$

$$= 1630.0 \cdot 684.98 + 0.0$$

$$= 1116.52 \text{ kgf-m.}$$

Eccentricity (e):

$$= Rmom / QO$$

$$= 1116.52 / 13110.48$$

$$= 85.16 \text{ mm} < Bplen / 6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$

$$= \max[0.0, 0.0967, 0.0]$$

$$= 0.0967 \text{ cm}^2$$



Review notes about nozzle loadings and supports in the Warnings report.

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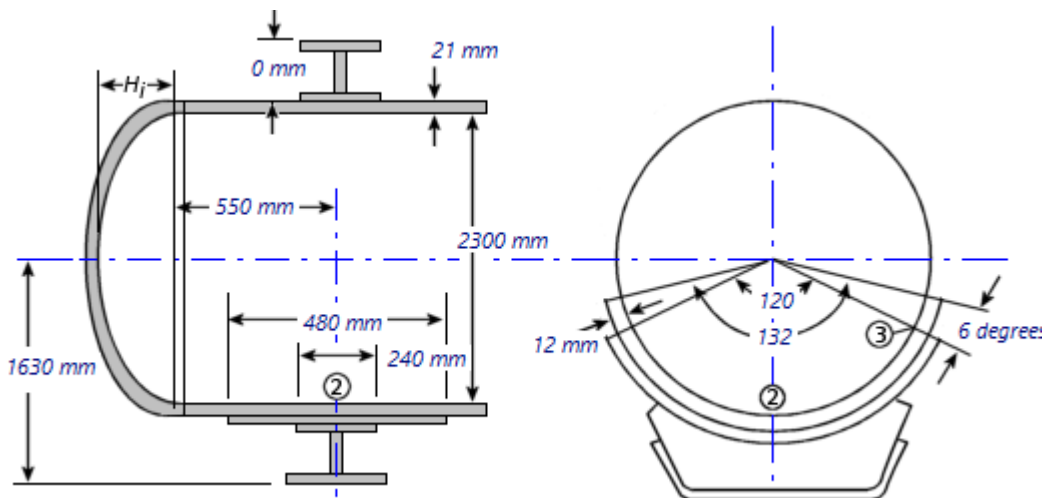


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	550.00	mm
Saddle Width	b1 or b	240.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	12.0	mm
Wear Plate Allowable Stress	fw or Sr	1650.41	kgf/cm ²
Shell Allowable Stress used in Calculation		1650.86	kgf/cm ²
Head Allowable Stress used in Calculation		1568.83	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	70954.81	kgf
Pressure used in Analysis	P	-1.055 kgf/cm ²

Horizontal Vessel Analysis Results:		Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan		-183.02	-2159.64
Long. Stress at Top of Midspan		183.02	1650.86
Long. Stress at Bottom of Midspan		120.22	1650.86
Long. Stress at Top of Saddles		-34.97	-2159.64
Long. Stress at Top of Saddles		34.97	1650.86
Long. Stress at Bottom of Saddles		-27.84	-2159.64
Long. Stress at Bottom of Saddles		27.84	1650.86
Tangential Shear in Shell		275.71	1320.68
Tangential Shear in Head		18.44	1252.34
Circ. Stress at Horn of Saddle		265.61	2063.58
Circ. Compressive Stress in Shell		36.08	1650.86



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(4151.4/2 + 0) \cdot 1630.0/2049.0164$
 $= 4953.7 \text{ kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(1233.62, 0) \cdot 1630.0/4250.0$
 $= 473.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(6317.98, 0) \cdot 1630.0/4250.0$
 $= 2423.1 \text{ kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(6318/2 + 0) \cdot 1630.0/2049.0164$
 $= 7539.0 \text{ kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 63416 + \max(473, 4954, 2423, 7539)$
 $= 70954.8 \text{ kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.0 \cdot 1616.989(0.0 + \pi/4(2.572 \cdot 1.2)^2)$
 $= 12097.753 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	71882.67	kgf
Transverse Shear Load	3158.99	kgf
Longitudinal Shear Load	6317.98	kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$
 $= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$
 $= 2049.016 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0132	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0109			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is less than $\text{Rm}/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (\text{Rm}^2 - \text{hm}^2)/(2a \cdot L))/(1 + (4\text{hm})/(3L))]$
 $= -70955 \cdot 55.0 [1 - (1 - 55.0/896.0 + (116.125^2 - 0.0^2)/(2 \cdot 55.0 \cdot 896.0))/(1 + (4 \cdot 0.0)/(3 \cdot 896.0))]$
 $= 2943.9 \text{ kgf-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(\text{Rm}^2 - \text{hm}^2)/(L^2))/(1 + (4\text{hm}^2)/(3L)) - 4a/L)$
 $= 70955 \cdot 896/4((1 + 2(116^2 - 0^2)/(896^2))/(1 + (4 \cdot 0)/(3 \cdot 896)) - 4 \cdot 55/896)$
 $= 125253.2 \text{ kgf-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot \text{Rm}/(2t) - \text{M2}/(\pi \cdot \text{Rm}^2 t)$
 $= -1.055 \cdot 1161.25/(2 \cdot 19.5) - 125253.2/(\pi \cdot 1161.2^2 \cdot 19.5)$



$$= -183.02 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) + 125253.2 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 120.22 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) - 2943.9 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -34.97 \text{ kgf/cm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25 / (2 \cdot 19.5) + 2943.9 / (\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -27.84 \text{ kgf/cm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 70955(896.0 - 2 \cdot 55.0) / (896.0 + (4 \cdot 0.0 / 3))$$

$$= 62243.8 \text{ kgf}$$

Shear Stress in the shell no rings, stiffened (4.15.13) [τ_3]:

$$= K_3 \cdot Q / (R_m \cdot t)$$

$$= 0.8799 \cdot 70955 / (1161.25 \cdot 19.5)$$

$$= 275.71 \text{ kgf/cm}^2$$

Shear Stress in the head, shell stiffened (4.15.14) [τ_3^*]:

$$= K_3 \cdot Q / (R_m \cdot t_h)$$

$$= 0.8799 \cdot 70955 / (1161.25 \cdot 291.5)$$

$$= 18.44 \text{ kgf/cm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$$

$$= 117.375 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(240.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 55.0)$$

$$= 474.75 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(1650.408 / 1650.863, 1)$$

$$= 0.9997$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.7603 \cdot 70955 \cdot 0.1 / (474.75(19.5 + 1 \cdot 12.0))$$

$$= -36.08 \text{ kgf/cm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot t_r)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot t_r)^2)$$

$$= -70955 / (4(19.5 + 1 \cdot 12.0)474.75) -$$

$$12 \cdot 0.013 \cdot 70955 \cdot 1161.25 / (896.0(19.5 + 1 \cdot 12.0)^2)$$

$$= -265.61 \text{ kgf/cm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	28.0000	mm
Web Thickness	Webtk	28.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		80.0	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	



Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)19.5^3)/12$$

$$= 44.104 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 44.104 + 139.184(169.914 - 9.75)^2$$

$$= 35748.230 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.357E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.120E+05
Web	28.0	419.0	241.0	117.3	2827412.5	0.231E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.657E+05
Totals	...	...	...	389.7	6621617.0	0.137E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 6621617/389.704$$

$$= 169.914 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 70954.812$$

$$= 14440.8418 \text{ kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 57.6435 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 1486.6764 \text{ kgf/cm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472$$

$$= 677.717 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 9786.8174 \text{ kgf-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 121.8113 \text{ kgf/cm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 1651.8627 \text{ kgf/cm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(70955 + 928)270.0 / (4 \cdot 2960.0 \cdot 1651.863))}$$

$$= 17.254 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0)$$

$$= 2049.016 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2049.0161 / (4 - 1)$$



= 683.005 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 683.0054 • 270.0/2
 = 922.057 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 71882.672/(2960.0 • 270.0)
 = 8.994 kgf/cm²

Axial Load [P]:

= Ap • Bp
 = 922.1 • 8.99
 = 8293.286 kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 59.36 + 95.621
 = 154.981 cm²

Compressive Stress [Sc]:

= P/Ar
 = 8293.3/154.9808
 = 53.512 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	28.0	240.0	...	67.2	...	0.323E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 240.0 - 28.0
 = 212.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/154.981
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 240.0/2
 = 120.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{3225.6/154.981}$
 = 45.621 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1630.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 28.0
 = 1010.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 2032632/2477.8}$
 = 127.251

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1010.5/45.621
 = 22.150



Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 6318.0 / (2 \cdot 2960.0) \cdot 683.005 \cdot 1010.5 / 2$$

$$= 368.288 \text{ kgf-m.}$$

Compressive Allowable, $KL/r < C_c$ (22.1498 < 127.2511) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.15)^2 / (2 \cdot 127.25^2)) 2478 / (5/3 + 3 \cdot (22.15) / (8 \cdot 127.25) - (22.15^3) / (8 \cdot 127.25^3))$$

$$= 1410 \text{ kgf/cm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 53.51 / 1409.51 + (368.29 \cdot 120.0 / 32256004) / 1651.86$$

$$= 0.121$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	28.0	212.0	0.0	59.4	0.0	0.322E+04
Web	683.0	28.0	0.0	191.2	0.0	125.
Totals	...	...	...	250.6	...	0.335E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 250.602$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 240.0 / 2$$

$$= 120.000 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (R_o + W_{pdthk})^2 - (Pitch/2)^2 } - B_{pthk}$$

$$= 1630.0 - \sqrt{ (1183.0 + 12.0)^2 - (683.005/2)^2 } - 28.0$$

$$= 469.364 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 3345.4 / 250.602 }$$

$$= 36.537 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 469.364 / 36.537$$

$$= 12.846$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 6317.981 / (2 \cdot 2960.0)$$

$$= 1.067 \text{ kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.067 \cdot 683.005 \cdot 469.364$$

$$= 342.130 \text{ kgf-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma_{B,base,c}$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 342.13 / 278785.188$$

$$= 122.722 \text{ kgf/cm}^2$$

Compressive Allowable, $KL/r < C_c$ (12.8462 < 127.2511) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (12.85)^2 / (2 \cdot 127.25^2)) 2478 / (5/3 + 3 \cdot (12.85) / (8 \cdot 127.25) - (12.85^3) / (8 \cdot 127.25^3))$$

$$= 1446 \text{ kgf/cm}^2$$



AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C/I/I)/Sba$$

$$= 65.85/1446.36 + (342.13 \cdot 120.0/3345.419)/1651.86$$

$$= 0.120$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (480.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ Rm \cdot t }, 2a)$$

$$= \min(270.0 + 1.56 \cdot \sqrt{ 1161.25 \cdot 19.5 }, 2 \cdot 1900.0)$$

$$= 504.7495 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	1161.25	mm
Shell Thickness used in this Case	t	21.000	mm
Stiffened Vessel Length per 4.15.6	L	896.00	cm
Distance from Saddle to Vessel tangent	a1 or a	1900.00	mm
Saddle Width	b1 or b	270.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	58.43	cm
Shell Allowable Stress used in Calculation		1650.86	kgf/cm ²
Head Allowable Stress used in Calculation		1650.86	kgf/cm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1630.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		15650.78	kgf
Pressure used in Analysis	P	-1.055	kgf/cm ²

Horizontal Vessel Analysis Results:	Actual kgf/cm ²	Allowable kgf/cm ²
Long. Stress at Top of Midspan	-35.43	-2159.64
Long. Stress at Top of Midspan	35.43	1650.86
Long. Stress at Bottom of Midspan	-27.38	-2159.64
Long. Stress at Bottom of Midspan	27.38	1650.86
Long. Stress at Top of Saddles	52.29	1650.86
Long. Stress at Bottom of Saddles	-77.79	-2159.64
Long. Stress at Bottom of Saddles	77.79	1650.86
Tangential Shear in Shell	42.87	1320.68
Circ. Stress at Horn of Saddle	378.07	2063.58
Circ. Compressive Stress in Shell	12.09	1650.86

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num \text{ of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(4151.4/2 + 0) \cdot 1630.0/2049.0164$$

$$= 4953.7 \text{ kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl,wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(1233.62, 0) \cdot 1630.0/4250.0$$

$$= 473.1 \text{ kgf}$$



Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq; Sum of X Forces}) \cdot B / L_s$$

$$= \max(6317.98, 0) \cdot 1630.0/4250.0$$

$$= 2423.1 \text{ kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(6318/2 + 0) \cdot 1630.0/2049.0164$$

$$= 7539.0 \text{ kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl, Fwt, Fsl, Fst})$$

$$= 8112 + \max(473, 4954, 2423, 7539)$$

$$= 15650.8 \text{ kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.0 \cdot 1616.989(0.0 + \pi/4(2.572 \cdot 1.2)^2)$$

$$= 12097.753 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	16466.02	kgf
Transverse Shear Load	3158.99	kgf
Longitudinal Shear Load	6317.98	kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(2300.0/2 + 21.0 + 12.0)\sin(120.0/2), 2 \cdot 1161.25)$$

$$= 2049.016 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -15651 \cdot 190.0 [1 - (1 - 190.0/896.0 + (116.125^2 - 58.425^2)/(2 \cdot 190.0 \cdot 896.0))/(1 + (4 \cdot 58.43)/(3 \cdot 896.0))]$$

$$= -7370.6 \text{ kgf-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 15651 \cdot 896/4((1 + 2(116^2 - 58^2)/(896^2))/(1 + (4 \cdot 58)/(3 \cdot 896)) - 4 \cdot 190/896)$$

$$= 3326.3 \text{ kgf-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) - 3326.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -35.43 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) + 3326.3/(\pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -27.38 \text{ kgf/cm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) - 7370.6/(0.1066 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= 52.29 \text{ kgf/cm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m/(2t) + M1/(K1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -1.055 \cdot 1161.25/(2 \cdot 19.5) + 7370.6/(0.1923 \cdot \pi \cdot 1161.2^2 \cdot 19.5)$$

$$= -77.79 \text{ kgf/cm}^2$$



Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a)/(L+(4 \cdot hm/3))$
 $= 15651(896.0 - 2 \cdot 190.0)/(896.0 + (4 \cdot 58.43/3))$
 $= 8292.2 \text{ kgf}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 8292.23 / (1161.25 \cdot 19.5)$
 $= 42.87 \text{ kgf/cm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(1161.25 \cdot 19.5)}$
 $= 117.375 \text{ mm}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 15651 \cdot 0.1 / (19.5(270.0 + 117.37 + 117.37))$
 $= -12.09 \text{ kgf/cm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:
 $= -Q/(4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$
 $= -15651 / (4 \cdot 19.5(270.0 + 117.375 + 117.375)) -$
 $12 \cdot 0.053 \cdot 15651 \cdot 1161.25 / (896.0 \cdot 19.5^2)$
 $= -378.07 \text{ kgf/cm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(270.0 + 1.56 \cdot \sqrt{(1161.25 \cdot 19.5)}, 2 \cdot 190.0)$
 $= 504.75 \text{ mm}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2960.0000	mm
Baseplate Thickness	Bpthk	28.0000	mm
Baseplate Width	Bpwid	270.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	22.0000	mm
Web Thickness	Webtk	22.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		80.0	°C
Saddle Yield Stress	Sy	2477.8	kgf/cm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1151.5 \cdot 19.5)} + 480.0)^3) / 12$
 $= 44.104 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 44.104 + 139.184(165.012 - 9.75)^2$
 $= 33596.133 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	713.8	19.5	9.8	139.2	135704.0	0.336E+05
Wearplate	480.0	12.0	25.5	57.6	146880.0	0.112E+05
Web	22.0	419.0	241.0	92.2	2221538.2	0.188E+05
BasePlate	270.0	28.0	464.5	75.6	3511620.5	0.679E+05
Totals	...	...	...	364.6	6015743.0	0.131E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 6015743/364.564$



$$= 165.012 \text{ mm}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 120.0/2 \\ &= 120.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0)) \\ &= 0.2035 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.204 \cdot 15650.775 \\ &= 3185.2722 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 14.1329 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 1486.6764 \text{ kgf/cm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\ &= 1630.0 - 1151.5 \cdot \sin(120.0/2) / 1.0472 \\ &= 677.717 \text{ mm} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2158.7161 \text{ kgf-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 27.0927 \text{ kgf/cm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 1651.8627 \text{ kgf/cm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(15651 + 815) \cdot 270.0) / (4 \cdot 2960.0 \cdot 1651.863))} \\ &= 8.258 \text{ mm} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 120.0/2) (1150.0 + 21.0 + 12.0) \\ &= 2049.016 \text{ mm} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 2049.0161 / (4 - 1) \\ &= 683.005 \text{ mm} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 683.0054 \cdot 270.0/2 \\ &= 922.057 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 16466.021 / (2960.0 \cdot 270.0) \\ &= 2.060 \text{ kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 922.1 \cdot 2.06 \\ &= 1899.727 \text{ kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 54.56 + 75.131 \\ &= 129.691 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P/Ar \\ &= 1899.7/129.6906 \end{aligned}$$



= 14.648 kgf/cm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	22.0	270.0	...	59.4	...	0.361E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 270.0 - 22.0
 = 248.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/129.691
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 270.0/2
 = 135.000 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{(3608.5/129.691)}$
 = 52.749 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 1630.0 - cos(120.0/2)(1150.0 + 21.0 + 12.0) - 28.0
 = 1010.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 2032632/2477.8)}$
 = 127.251

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1010.5/52.749
 = 19.157

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 6318.0 / (2 • 2960.0) • 683.005 • 1010.5/2
 = 368.288 kgf-m.

Compressive Allowable, KL/r < Cc (19.1569 < 127.2511) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (19.16)^2 / (2 \cdot 127.25^2)) 2478 / (5/3 + 3 \cdot (19.16) / (8 \cdot 127.25) - (19.16^3) / (8 \cdot 127.25^3))$
 = 1422 kgf/cm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I)/Sba
 = 14.65/1422.03 + (368.29 • 135.0/36085492)/1651.86
 = 0.094

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	22.0	248.0	0.0	54.6	0.0	0.361E+04
Web	683.0	22.0	0.0	150.3	0.0	60.6
Totals	...	...	...	204.8	...	0.367E+04



Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0/204.821 \\ &= 0.000 \text{ mm} \end{aligned}$$

Distance to Centroid [C1]:

$$\begin{aligned} &= \text{Saddle Width} / 2 \\ &= 270.0/2 \\ &= 135.000 \text{ mm} \end{aligned}$$

Length of Inner Rib [L]:

$$\begin{aligned} &= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk} \\ &= 1630.0 - \sqrt{(1183.0 + 12.0)^2 - (683.005/2)^2} - 28.0 \\ &= 469.364 \text{ mm} \end{aligned}$$

Radius of Gyration [r]:

$$\begin{aligned} &= \sqrt{(\text{Total Inertia} / \text{Total Area})} \\ &= \sqrt{(3667.2/204.821)} \\ &= 42.314 \text{ mm} \end{aligned}$$

Slenderness ratio [KL/r]:

$$\begin{aligned} &= KL/r \\ &= 1 \cdot 469.364/42.314 \\ &= 11.093 \end{aligned}$$

Unit Force [Force,u]:

$$\begin{aligned} &= FI / (2 \cdot \text{Baseplate Length}) \\ &= 6317.981/(2 \cdot 2960.0) \\ &= 1.067 \text{ kgf/mm} \end{aligned}$$

Moment at base of inner Rib [Mbase,c]:

$$\begin{aligned} &= \text{Unit Force} \cdot e \cdot L \\ &= 1.067 \cdot 683.005 \cdot 469.364 \\ &= 342.130 \text{ kgf-m.} \end{aligned}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$\begin{aligned} &= \text{Bending Moment} / \text{Section Modulus} \\ &= 342.13/271644.625 \\ &= 125.948 \text{ kgf/cm}^2 \end{aligned}$$

Compressive Allowable, $KL/r < C_c$ ($11.0925 < 127.2511$) per AISC E2-1 [Sca]:

$$\begin{aligned} &= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3)) \\ &= (1 - (11.09)^2 / (2 \cdot 127.25^2)) 2478 / \\ &\quad (5/3 + 3 \cdot (11.09) / (8 \cdot 127.25) - (11.09^3) / (8 \cdot 127.25^3)) \\ &= 1453 \text{ kgf/cm}^2 \end{aligned}$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$\begin{aligned} &= S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba} \\ &= 19.07/1452.61 + (342.13 \cdot 135.0/3667.199)/1651.86 \\ &= 0.089 \end{aligned}$$

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Calculations per ASME VIII-2, 4.3.11

Input Values:

SH.2

Design Internal Pressure, Small End	23.601	kgf/cm ²
Design Internal Pressure, Large End	23.601	kgf/cm ²
Temperature for Internal Pressure	80.0	°C
Design External Pressure	1.055	kgf/cm ²
Temperature for External Pressure	37.0	°C

CONE Properties:

Cone Material		SA-516 70	+N+IT
Cone Allowable Stress at Temperature	Ss	1650.86	kgf/cm ²
Cone Allowable Stress At Ambient		1784.44	kgf/cm ²
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	21.0000	mm
Corrosion Allowance for Cone	Cac	1.5000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	1200.000	mm
Diameter of Large End of Cone	Dl	2300.000	mm
Half Apex Angle for Cone		29.94	degrees
Axial Length of Cone	Lc	1910.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 70	+N+IT
Operating Allowable Stress	Ss	1784.41	kgf/cm ²
Ambient Allowable Stress		1784.41	kgf/cm ²
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	16.0000	mm
Corrosion Allowance	Cas	1.5000	mm
Axial Length of Attached Element	Ls	300.031	mm

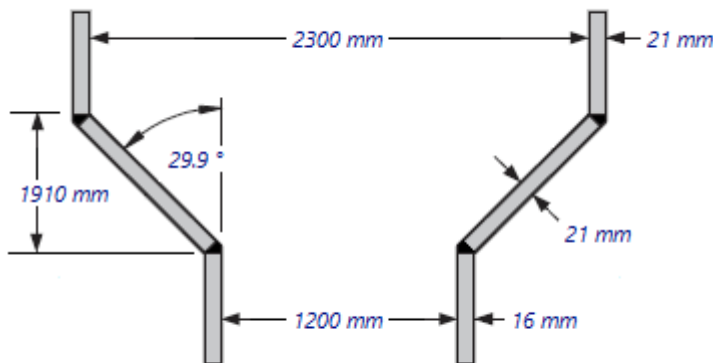
LARGE-END Properties:

Attached Large End Material Specification		SA-516 70	+N+IT
Operating Allowable Stress	Sl	1650.86	kgf/cm ²
Ambient Allowable Stress		1784.44	kgf/cm ²
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	21.0000	mm
Corrosion Allowance	Cal	1.5000	mm
Axial Length of Attached Element	Ll	6941.667	mm

Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: None

Warning: rtk -1.500mm, must be > Tl, 19.500mm, see 4.3.12



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 37 °C

Elastic Modulus of Cone Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Small Cylinder Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Large Cylinder Material	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Large End Reinforcement	0.206E+07 kgf/cm ² at 37 °C
Elastic Modulus of Small End Reinforcement	0.206E+07 kgf/cm ² at 37 °C

Elastic Modulus Data from ASME Section II Part D at 80 °C

Elastic Modulus of Cone Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Small Cylinder Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Large Cylinder Material	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Large End Reinforcement	0.203E+07 kgf/cm ² at 80 °C
Elastic Modulus of Small End Reinforcement	0.203E+07 kgf/cm ² at 80 °C

Axial forces and moments are not computed for Horizontal geometries.
 Compute them manually and enter the loads using the Force/Moment dialog.
 Place the loads at the ends of the cone as needed.

Maximum Centroid Reinforcement Distance Large End	25.4000 mm
Maximum Centroid Reinforcement Distance Small End	25.4000 mm

Note: No ring was found close enough to the large end to be considered.

Note: No ring was found close enough to the small end to be considered.

Junction Calculations per 4.3.11:

Calculations for the Large End Cone/Shell Junction - Without Knuckle

F_{he} per 4.1.3 or 4.3.11.2 Step 6 [F_{he}]:
 $= 0.4 \cdot E(tc/D)$
 $= 0.4 \cdot 2057748(19.5/2342.0)$
 $= 6853.3 \text{ kgf/cm}^2$

Factor of Safety [FS]:
 Since F_{ic} [2477.82] >= F_y[2477.82], FS = 1.667

Allowable Stress per 4.3.11 Step 6 [F_{ha}]:
 $= F_{ic}/FS$
 $= 2477.8247/1.667$
 $= 1486.398 \text{ kgf/cm}^2$

Large End under Internal Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:
 $= 2 \cdot \sqrt{(RL \cdot Tc / \cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc / \cos(\alpha))}$
 $= 2 \cdot \sqrt{(1151.731 \cdot 19.5 / \cos(29.94)) + 1.4 \cdot \sqrt{(601.731 \cdot 19.5 / \cos(29.94))}$
 $= 484.854 \text{ mm}$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	696.982	2476.295	Pass
Sigma sm-	696.982	2476.295	Pass
Sigma thetam+	-443.907	2476.295	Pass
Sigma thetam-	-443.907	2476.295	Pass
Sigma sm+ + Sigma	-2664.997	4955.649	Pass
Sigma sm+ - Sigma	4058.961	4955.649	Pass
Sigma sm- + Sigma	-2664.997	4955.649	Pass
Sigma sm- - Sigma	4058.961	4955.649	Pass
Sigma thetam+ +	-1452.501	4955.649	Pass
Sigma thetam+ -	564.687	4955.649	Pass
Sigma thetam- +	-1452.501	4955.649	Pass
Sigma thetam- -	564.687	4955.649	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	697.175	2476.295	Pass
Sigma sm-	697.175	2476.295	Pass
Sigma thetam+	-487.826	2476.295	Pass
Sigma thetam-	-487.826	2476.295	Pass
Sigma sm+ + Sigma	-2664.804	4955.649	Pass
Sigma sm+ - Sigma	4059.154	4955.649	Pass
Sigma sm- + Sigma	-2664.804	4955.649	Pass
Sigma sm- - Sigma	4059.154	4955.649	Pass
Sigma thetam+ +	-1496.420	4955.649	Pass
Sigma thetam+ -	520.768	4955.649	Pass
Sigma thetam- +	-1496.420	4955.649	Pass
Sigma thetam- -	520.768	4955.649	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	697.079	-1191.225	Pass
Sigma _{smCyl} - + S	697.079	-1191.225	Pass
Sigma _{thetamCyl} +	-465.867	-1486.398	Pass
Sigma _{thetamCyl} -	-465.867	-1486.398	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.141 \cdot 1151.731) + 0.0 / (3.142 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 1151.731) - 0.0 / (3.141 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Thickness Ratio [n]:
 $= tc/tL = 19.5/19.5 = 1.0$

Curve Fitting Geometry [H and B]:
 $H = \sqrt{0.5 \cdot D/tL} = \sqrt{0.5 \cdot 2303.462/19.5} = 7.6853$
 $B = \tan(29.938) = 0.5759$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.0) < 1.25$



Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 = -3.065534 C2 = 3.642747 C3 = 0.810048 C4 = -0.221192
 C5 = -0.081824 C6 = 0.035052 C7 = 0.025775 C8 = -0.015413
 C9 = 0.002102 C10 = -0.005587 C11 = 0.000000

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 3.1672, MsN1 = -\exp(X) = -23.7415$$

Table 4.3.3 [Qn1]:

C1 = -1.983852 C2 = 2.410703 C3 = 0.626443 C4 = -0.119151
 C5 = -0.115841 C6 = 0.122993 C7 = 0.012160 C8 = -0.016987
 C9 = 0.010919 C10 = -0.016653 C11 = 0.000000

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B) \\ = C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576) \\ X = 2.0684, Qn = -\exp(X) = -7.9123$$

Table 4.3.4 MsN2:

C1 = -5.697151 C2 = 0.003838 C3 = 0.476317 C4 = -0.213157
 C5 = 2.233703 C6 = 0.000032 C7 = 0.002506 C8 = -0.001663
 C9 = -0.212965 C10 = 0.000138 C11 = -0.106203

Compute using Table 4.3.4 [MsN2]:

$$X1 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\ X2 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + C10 \cdot \ln(7.69^2) \cdot \ln(0.52) = 0.2743 \\ MsN2 = \exp(X1/X2) = \exp(-0.053/0.274) = -0.8241$$

Table 4.3.4 QN2:

C1 = -4.774616 C2 = 0.000461 C3 = -0.002831 C4 = -0.197117
 C5 = 1.982132 C6 = 0.000069 C7 = -0.000234 C8 = -0.003536
 C9 = -0.202493 C10 = -0.000088 C11 = 0.001365

Compute using Table 4.3.4 [Qn2]:

$$X3 = C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\ = C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\ X4 = 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\ = 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + C10 \cdot \ln(7.685^2) \cdot \ln(0.523) = 0.2743 \\ Qn2 = \exp(X3/X4) = \exp(-0.373/0.291) = -0.2772$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = 23.601 \cdot 19.5^2 \cdot -23.742 = -2130.6509 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = -2130.651 + -0.0 = -2130.6509 \text{ kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = -2130.651 + -0.0 = -2130.6509 \text{ kgf}$$



Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = 23.601 \cdot 19.5 \cdot -7.912 = -36.4144 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = -36.414 + -0.0 = -36.4144 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = -36.414 + -0.0 = -36.4144 \text{ kgf/mm}$$

Compute [βcy]:

$$= 3 \cdot (1 - v^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\ = (3 \cdot (1 - 0.3^2) / (1151.731^2 \cdot 19.5^2)) \cdot 0.25 = 0.0086 \text{ 1/mm}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = 23.601 \cdot 1151.731/2 + 0.0 = 135.9113 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = 23.601 \cdot 1151.731/2 + 0.0 = 135.9113 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 23.601 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\ (-2130.651 \cdot 0.009 + -36.414) = -86.5618 \text{ kgf/mm}$$

Compute with XL- [Ntheta-]:

$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\ = 23.601 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\ (-2130.651 \cdot 0.009 + -36.414) = -86.5618 \text{ kgf/mm}$$

Note: Kpc has been set to 1.0

σSm with XL+ [σsm+]:

$$= Ns / tL \\ = 135.911/19.5 \\ = 696.982 \text{ kgf/cm}^2$$

σSm with XL- [σsm-]:

$$= Ns / tL \\ = 135.911/19.5 \\ = 696.982 \text{ kgf/cm}^2$$

σSb with XL+ [σsb+]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -2130.651 / (19.5^2 \cdot 1.0) \\ = -3361.979 \text{ kgf/cm}^2$$

σSb with XL- [σsb-]:

$$= 6 \cdot Ms / (tL^2 \cdot Kpc) \\ = 6 \cdot -2130.651 / (19.5^2 \cdot 1.0) \\ = -3361.979 \text{ kgf/cm}^2$$

σThetam with XL+ [σthetam+]:

$$= Ntheta / tL \\ = -86.562/19.5 \text{ kgf/cm}^2 \\ = -443.907 \text{ kgf/cm}^2$$

σThetam with XL- [σthetam-]:

$$= Ntheta / tL \\ = -86.562/19.5 \\ = -443.907 \text{ kgf/cm}^2$$

σThetab with XL+ [σthetab+]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$



$$= 6 \cdot 0.3 \cdot -2130.651 / (19.5^2 \cdot 1.0)$$

$$= -1008.594 \text{ kgf/cm}^2$$

σ_{Thetab} with XL- [σ_{thetab}]:

$$= 6 \cdot v \cdot Ms / (tL^2 \cdot Kpc)$$

$$= 6 \cdot 0.3 \cdot -2130.651 / (19.5^2 \cdot 1.0)$$

$$= -1008.594 \text{ kgf/cm}^2$$

For the Large End Cone Junction per Table 4.3.1

Compute [McsP]:

$$= Msp = Msp = -2130.6509 \text{ kgf}$$

McsX with XL+ [McsX+]:

$$= MsX = -0.0 \text{ kgf}$$

McsX with XL- [McsX-]:

$$= MsX = -0.0 \text{ kgf}$$

Mcs with XL+ [Mcs+]:

$$= McsP + MsX = -2130.651 + -0.0$$

$$= -2130.651 \text{ kgf}$$

Mcs with XL- [Mcs-]:

$$= McsP + MsX = -2130.651 + -0.0$$

$$= -2130.651 \text{ kgf}$$

Compute with XL+ [McsX+]:

$$= MsX = -0.0 \text{ kgf}$$

Compute with XL- [McsX-]:

$$= MsX = -0.0 \text{ kgf}$$

Compute with XL+ [Qc+]:

$$= Q \cdot \cos(a) + Ns \cdot \sin(a)$$

$$= -36.414 \cdot \cos(29.938) + 135.911 \cdot \sin(29.938)$$

$$= 36.274 \text{ kgf/mm}$$

Compute with XL- [Qc-]:

$$= Q \cdot \cos(a) - Ns \cdot \sin(a)$$

$$= -36.414 \cdot \cos(29.938) - 135.911 \cdot \sin(29.938)$$

$$= 36.274 \text{ kgf/mm}$$

Compute [Rc]:

$$= RL / \cos(a) = 1151.731 / \cos(29.938) = 1329.0795 \text{ mm}$$

Compute [β -co]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$$

$$= (3 \cdot (1 - 0.3^2) / (1329.079^2 \cdot 19.5^2))^{1/4}$$

$$= 0.008 \text{ 1/mm}$$

Compute with XL +ve [Ncs+]:

$$= Ns \cdot \cos(a) - Q \cdot \sin(a)$$

$$= 135.911 \cdot \cos(29.938) - -36.414 \cdot \sin(29.938)$$

$$= 135.949 \text{ kgf/mm}$$

Compute with XL -ve [Ncs-]:

$$= Ns \cdot \cos(a) + Q \cdot \sin(a)$$

$$= 135.911 \cdot \cos(29.938) + -36.414 \cdot \sin(29.938)$$

$$= 135.949 \text{ kgf/mm}$$

Compute with XL +ve [NcTheta+]:

$$= P \cdot RL / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 23.601 \cdot 1151.731 / \cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-2130.651 \cdot 0.008 - 36.274)$$

$$= -95.126 \text{ kgf/mm}$$

Compute with XL -ve [NcTheta-]:

$$= P \cdot RL / \cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$$

$$= 23.601 \cdot 1151.731 / \cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-2130.651 \cdot 0.008 - 36.274)$$



$$(-2130.651 \cdot 0.008 - 36.274)$$

$$= -95.126 \text{ kgf/mm}$$

Note: $K_{cpc} = 1.0$

Compute with XL +ve $[\sigma_{sm+}]$:

$$= N_{cs}/tC = 135.949/19.5 = 697.175 \text{ kgf/cm}^2$$

Compute with XL -ve $[\sigma_{sm-}]$:

$$= N_{cs}/tC = 135.949/19.5 = 697.175 \text{ kgf/cm}^2$$

Compute with XL +ve $[\sigma_{sb+}]$

$$= 6 \cdot M_{cs}/(tC^2 \cdot K_{cpc})$$

$$= 6 \cdot -2130.651/(19.5^2 \cdot 1.0)$$

$$= -3361.979 \text{ kgf/cm}^2$$

Compute with XL -ve $[\sigma_{sb-}]$

$$= 6 \cdot M_{cs}/(tC^2 \cdot K_{cpc})$$

$$= 6 \cdot -2130.651/(19.5^2 \cdot 1.0)$$

$$= -3361.979 \text{ kgf/cm}^2$$

Compute with XL +ve $[\sigma_{\theta_{etam+}}]$:

$$= N_{C\theta_{etam}}/tC = -95.126/19.5 = -487.83 \text{ kgf/cm}^2$$

Compute with XL -ve $[\sigma_{\theta_{etam-}}]$:

$$= N_{C\theta_{etam}}/tC = -95.126/19.5 = -487.83 \text{ kgf/cm}^2$$

Compute with XL +ve $[\sigma_{\theta_{etab+}}]$:

$$= 6 \cdot v \cdot M_{cs}/(tC^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot -2130.651/(19.5^2 \cdot 1.0)$$

$$= -1008.594 \text{ kgf/cm}^2$$

Compute with XL -ve $[\sigma_{\theta_{etab-}}]$:

$$= 6 \cdot v \cdot M_{cs} / (tC^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot -2130.651/(19.5^2 \cdot 1.0)$$

$$= -1008.594 \text{ kgf/cm}^2$$

Large End under External Pressure:

Cone Length along Cone surface per 4.3.46 [ConeMinLen]:

$$= 2 \cdot \sqrt{(RL \cdot Tc/\cos(\alpha)) + 1.4 \cdot \sqrt{(Rs \cdot tc/\cos(\alpha))}}$$

$$= 2 \cdot \sqrt{(1151.731 \cdot 19.5/\cos(29.94)) + 1.4 \cdot \sqrt{(601.731 \cdot 19.5/\cos(29.94))}}$$

$$= 484.854 \text{ mm}$$

Results for the Large End Cylinder Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-31.144	2476.295	Pass
Sigma sm-	31.144	2476.295	Pass
Sigma thetam+	19.836	2476.295	Pass
Sigma thetam-	19.836	2476.295	Pass
Sigma sm+ + Sigma	119.083	4955.649	Pass
Sigma sm+ - Sigma	-181.372	4955.649	Pass
Sigma sm- + Sigma	119.083	4955.649	Pass
Sigma sm- - Sigma	-181.372	4955.649	Pass
Sigma thetam+ +	64.904	4955.649	Pass
Sigma thetam+ -	-25.233	4955.649	Pass
Sigma thetam- +	64.904	4955.649	Pass
Sigma thetam- -	-25.233	4955.649	Pass

Results for the Large End Cone Without a Knuckle

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-31.153	2476.295	Pass
Sigma sm-	31.153	2476.295	Pass
Sigma thetam+	21.798	2476.295	Pass
Sigma thetam-	21.798	2476.295	Pass
Sigma sm+ + Sigma	119.075	4955.649	Pass



Sigma sm+ - Sig	-181.380	4955.649	Pass
Sigma sm- + Sig	119.075	4955.649	Pass
Sigma sm- - Sig	-181.380	4955.649	Pass
Sigma thetam+ +	66.866	4955.649	Pass
Sigma thetam+ -	-23.270	4955.649	Pass
Sigma thetam- +	66.866	4955.649	Pass
Sigma thetam- -	-23.270	4955.649	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

SigmaσmCyl+ + S	-31.148	-1191.225	Pass
SigmaσmCyl- + S	-31.148	-1191.225	Pass
SigmaThetamCyl+	20.817	-1486.398	Pass
SigmaThetamCyl-	20.817	-1486.398	Pass

Cone Large end without a Knuckle Cylinder Section:

Net Axial Force ML +ve [XL+]:
 $= FL / (2 \cdot \pi \cdot RL) + ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.141 \cdot 1151.731) + 0.0 / (3.142 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Net Axial Force ML -ve [XL-]:
 $= FL / (2 \cdot \pi \cdot RL) - ML / (\pi \cdot RL^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 1151.731) - 0.0 / (3.141 \cdot 1151.731^2)$
 $= 0.0 \text{ kgf/mm}$

Thickness Ratio [n]:
 $= tc/tL = 19.5/19.5 = 1.0$

Curve Fitting Geometry [H and B]:
 $H = \text{Sqrt}(0.5 \cdot D/tL) = \text{Sqrt}(0.5 \cdot 2303.462/19.5) = 7.6853$
 $B = \tan(29.938) = 0.5759$

This analysis is preformed per Table 4.3.1 for the Large End:

Using Data from Table 4.3.3 and 4.3.4 knowing n
 Table 4.3.3 - n: $1.0 < (n=1.0) < 1.25$

Use n to construct a table: C1, C2, C3..... by interpolation

Table 4.3.3 [MsN1]:

C1 =	-3.065534	C2 =	3.642747	C3 =	0.810048	C4 =	-0.221192
C5 =	-0.081824	C6 =	0.035052	C7 =	0.025775	C8 =	-0.015413
C9 =	0.002102	C10 =	-0.005587	C11 =	0.000000		

Compute using Table 4.3.3 [MsN1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B)$$

$$= C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576)$$

$$X = 3.1672, \text{ MsN1} = -\exp(X) = -23.7415$$

Table 4.3.3 [Qn1]:

C1 =	-1.983852	C2 =	2.410703	C3 =	0.626443	C4 =	-0.119151
C5 =	-0.115841	C6 =	0.122993	C7 =	0.012160	C8 =	-0.016987
C9 =	0.010919	C10 =	-0.016653	C11 =	0.000000		

Compute using Table 4.3.3 [Qn1]:

$$X = C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot (\ln(H))^2 + C5 \cdot (\ln(B))^2 + C6 \cdot \ln(H) \cdot \ln(B) + C7 \cdot (\ln(H))^3 + C8 \cdot (\ln(B))^3 + C9 \cdot \ln(H) \cdot (\ln(B))^2 + C10 \cdot (\ln(H))^2 \cdot \ln(B)$$

$$= C1 + C2 \cdot \ln(7.685) + C3 \cdot \ln(0.576) + C4 \cdot (\ln(7.685))^2 + C5 \cdot (\ln(0.576))^2 + C6 \cdot \ln(7.685) \cdot \ln(0.576) + C7 \cdot (\ln(7.685))^3 + C8 \cdot (\ln(0.576))^3 + C9 \cdot \ln(7.685) \cdot (\ln(0.576))^2 + C10 \cdot (\ln(7.685))^2 \cdot \ln(0.576)$$

$$X = 2.0684, \text{ Qn} = -\exp(X) = -7.9123$$

Table 4.3.4 MsN2:

C1 =	-5.697151	C2 =	0.003838	C3 =	0.476317	C4 =	-0.213157
C5 =	2.233703	C6 =	0.000032	C7 =	0.002506	C8 =	-0.001663
C9 =	-0.212965	C10 =	0.000138	C11 =	-0.106203		

Compute using Table 4.3.4 [MsN2]:



$$\begin{aligned}
 X1 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\
 &= C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + \\
 &\quad C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\
 X2 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\
 &= 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + \\
 &\quad C10 \cdot \ln(7.69^2) \cdot \ln(0.52) = 0.2743 \\
 MsN2 &= \exp(X1/X2) = \exp(-0.053/0.274) = -0.8241
 \end{aligned}$$

Table 4.3.4 QN2:

$$\begin{aligned}
 C1 &= -4.774616 & C2 &= 0.000461 & C3 &= -0.002831 & C4 &= -0.197117 \\
 C5 &= 1.982132 & C6 &= 0.000069 & C7 &= -0.000234 & C8 &= -0.003536 \\
 C9 &= -0.202493 & C10 &= -0.000088 & C11 &= 0.001365
 \end{aligned}$$

Compute using Table 4.3.4 [Qn2]:

$$\begin{aligned}
 X3 &= C1 + C3 \cdot \ln(H^2) + C5 \cdot \ln(A) + C7 \cdot (\ln(H^2))^2 + C9 \cdot (\ln(A))^2 + C11 \cdot \ln(H^2) \cdot \ln(A) \\
 &= C1 + C3 \cdot \ln(7.685^2) + C5 \cdot \ln(0.523) + C7 \cdot (\ln(7.685^2))^2 + C9 \cdot (\ln(0.523))^2 + \\
 &\quad C11 \cdot \ln(7.685^2) \cdot \ln(0.523) = -0.0531 \\
 X4 &= 1 + C2 \cdot \ln(H^2) + C4 \cdot \ln(AR) + C6 \cdot (\ln(H^2))^2 + C8 \cdot (\ln(AR))^2 + C10 \cdot \ln(H^2) \cdot \ln(AR) \\
 &= 1 + C2 \cdot \ln(7.685) + C4 \cdot \ln(0.523) + C6 \cdot (\ln(7.685^2))^2 + C8 \cdot (\ln(0.523))^2 + \\
 &\quad C10 \cdot \ln(7.685^2) \cdot \ln(0.523) = 0.2743 \\
 Qn2 &= \exp(X3/X4) = \exp(-0.373/0.291) = -0.2772
 \end{aligned}$$

Stresses for the Large End Cylinder per Table 4.3.1

Stress Resultant Value [Msp]:

$$= P \cdot tL^2 \cdot MsN1 = -1.055 \cdot 19.5^2 \cdot -23.742 = 95.2066 \text{ kgf}$$

Compute with XL+ [Msx+]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL- [Msx-]:

$$= XL \cdot tL \cdot MsN2 = 0.0 \cdot 19.5 \cdot -0.824 = -0.0 \text{ kgf}$$

Compute with XL+ [Ms+]:

$$= MsP + Msx = 95.207 + -0.0 = 95.2066 \text{ kgf}$$

Compute with XL- [Ms-]:

$$= MsP + Msx = 95.207 + -0.0 = 95.2066 \text{ kgf}$$

Compute [Qp]:

$$= P \cdot tL \cdot Qn1 = -1.055 \cdot 19.5 \cdot -7.912 = 1.6272 \text{ kgf/mm}$$

Compute with XL+ [Qx+]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL- [Qx-]:

$$= XL \cdot QN2 = 0.0 \cdot -0.005 = -0.0 \text{ kgf/mm}$$

Compute with XL+ [Q+]:

$$= Qp + Qx = 1.627 + -0.0 = 1.6272 \text{ kgf/mm}$$

Compute with XL- [Q-]:

$$= Qp + Qx = 1.627 + -0.0 = 1.6272 \text{ kgf/mm}$$

Compute [βcy]:

$$\begin{aligned}
 &= 3 \cdot (1 - \nu^2) / (RL^2 \cdot tL^2) \cdot 0.25 \\
 &= (3 \cdot (1 - 0.3^2) / (1151.731^2 \cdot 19.5^2)) \cdot 0.25 = 0.0086 \text{ 1/mm}
 \end{aligned}$$

Compute with XL+ [Ns+]:

$$= P \cdot RL/2 + XL = -1.055 \cdot 1151.731/2 + 0.0 = -6.0731 \text{ kgf/mm}$$

Compute with XL- [Ns-]:

$$= P \cdot RL/2 + XL = -1.055 \cdot 1151.731/2 + 0.0 = -6.0731 \text{ kgf/mm}$$

Compute with XL+ [Ntheta+]:

$$\begin{aligned}
 &= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q) \\
 &= -1.055 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot \\
 &\quad (-95.207 \cdot 0.009 + 1.627) = 3.8679 \text{ kgf/mm}
 \end{aligned}$$

Compute with XL- [Ntheta-]:



$$= P \cdot RL + 2 \cdot \beta_{cy} \cdot RL \cdot (-Ms \cdot \beta_{cy} + Q)$$

$$= -1.055 \cdot 1151.731 + 2 \cdot 0.0086 \cdot 1151.731 \cdot (-95.207 \cdot 0.009 + 1.627) = 3.8679 \text{ kgf/mm}$$

Note: Kpc has been set to 1.0

$$\sigma_{Sm} \text{ with XL+ } [\sigma_{sm+}]:$$

$$= N_s / t_L$$

$$= -6.073/19.5$$

$$= -31.144 \text{ kgf/cm}^2$$

$$\sigma_{Sm} \text{ with XL- } [\sigma_{sm-}]:$$

$$= N_s / t_L$$

$$= -6.073/19.5$$

$$= -31.144 \text{ kgf/cm}^2$$

$$\sigma_{Sb} \text{ with XL+ } [\sigma_{sb+}]:$$

$$= 6 \cdot Ms / (t_L^2 \cdot K_{pc})$$

$$= 6 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 150.228 \text{ kgf/cm}^2$$

$$\sigma_{Sb} \text{ with XL- } [\sigma_{sb-}]:$$

$$= 6 \cdot Ms / (t_L^2 \cdot K_{pc})$$

$$= 6 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 150.228 \text{ kgf/cm}^2$$

$$\sigma_{Thetam} \text{ with XL+ } [\sigma_{thetam+}]:$$

$$= N_{theta} / t_L$$

$$= 3.868/19.5 \text{ kgf/cm}^2$$

$$= 19.836 \text{ kgf/cm}^2$$

$$\sigma_{Thetam} \text{ with XL- } [\sigma_{thetam-}]:$$

$$= N_{theta} / t_L$$

$$= 3.868/19.5$$

$$= 19.836 \text{ kgf/cm}^2$$

$$\sigma_{Thetab} \text{ with XL+ } [\sigma_{thetab+}]:$$

$$= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 45.068 \text{ kgf/cm}^2$$

$$\sigma_{Thetab} \text{ with XL- } [\sigma_{thetab-}]:$$

$$= 6 \cdot v \cdot Ms / (t_L^2 \cdot K_{pc})$$

$$= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 45.068 \text{ kgf/cm}^2$$

For the Large End Cone Junction per Table 4.3.1

$$\text{Compute } [McsP]:$$

$$= M_{sp} = M_{sp} = 95.2066 \text{ kgf}$$

$$McsX \text{ with XL+ } [McsX+]:$$

$$= M_{sX} = -0.0 \text{ kgf}$$

$$McsX \text{ with XL- } [McsX-]:$$

$$= M_{sX} = -0.0 \text{ kgf}$$

$$Mcs \text{ with XL+ } [Mcs+]:$$

$$= M_{csP} + M_{sX} = 95.207 + -0.0$$

$$= 95.207 \text{ kgf}$$

$$Mcs \text{ with XL- } [Mcs-]:$$

$$= M_{csP} + M_{sX} = 95.207 + -0.0$$

$$= 95.207 \text{ kgf}$$

$$\text{Compute with XL+ } [McsX+]:$$

$$= M_{sX} = -0.0 \text{ kgf}$$

$$\text{Compute with XL- } [McsX-]:$$

$$= M_{sX} = -0.0 \text{ kgf}$$



Compute with XL+ [Qc+]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a)$
 $= 1.627 \cdot \cos(29.938) + -6.073 \cdot \sin(29.938)$
 $= -1.621 \text{ kgf/mm}$

Compute with XL- [Qc-]:
 $= Q \cdot \cos(a) + Ns \cdot \sin(a)$
 $= 1.627 \cdot \cos(29.938) + -6.073 \cdot \sin(29.938)$
 $= -1.621 \text{ kgf/mm}$

Compute [Rc]:
 $= RL/\cos(a) = 1151.731/\cos(29.938) = 1329.0795 \text{ mm}$

Compute [β -co]:
 $= (3 \cdot (1 - v^2) / (Rc^2 \cdot tC^2))^{1/4}$
 $= (3 \cdot (1 - 0.3^2) / (1329.079^2 \cdot 19.5^2))^{1/4}$
 $= 0.008 \text{ 1/mm}$

Compute with XL +ve [Ncs+]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -6.073 \cdot \cos(29.938) - 1.627 \cdot \sin(29.938)$
 $= -6.075 \text{ kgf/mm}$

Compute with XL -ve [Ncs-]:
 $= Ns \cdot \cos(a) - Q \cdot \sin(a)$
 $= -6.073 \cdot \cos(29.938) - 1.627 \cdot \sin(29.938)$
 $= -6.075 \text{ kgf/mm}$

Compute with XL +ve [NcTheta+]:
 $= P \cdot RL/\cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$
 $= -1.055 \cdot 1151.731/\cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-95.207 \cdot 0.008 - -1.621)$
 $= 4.251 \text{ kgf/mm}$

Compute with XL -ve [NcTheta-]:
 $= P \cdot RL/\cos(A) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} - Qc)$
 $= -1.055 \cdot 1151.731/\cos(29.938) + 2 \cdot 0.008 \cdot 1329.079 \cdot (-95.207 \cdot 0.008 - -1.621)$
 $= 4.251 \text{ kgf/mm}$

Note: $Kcpc = 1.0$

Compute with XL +ve [σ_{sm}]:
 $= Ncs/tC = -6.075/19.5 = -31.1528 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{sm}]:
 $= Ncs/tC = -6.075/19.5 = -31.1528 \text{ kgf/cm}^2$

Compute with XL +ve [σ_{sb}]:
 $= 6 \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 150.228 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{sb}]:
 $= 6 \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 150.228 \text{ kgf/cm}^2$

Compute with XL +ve [σ_{Theta+}]:
 $= NTheta/tC = 4.251/19.5 = 21.8 \text{ kgf/cm}^2$

Compute with XL -ve [σ_{Theta-}]:
 $= NTheta/tC = 4.251/19.5 = 21.8 \text{ kgf/cm}^2$

Compute with XL +ve [$\sigma_{thetab+}$]:
 $= 6 \cdot v \cdot Mcs/(tC^2 \cdot Kcpc)$
 $= 6 \cdot 0.3 \cdot 95.207/(19.5^2 \cdot 1.0)$
 $= 45.068 \text{ kgf/cm}^2$



Compute with XL -ve [σ_{thetab}]:

$$= 6 \cdot v \cdot M_{cs} / (tC^2 \cdot K_{cpc})$$

$$= 6 \cdot 0.3 \cdot 95.207 / (19.5^2 \cdot 1.0)$$

$$= 45.068 \text{ kgf/cm}^2$$

Calculations for the Small End Cone/Shell Junction

Fhe per 4.1.3 or 4.3.11.2 Step 6 [Fhe]:

$$= 0.4 \cdot E(tc/D)$$

$$= 0.4 \cdot 2057748(19.5/1232.0)$$

$$= 13027.9 \text{ kgf/cm}^2$$

Factor of Safety [FS]:

$$\text{Since } F_{ic} [2477.82] \geq F_y [2477.82], FS = 1.667$$

Allowable Stress per 4.3.11 Step 6 [Fha]:

$$= F_{ic}/FS$$

$$= 2477.8247/1.667$$

$$= 1486.398 \text{ kgf/cm}^2$$

Small End under Internal Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	489.711	2676.616	Pass
Sigma sm-	489.711	2676.616	Pass
Sigma thetam+	1901.522	2676.616	Pass
Sigma thetam-	1901.522	2676.616	Pass
Sigma sm+ + Sigma	2460.200	5353.233	Pass
Sigma sm- + Sigma	2460.200	5353.233	Pass
Sigma sm+ - Sigma	-1480.779	5353.233	Pass
Sigma sm- - Sigma	-1480.779	5353.233	Pass
Sigma thetam+ +	2492.669	5353.233	Pass
Sigma thetam- +	2492.669	5353.233	Pass
Sigma thetam+ -	1310.375	5353.233	Pass
Sigma thetam- -	1310.375	5353.233	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	360.535	2476.295	Pass
Sigma sm-	360.535	2476.295	Pass
Sigma thetam+	1830.135	2476.295	Pass
Sigma thetam-	1830.135	2476.295	Pass
Sigma sm+ + Sigma	1450.069	4955.649	Pass
Sigma sm- + Sigma	1450.069	4955.649	Pass
Sigma sm+ - Sigma	-729.000	4955.649	Pass
Sigma sm- - Sigma	-729.000	4955.649	Pass
Sigma thetam+ +	2156.995	4955.649	Pass
Sigma thetam- +	2156.995	4955.649	Pass
Sigma thetam+ -	1503.274	4955.649	Pass
Sigma thetam- -	1503.274	4955.649	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:

$$= tc/ts = 19.5/14.5 = 1.3448$$

Compute the ratio equation 4.3.61 [H]:

$$= \sqrt{(R_s/ts)} = \sqrt{(601.731/14.5)} = 6.4419$$

Compute the ratio equation 4.3.62 [B]:

$$= \tan(a) = \tan(29.938) = 0.5759$$

Compute Ms+ from equation 4.3.59 [Xs+]:

$$= F_s / (2 \cdot \pi \cdot R_s) + M_s / (\pi \cdot R_s^2)$$



$$= 0.0 / (2 \cdot 3.142 \cdot 601.731) + 0.0 / (3.141 \cdot 601.731^2)$$

$$= 0.0000 \text{ kgf/mm}$$

Compute Ms- from equation 4.3.59 [Xs-]:

$$= F_s / (2 \cdot \pi \cdot R_s) - M_s / (\pi \cdot R_s^2) + X_s'$$

$$= 0.0 / (2 \cdot 3.142 \cdot 601.731) - 0.0 / (3.141 \cdot 601.731^2) + 0.0$$

$$= 0.0000 \text{ kgf/mm}$$

From Table 4.3.5 MsN:

$$\begin{aligned} C1 &= -10.652290 & C2 &= 1.976683 & C3 &= 4.469143 & C4 &= -0.063541 \\ C5 &= -1.143284 & C6 &= -0.067321 & C7 &= 0.004729 & C8 &= 0.124486 \\ C9 &= 0.017859 & C10 &= -0.004846 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.5 [MsN]:

$$= \exp[C1 + C2 \cdot \ln(H^2) + C3 \cdot \ln(a) + C4 \cdot (\ln(H^2))^2 + C5 \cdot \ln(a)^2 + C6 \cdot \ln(H^2) \cdot \ln(a) + C7 \cdot \ln(H^2)^3 + C8 \cdot \ln(a)^3 + C9 \cdot \ln(H^2) \cdot \ln(a)^2 + C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$$

$$= \exp[-10.652 + 1.977 \cdot \ln(6.442^2) + 4.469 \cdot \ln(29.938) + 0.064 \cdot (\ln(6.442^2))^2 - 1.143 \cdot \ln(29.938)^2 + 0.067 \cdot \ln(6.442^2) \cdot \ln(29.938) + 0.005 \cdot \ln(6.442^2)^3 + 0.124 \cdot \ln(29.938)^3 + 0.018 \cdot \ln(6.442^2) \cdot \ln(29.938)^2 + 0.005 \cdot \ln(6.442^2)^2 \cdot \ln(29.938)] = 13.9152$$

Compute from Table 4.3.2 [Msp]:

$$= p \cdot tL^2 \cdot M_{sn} = 23.601 \cdot 14.5^2 \cdot 13.91515 = 690.4915 \text{ kgf}$$

From table 4.3.6 MsN:

$$\begin{aligned} C1 &= 0.000941 & C2 &= -0.000215 & C3 &= -0.001147 & C4 &= 0.129940 \\ C5 &= 0.009760 & C6 &= 0.000003 & C7 &= -0.000003 & C8 &= 0.008361 \\ C9 &= 0.004813 & C10 &= 0.000040 & C11 &= 0.207115 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B)$$

$$= (0.00094 + -0.00115 \cdot 6.442 + 0.00976 \cdot 29.938 + -0.6 \cdot 4.42^2 + 0.00481 \cdot 29.938^2 + 0.20711 \cdot 6.442 \cdot 29.938) / (1 + -0.00022 \cdot 6.442 + 0.12994 \cdot 29.938 + 0.6 \cdot 4.42^2 + 0.00836 \cdot 29.938^2 + 0.00004 \cdot 6.442 \cdot 29.938)$$

$$= 0.7144$$

Compute from Table 4.3.2 [Msx+]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= X_s \cdot t_s \cdot M_{sN} = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= M_s + M_{sx+} = 690.491 + 0.0 = 690.4915 \text{ kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= M_s + M_{sx-} = 690.491 + 0.0 = 690.4915 \text{ kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.271511 & C2 &= 0.000001 & C3 &= 0.004638 & C4 &= 0.002846 \\ C5 &= -0.000906 & C6 &= 0.000000 & C7 &= -0.000002 & C8 &= -0.000119 \\ C9 &= -0.000126 & C10 &= 0.000001 & C11 &= -0.004307 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha)$$

$$= (0.27151 + 0.00464 \cdot 6.442^2 + -0.00091 \cdot 29.938 - 0.6 \cdot 4.42^4 + -0.00013 \cdot 29.938^2 + -0.00431 \cdot 6.442^2 \cdot 29.938) / (1 + 0.6 \cdot 4.42^2 + 0.00285 \cdot 29.938 + 0.6 \cdot 4.42^4 + -0.00012 \cdot 29.938^2 + 0.000001 \cdot 6.442^2 \cdot 29.938)$$

$$= -5.136$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot t_s \cdot Q_n = 23.601 \cdot 14.5 \cdot -5.136 = -17.5754 \text{ kgf/mm}$$

From table 4.3.6 Qn:



C1 = -0.438658 C2 = 0.009244 C3 = -0.367518 C4 = -0.001240
 C5 = -0.126278 C6 = 0.002049 C7 = 0.000023 C8 = -0.016622
 C9 = 0.000664 C10 = 0.000018 C11 = 0.000000

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned}
 &= C1+C2\cdot\ln(H)+C3\cdot\ln(B)+C4\cdot\ln(H)^2+C5\cdot\ln(B)^2+C6\cdot\ln(H)\cdot\ln(B)+ \\
 &C7\cdot\ln(H)^3+C8\cdot\ln(B)^3+C9\cdot\ln(H)\cdot\ln(B)^2+C10\cdot\ln(H)^2\cdot\ln(B) \\
 &= -0.43866 + 0.00924\cdot\ln(6.442) + -0.36752\cdot\ln(0.576) + \\
 &-0.00124\cdot\ln(6.442)^2 + -0.12628\cdot(\ln(0.576))^2 + \\
 &0.00205\cdot\ln(6.442)\cdot\ln(0.576) + 0.00002\cdot\ln(6.442)^3 + \\
 &-0.01662\cdot\ln(0.576)^3 + 0.00066\cdot\ln(6.442) \cdot \ln(0.576)^2 + \\
 &0.00002\cdot\ln(6.442)^2\cdot\ln(0.576) \\
 &= -0.2602
 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = -17.575 + -0.0 = -17.5754 \text{ kgf/mm}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = -17.575 + -0.0 = -17.5754 \text{ kgf/mm}$$

Compute angle [β_{cy}]:

$$\begin{aligned}
 &= (3\cdot(1-\nu^2)/(Rs^2\cdot ts^2))^{0.25} = (3\cdot(1-0.3^2)/(601.731^2\cdot 14.5^2))^{0.25} \\
 &= 0.0138 \text{ 1/mm}
 \end{aligned}$$

Compute with Xs+ [Ns+]:

$$= P\cdot Rs/2 + Xs+ = 23.601\cdot 601.731/2 + 0.0+ = 71.0079 \text{ kgf/mm}$$

Compute with Xs- [Ns-]:

$$= P\cdot Rs/2 + Xs- = 23.601\cdot 601.731/2 + 0.0- = 71.0079 \text{ kgf/mm}$$

Compute with Xs+ [NTheta+]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 23.601 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\
 &(-690.491 \cdot 0.014 - 1.627) = 275.7203 \text{ kgf/mm}
 \end{aligned}$$

Compute with Xs- [NTheta-]:

$$\begin{aligned}
 &= P \cdot Rs + 2 \cdot \beta_{cy} \cdot Rs \cdot (-Ms \cdot \beta_{cy} - Q) \\
 &= 23.601 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\
 &(-690.491 \cdot 0.014 - 1.627) = 275.7203 \text{ kgf/mm}
 \end{aligned}$$

Compute Stress with Xs+ [$\sigma_{sm}+$]:

$$= Ns(+)/ts = 71.008/14.5 = 489.7105 \text{ kgf/cm}^2$$

Compute Stress with Xs- [$\sigma_{sm}-$]:

$$= Ns(-)/ts = 71.008/14.5 = 489.7105 \text{ kgf/cm}^2$$

Compute stress with Xs+ [$\sigma_{sb}+$]:

$$= 6\cdot Ms(+)/(ts^2\cdot Kpc) = 6\cdot 690.491/(14.5^2\cdot 1.0) = 1970.4897 \text{ kgf/cm}^2$$

Compute stress with Xs- [$\sigma_{sb}-$]:

$$= 6\cdot Ms(-)/(ts^2\cdot Kpc) = 6\cdot 690.491/(14.5^2\cdot 1.0) = 1970.4897 \text{ kgf/cm}^2$$

Compute stress with Xs+ [$\sigma_{\theta\eta\eta\eta}+$]:

$$= N\theta\eta\eta(+)/ts = 275.72/14.5 = 1901.5222 \text{ kgf/cm}^2$$

Compute stress with Xs- [$\sigma_{\theta\eta\eta\eta}-$]:

$$= N\theta\eta\eta(-)/ts = 275.72/14.5 = 1901.5222 \text{ kgf/cm}^2$$

Compute stress with Xs+ [$\sigma_{\theta\eta\eta\eta\eta}+$]:

$$= 6\cdot v\cdot Ms(+)/(ts^2\cdot Kpc) = 6\cdot 0.3\cdot 690.491/(14.5^2\cdot 1.0) = 591.147 \text{ kgf/cm}^2$$

Compute stress with Xs- [$\sigma_{\theta\eta\eta\eta\eta}-$]:

$$= 6\cdot v\cdot Ms(-)/(ts^2\cdot Kpc) = 6\cdot 0.3\cdot 690.491/(14.5^2\cdot 1.0) = 591.147 \text{ kgf/cm}^2$$



Small End Cone Without a Flare

Compute with Xs+ [Qc+]:

$$= Q + \cos(a) + Ns \sin(a) = -17.6 \cos(29.94) + 71.0 \sin(29.94) \\ = 20.2076 \text{ kgf/mm}$$

Compute with Xs- [Qc-]:

$$= Q - \cos(a) + Ns \sin(a) = -17.6 \cos(29.94) + 71.0 \sin(29.94) \\ = 20.2076 \text{ kgf/mm}$$

Compute effective radius [Rc]:

$$= Rs / \cos(a) = 601.731 / \cos(29.938) = 694.3881 \text{ mm}$$

Compute the angle [βco]:

$$= (3 \cdot (1 - v^2) / (Rc^2 \cdot tc^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (694.388^2 \cdot 19.5^2))^{0.25} \\ = 0.0110 \text{ 1/mm}$$

Compute with Ms+ [Ncs+]:

$$= Ns \cos(a) - Q \sin(a) \\ = 71.008 \cos(29.94) - 17.575 \sin(29.94) = 70.3042 \text{ kgf/mm}$$

Compute with Ms- [Ncs-]:

$$= Ns \cos(a) - Q \sin(a) \\ = 71.008 \cos(29.94) - 17.575 \sin(29.94) = 70.3042 \text{ kgf/mm}$$

Compute with Ms+ [Nctheta+]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc) \\ = 23.601 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot 694.388 \cdot (-690.491 \cdot 0.011 + 20.208) = 356.8757 \text{ kgf/mm}$$

Compute with Ms- [Nctheta-]:

$$= P \cdot Rs / \cos(a) + 2 \cdot \beta_{co} \cdot Rc \cdot (-Mcs \cdot \beta_{co} + Qc) \\ = 23.601 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot (-690.491 \cdot 0.011 + 20.208) = 356.8757 \text{ kgf/mm}$$

Compute the stress with Ms+ [σsm+]:

$$= Ncs / tc = 70.304 / 19.5 = 360.5346 \text{ kgf/cm}^2$$

Compute the stress with Ms- [σsm-]:

$$= Ncs / tc = 70.304 / 19.5 = 360.5346 \text{ kgf/cm}^2$$

Compute the stress with Ms+ [σsb+]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 690.491 / (19.5^2 \cdot 1.0) = 1089.5344 \text{ kgf/cm}^2$$

Compute the stress with Ms- [σsb-]:

$$= 6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 690.491 / (19.5^2 \cdot 1.0) = 1089.5344 \text{ kgf/cm}^2$$

Compute stress with Ms+ [σthetam+]:

$$= Nctheta / tc = 356.876 / 19.5 = 1830.135 \text{ kgf/cm}^2$$

Compute stress with Ms- [σthetam-]:

$$= Nctheta / tc = 356.876 / 19.5 = 1830.135 \text{ kgf/cm}^2$$

Compute stress with Ms+ [σthetab+]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 690.491 / (19.5^2 \cdot 1.0) = 326.8604 \text{ kgf/cm}^2$$

Compute stress with Ms- [σthetab-]:

$$= 6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot 690.491 / (19.5^2 \cdot 1.0) = 326.8604 \text{ kgf/cm}^2$$

Small End under External Pressure:

Results for Cylinder Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-21.882	2676.616	Pass
Sigma sm-	-21.882	2676.616	Pass
Sigma thetam+	-84.968	2676.616	Pass
Sigma thetam-	-84.968	2676.616	Pass
Sigma sm+ + Sigm	-109.932	5353.233	Pass



Sigma sm- + Sigm	-109.932	5353.233	Pass
Sigma sm+ - Sigm	66.168	5353.233	Pass
Sigma sm- - Sigm	66.168	5353.233	Pass
Sigma thetam+ +	-111.383	5353.233	Pass
Sigma thetam- +	-111.383	5353.233	Pass
Sigma thetam+ -	-58.553	5353.233	Pass
Sigma thetam- -	-58.553	5353.233	Pass

Results for the Cone Small End Without a Flare:

Stress Type	Stress	Allowable	Pass/Fail
Sigma sm+	-16.110	2476.295	Pass
Sigma sm-	-16.110	2476.295	Pass
Sigma thetam+	-81.778	2476.295	Pass
Sigma thetam-	-81.778	2476.295	Pass
Sigma sm+ + Sigm	-64.795	4955.649	Pass
Sigma sm- + Sigm	-64.795	4955.649	Pass
Sigma sm+ - Sigm	32.575	4955.649	Pass
Sigma sm- - Sigm	32.575	4955.649	Pass
Sigma thetam+ +	-96.384	4955.649	Pass
Sigma thetam- +	-96.384	4955.649	Pass
Sigma thetam+ -	-67.173	4955.649	Pass
Sigma thetam- -	-67.173	4955.649	Pass

Average Compressive Stresses and comparisons with Fha and Fxa:

Sigma _{smCyl} + + S	-18.996	-1191.225	Pass
Sigma _{smCyl} - + S	-18.996	-1191.225	Pass
Sigma _{thetamCyl} +	-83.373	-1486.398	Pass
Sigma _{thetamCyl} -	-83.373	-1486.398	Pass

Small End Cylinder Without a Flare

Compute the ratio equation 4.3.60 [n]:
 $= tc/ts = 19.5/14.5 = 1.3448$

Compute the ratio equation 4.3.61 [H]:
 $= \sqrt{(Rs/ts)} = \sqrt{(601.731/14.5)} = 6.4419$

Compute the ratio equation 4.3.62 [B]:
 $= \tan(a) = \tan(29.938) = 0.5759$

Compute Ms+ from equation 4.3.59 [Xs+]:
 $= Fs / (2 \cdot \pi \cdot Rs) + Ms / (\pi \cdot Rs^2)$
 $= 0.0 / (2 \cdot 3.142 \cdot 601.731) + 0.0 / (3.141 \cdot 601.731^2)$
 $= 0.0000 \text{ kgf/mm}$

Compute Ms- from equation 4.3.59 [Xs-]:
 $= Fs / (2 \cdot \pi \cdot Rs) - Ms / (\pi \cdot Rs^2) + XS'$
 $= 0.0 / (2 \cdot 3.142 \cdot 601.731) - 0.0 / (3.141 \cdot 601.731^2) + 0.0$
 $= 0.0000 \text{ kgf/mm}$

From Table 4.3.5 MsN:

C1 = -10.652290 C2 = 1.976683 C3 = 4.469143 C4 = -0.063541
 C5 = -1.143284 C6 = -0.067321 C7 = 0.004729 C8 = 0.124486
 C9 = 0.017859 C10 = -0.004846 C11 = 0.000000

Compute from Table 4.3.5 [MsN]:
 $= \exp[C1+C2 \cdot \ln(H^2)+C3 \cdot \ln(a)+C4 \cdot (\ln(H^2))^2+C5 \cdot \ln(a)^2+C6 \cdot \ln(H^2) \cdot \ln(a)+$
 $C7 \cdot \ln(H^2)^3+C8 \cdot \ln(a)^3+C9 \cdot \ln(H^2) \cdot \ln(a)^2+C10 \cdot \ln(H^2)^2 \cdot \ln(a)]$
 $= \exp[-10.652+1.977 \cdot \ln(6.442^2)+4.469 \cdot \ln(29.938)+$
 $-0.064 \cdot (\ln(6.442^2))^2+1.143 \cdot \ln(29.938)^2+$
 $-0.067 \cdot \ln(6.442^2) \cdot \ln(29.938)+ 0.005 \cdot \ln(6.442^2)^3$
 $0.124 \cdot \ln(29.938)^3+ 0.018 \cdot \ln(6.442^2) \cdot \ln(29.938)^2+$
 $-0.005 \cdot \ln(6.442^2)^2 \cdot \ln(29.938)] = 13.9152$

Compute from Table 4.3.2 [Msp]:
 $= p \cdot tL^2 \cdot Msn = -1.055 \cdot 14.5^2 \cdot 13.91515 = -30.8541 \text{ kgf}$



From table 4.3.6 MsN:

$$\begin{aligned} C1 &= 0.000941 & C2 &= -0.000215 & C3 &= -0.001147 & C4 &= 0.129940 \\ C5 &= 0.009760 & C6 &= 0.000003 & C7 &= -0.000003 & C8 &= 0.008361 \\ C9 &= 0.004813 & C10 &= 0.000040 & C11 &= 0.207115 \end{aligned}$$

Compute from Table 4.3.6 [MsN]:

$$\begin{aligned} &= (C1 + C3 \cdot H + C5 \cdot B + C7 \cdot H^2 + C9 \cdot B^2 + C11 \cdot H \cdot B) / \\ &\quad (1 + C2 \cdot H + C4 \cdot B + C6 \cdot H^2 + C8 \cdot B^2 + C10 \cdot H \cdot B) \\ &= (0.00094 + -0.00115 \cdot 6.442 + 0.00976 \cdot 29.938 + \\ &\quad -0.6.442^2 + 0.00481 \cdot 29.938^2 + \\ &\quad 0.20711 \cdot 6.442 \cdot 29.938) / (1 + -0.00022 \cdot 6.442 + \\ &\quad 0.12994 \cdot 29.938 + 0.6.442^2 + 0.00836 \cdot 29.938^2 + \\ &\quad 0.00004 \cdot 6.442 \cdot 29.938) \\ &= 0.7144 \end{aligned}$$

Compute from Table 4.3.2 [Msx+]:

$$= Xs \cdot ts \cdot MsN = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Msx-]:

$$= Xs \cdot ts \cdot MsN = 0.0 \cdot 14.5 \cdot 0.714 = 0.0 \text{ kgf}$$

Compute from Table 4.3.2 [Ms+]:

$$= Ms + Msx+ = -30.854 + 0.0 = -30.8541 \text{ kgf}$$

Compute from Table 4.3.2 [Ms-]:

$$= Ms + Msx- = -30.854 + 0.0 = -30.8541 \text{ kgf}$$

From table 4.3.5 Qn:

$$\begin{aligned} C1 &= 0.271511 & C2 &= 0.000001 & C3 &= 0.004638 & C4 &= 0.002846 \\ C5 &= -0.000906 & C6 &= 0.000000 & C7 &= -0.000002 & C8 &= -0.000119 \\ C9 &= -0.000126 & C10 &= 0.000001 & C11 &= -0.004307 \end{aligned}$$

Compute from Table 4.3.5 [Qn]:

$$\begin{aligned} &= (C1 + C3 \cdot H^2 + C5 \cdot \alpha + C7 \cdot H^4 + C9 \cdot \alpha^2 + C11 \cdot H^2 \cdot \alpha) / \\ &\quad (1 + C2 \cdot H^2 + C4 \cdot \alpha + C6 \cdot H^4 + C8 \cdot \alpha^2 + C10 \cdot H^2 \cdot \alpha) \\ &= (0.27151 + 0.00464 \cdot 6.442^2 + -0.00091 \cdot 29.938 \\ &\quad -0.6.442^4 + -0.00013 \cdot 29.938^2 + -0.00431 \cdot 6.442^2 \\ &\quad \cdot 29.938) / (1 + 0.6.442^2 + 0.00285 \cdot 29.938 + \\ &\quad 0.6.442^4 + -0.00012 \cdot 29.938^2 + 0.000001 \cdot 6.442^2 \cdot 29.938) \\ &= -5.136 \end{aligned}$$

Compute from Table 4.3.2 [Qp]:

$$= P \cdot ts \cdot Qn = -1.055 \cdot 14.5 \cdot -5.136 = 0.7853 \text{ kgf/mm}$$

From table 4.3.6 Qn:

$$\begin{aligned} C1 &= -0.438658 & C2 &= 0.009244 & C3 &= -0.367518 & C4 &= -0.001240 \\ C5 &= -0.126278 & C6 &= 0.002049 & C7 &= 0.000023 & C8 &= -0.016622 \\ C9 &= 0.000664 & C10 &= 0.000018 & C11 &= 0.000000 \end{aligned}$$

Compute from Table 4.3.6 [Qn]:

$$\begin{aligned} &= C1 + C2 \cdot \ln(H) + C3 \cdot \ln(B) + C4 \cdot \ln(H)^2 + C5 \cdot \ln(B)^2 + C6 \cdot \ln(H) \cdot \ln(B) + \\ &\quad C7 \cdot \ln(H)^3 + C8 \cdot \ln(B)^3 + C9 \cdot \ln(H) \cdot \ln(B)^2 + C10 \cdot \ln(H)^2 \cdot \ln(B) \\ &= -0.43866 + 0.00924 \cdot \ln(6.442) + -0.36752 \cdot \ln(0.576) + \\ &\quad -0.00124 \cdot \ln(6.442)^2 + -0.12628 \cdot (\ln(0.576))^2 + \\ &\quad 0.00205 \cdot \ln(6.442) \cdot \ln(0.576) + 0.00002 \cdot \ln(6.442)^3 + \\ &\quad -0.01662 \cdot \ln(0.576)^3 + 0.00066 \cdot \ln(6.442) \cdot \ln(0.576)^2 + \\ &\quad 0.00002 \cdot \ln(6.442)^2 \cdot \ln(0.576) \\ &= -0.2602 \end{aligned}$$

Compute with Xs+ [Qx+]:

$$= Xs+ \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs- [Qx-]:

$$= Xs- \cdot Qn = 0.0 \cdot -0.26 = -0.0 \text{ kgf/mm}$$

Compute with Xs+ [Q+]:

$$= Qp + Qx+ = 0.785 + -0.0 = 0.7853 \text{ kgf/mm}$$

Compute with Xs- [Q-]:

$$= Qp + Qx- = 0.785 + -0.0 = 0.7853 \text{ kgf/mm}$$



Compute angle [β_{cy}]:

$$= (3 \cdot (1 - v^2) / (R_s^2 \cdot t_s^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (601.731^2 \cdot 14.5^2))^{0.25} = 0.0138 \text{ 1/mm}$$

Compute with X_s+ [N_s+]:

$$= P \cdot R_s / 2 + X_s+ = -1.055 \cdot 601.731 / 2 + 0.0+ = -3.1729 \text{ kgf/mm}$$

Compute with X_s- [N_s-]:

$$= P \cdot R_s / 2 + X_s- = -1.055 \cdot 601.731 / 2 + 0.0- = -3.1729 \text{ kgf/mm}$$

Compute with X_s+ [$N_{\theta}+$]:

$$= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q) \\ = -1.055 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\ (-30.854 \cdot 0.014 - 1.627) = -12.3204 \text{ kgf/mm}$$

Compute with X_s- [$N_{\theta}-$]:

$$= P \cdot R_s + 2 \cdot \beta_{cy} \cdot R_s \cdot (-M_s \cdot \beta_{cy} - Q) \\ = -1.055 \cdot 601.731 + 2 \cdot 0.014 \cdot 601.731 \cdot \\ (-30.854 \cdot 0.014 - 1.627) = -12.3204 \text{ kgf/mm}$$

Compute Stress with X_s+ [$\sigma_{sm}+$]:

$$= N_s(+)/t_s = -3.173/14.5 = -21.8824 \text{ kgf/cm}^2$$

Compute Stress with X_s- [$\sigma_{sm}-$]:

$$= N_s(-)/t_s = -3.173/14.5 = -21.8824 \text{ kgf/cm}^2$$

Compute stress with X_s+ [$\sigma_{sb}+$]:

$$= 6 \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot -30.854/(14.5^2 \cdot 1.0) = -88.0499 \text{ kgf/cm}^2$$

Compute stress with X_s- [$\sigma_{sb}-$]:

$$= 6 \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot -30.854/(14.5^2 \cdot 1.0) = -88.0499 \text{ kgf/cm}^2$$

Compute stress with X_s+ [$\sigma_{\theta}+$]:

$$= N_{\theta}(+)/t_s = -12.32/14.5 = -84.9682 \text{ kgf/cm}^2$$

Compute stress with X_s- [$\sigma_{\theta}-$]:

$$= N_{\theta}(-)/t_s = -12.32/14.5 = -84.9682 \text{ kgf/cm}^2$$

Compute stress with X_s+ [$\sigma_{\theta}+$]:

$$= 6 \cdot v \cdot M_s(+)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot -30.854/(14.5^2 \cdot 1.0) = -26.415 \text{ kgf/cm}^2$$

Compute stress with X_s- [$\sigma_{\theta}-$]:

$$= 6 \cdot v \cdot M_s(-)/(t_s^2 \cdot K_{pc}) = 6 \cdot 0.3 \cdot -30.854/(14.5^2 \cdot 1.0) = -26.415 \text{ kgf/cm}^2$$

Small End Cone Without a Flare

Compute with X_s+ [Q_c+]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = 0.8 \cdot \cos(29.94) + -3.2 \cdot \sin(29.94) \\ = -0.9030 \text{ kgf/mm}$$

Compute with X_s- [Q_c-]:

$$= Q \cdot \cos(a) + N_s \cdot \sin(a) = 0.8 \cdot \cos(29.94) + -3.2 \cdot \sin(29.94) \\ = -0.9030 \text{ kgf/mm}$$

Compute effective radius [R_c]:

$$= R_s / \cos(a) = 601.731 / \cos(29.938) = 694.3881 \text{ mm}$$

Compute the angle [β_{co}]:

$$= (3 \cdot (1 - v^2) / (R_c^2 \cdot t_c^2))^{0.25} = (3 \cdot (1 - 0.3^2) / (694.388^2 \cdot 19.5^2))^{0.25} = 0.0110 \text{ 1/mm}$$

Compute with M_s+ [$N_{cs}+$]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = -3.173 \cdot \cos(29.94) - 0.785 \cdot \sin(29.94) = -3.1415 \text{ kgf/mm}$$

Compute with M_s- [$N_{cs}-$]:

$$= N_s \cdot \cos(a) - Q \cdot \sin(a) \\ = -3.173 \cdot \cos(29.94) - 0.785 \cdot \sin(29.94) = -3.1415 \text{ kgf/mm}$$



Compute with Ms+ [Ncttheta+]:
= $P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c (-Mcs \cdot \beta_{co} + Qc)$
= $-1.055 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot 694.388 \cdot$
 $(-30.854 \cdot 0.011 + -0.903) = -15.9467 \text{ kgf/mm}$

Compute with Ms- [Ncttheta-]:
= $P \cdot R_s / \cos(a) + 2 \cdot \beta_{co} \cdot R_c (-Mcs \cdot \beta_{co} + Qc)$
= $-1.055 \cdot 601.731 / \cos(29.94) + 2 \cdot 0.011 \cdot$
 $(-30.854 \cdot 0.011 + -0.903) = -15.9467 \text{ kgf/mm}$

Compute the stress with Ms+ [σ_{sm+}]:
= $Ncs / tc = -3.141 / 19.5 = -16.1102 \text{ kgf/cm}^2$

Compute the stress with Ms- [σ_{sm-}]:
= $Ncs / tc = -3.141 / 19.5 = -16.1102 \text{ kgf/cm}^2$

Compute the stress with Ms+ [σ_{sb+}]:
= $6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -30.854 / (19.5^2 \cdot 1.0) = -48.6851 \text{ kgf/cm}^2$

Compute the stress with Ms- [σ_{sb-}]:
= $6 \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot -30.854 / (19.5^2 \cdot 1.0) = -48.6851 \text{ kgf/cm}^2$

Compute stress with Ms+ [$\sigma_{thetam+}$]:
= $Ncttheta / tc = -15.947 / 19.5 = -81.778 \text{ kgf/cm}^2$

Compute stress with Ms- [$\sigma_{thetam-}$]:
= $Ncttheta / tc = -15.947 / 19.5 = -81.778 \text{ kgf/cm}^2$

Compute stress with Ms+ [$\sigma_{thetab+}$]:
= $6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -30.854 / (19.5^2 \cdot 1.0) = -14.6055 \text{ kgf/cm}^2$

Compute stress with Ms- [$\sigma_{thetab-}$]:
= $6 \cdot v \cdot Mcs / (tc^2 \cdot Kcpc) = 6 \cdot 0.3 \cdot -30.854 / (19.5^2 \cdot 1.0) = -14.6055 \text{ kgf/cm}^2$



Input, Nozzle Desc: NOZZLE N1

From: 40

Pressure for Reinforcement Calculations	P	163.158	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.40	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		132.92	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

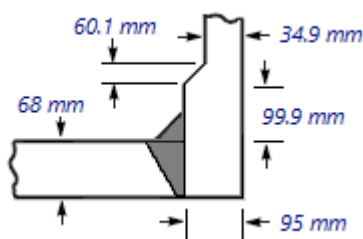
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1568.83	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.44	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	99.9250	mm
Height of Beveled Transition	L'	60.0750	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	582.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NOZZLE N1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.250	in.
Actual Thickness Used in Calculation	1.375	in.



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 390.35 \cdot (\exp(163.158/(1568.835 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 21.3912 + 1.5 + 0.0 = 22.8912 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1568.835 \cdot 1.0 \cdot \ln((2 \cdot 33.425 + 390.35) / 390.35) - 0.0 \\ &= 247.997 - 0.0 = 247.9969 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 64.056 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 30.556 mm, is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 492.9,	Allowable	: 1568.8 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3571.2 kgf/cm ²	Passed
Occasional	: 510.0,	Allowable	: 2086.6 kgf/cm ²	Passed
Shear	: 43.9,	Allowable	: 1098.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1568.8/1650.4, 1] \\ &= 0.951 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 95.0 - 1.5 \\ &= 93.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 34.925 - 1.5 \\ &= 33.425 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1203.0/66.5 \\ &= 18.090 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1200.0/2 + 1.5 \\ &= 601.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{601.5 \cdot 66.5}, 2 \cdot 195.175) \\ &= 199.999 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$



$$= \min[1.5 \cdot 66.5, 0.0] + \sqrt{(195.175 \cdot 93.5)} \\ = 135.088 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1} \\ = 582.0 \\ = 582.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e) \\ = 8(66.5 + 0.0) \\ = 532.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ = \min[135.088, 582.0, 532.0] + 66.5 \\ = 201.588 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot frp) / LR) \\ = 66.5 + ((0.0 \cdot 1.0) / 200.0) \\ = 66.500 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ = \min(12, (2 \cdot 195.18 + 93.5) / (\sqrt{(1203.0 + 66.5) \cdot 66.5})) \\ = 1.665$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ = 66.5 \cdot 200.0 \cdot \max((1.665/5)^{0.85}, 1) \\ = 133.000 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= A2A + A2B \\ = 155.607 + 26.695 \\ = 182.303 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}^2 \\ = 0.5 \cdot 10.0^2 \\ = 0.500 \text{ cm}^2$$

The total area contributed by A1 through A43 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ = 133.0 + 0.951(182.303 + 0.0) + 0.5 + 0.0 + 0.0 \\ = 306.792 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n] \\ = 93.5 / \ln[1 + 93.5/195.175] \\ = 238.883 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ = 66.5 / \ln[1 + 66.5/601.5] \\ = 634.169 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E \\ = 1.5 \cdot 1650.405 \cdot 1.0 \\ = 2475.6 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} \cdot L_H \\ = 163.158 \cdot 238.883 \cdot 201.588 \\ = 78570.5 \text{ kgf}$$



Force acting on the Shell [fS]:

$$\begin{aligned} &= P \cdot Rxs(LR + tn) \\ &= 163.158 \cdot 634.169(200.0 + 93.5) \\ &= 303683.4 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163.158 \cdot 634.169 \cdot 195.175 \\ &= 201947.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (303683.4 + 201947.3 + 78570.5)/163.158 \\ &= 3580.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.607 / (2 \cdot 3580.581/306.792 - 634.169/66.5) \\ &= 179.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.405[66.5/634.169] \\ &= 173.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(179.318, 173.064) \\ &= 173.064 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (78570.516 + 303683.375 + 201947.281)/306.792 \\ &= 1904.227 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163.158 \cdot 634.169/66.5 \\ &= 1555.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1904.227 - 1555.94, 1555.94) \\ &= 2252.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2252.51	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	173.06	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$\begin{aligned} &= (Rnc + tn) / Rnc \\ &= (195.175 + 93.5)/195.175 \\ &= 1.479 \text{ For set-in Nozzles} \end{aligned}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (195.175 + 93.5) \\ &= 453.450 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 93.500, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2)$$



$$= \min(201947 \cdot 1.48, 1.5 \cdot 1568.8(182.303 + 0.0), \pi/4 \cdot 163.2 \cdot 195.18^2 \cdot 1.48^2)$$

$$= 106786.695 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 106786.695 / (453.45(0.49 \cdot 7.071 + 0.6 \cdot 66.0 + 0.49 \cdot 0.0))$$

$$= 546.848 < \text{or} = \text{to } 1650.405 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 74.0088 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 724.0088 mm

Input Echo, WRC297 Item 1, Description: NOZZLE N1 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1200.000	mm	
Shell Thickness	68.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.40	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	387.350	mm	
Nozzle Thickness	95.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	1682.44	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1568.83	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	163.16	kgf/cm ²
Radial Load	P	-6653.97	kgf
Circumferential Shear	Vc	5506.74	kgf
Longitudinal Shear	Vl	5506.74	kgf
Circumferential Moment	Mc	7.84	kgf-m.
Longitudinal Moment	Ml	7.84	kgf-m.
Torsional Moment	Mt	10.32	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

*Warning - The ratio, Dv/Tv (19.647) is < 20
 check the Limitations of WRC bulletin 297.*

*Warning - The ratio, Dn/Tn (6.077) is < 10
 check the Limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$



$$= 577.35 + 2 \cdot 1.65 \cdot \sqrt{634.75(68.0 - 1.5)}$$

$$= 1255.344 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1269.500 mm
 Nozzle Outside Diameter (d) = 577.350 mm
 Vessel Thickness used (T) = 66.500 mm
 Nozzle Thickness used (t) = 93.500 mm
 T / t = 0.711
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 1.987$$

Nr/P = 0.076
 Mr/P = 0.132
 M0/P = 0.040
 N0/P = 0.101
 MrD/Mc = 0.269
 NrDL/Mc = 0.118
 M0d/Mc = 0.088
 N0DL/Mc = 0.063
 MrD/M1 = 0.113
 NrDL/M1 = 0.068
 M0D/M1 = 0.039
 N0DL/M1 = 0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1398	1561	1398	1561
Outplane Stress Summary	1449	1540	1449	1540

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	119	-119	119	-119
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	699	699	699	699
Inplane Stress Summary	829	591	829	591

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	9	9	-9	-9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	1449	1540	1449	1540
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	699	699	699	699

Outplane Stress Summary	750	678	750	678
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	119	-119	119	-119
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1398	1561	1398	1561

Inplane Stress Summary	1528	1454	1529	1453
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-9	-9	9	9
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	1528	1454	1529	1453
------------------------	------	------	------	------

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525

Hoop Stress Summary	344	540	344	540
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164
Axial Stress Summary	212	126	212	126

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	6	6	-6	-6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	345	540	345	540

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525
Hoop Stress Summary	344	540	344	540

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164
Axial Stress Summary	212	126	212	126

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-6	-6	6	6

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	345	540	345	540

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1398	1561	1398	1561	1398	1561	1398	1561
Circ. Pl (SUS)	15	15	15	15	11	11	11	11
Circ. Q (SUS)	36	-36	36	-36	119	-119	119	-119
Long. Pm (SUS)	699	699	699	699	699	699	699	699
Long. Pl (SUS)	11	11	11	11	15	15	15	15
Long. Q (SUS)	119	-119	119	-119	36	-36	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1398	1561	1398	1561	1398	1561	1398	1561
Pm+Pl (SUS)	1413	1576	1413	1576	1409	1572	1409	1572
Pm+Pl+Q (Total)	1449	1540	1449	1540	1528	1453	1528	1453

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1561.00	1650.40	Passed
Pm+Pl (SUS)	1576.09	2475.61	Passed
Pm+Pl+Q (TOTAL)	1540.09	4951.21	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	329	525	329	525	329	525	329	525
Circ. Pl (SUS)	15	15	15	15	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	164	164	164	164	164	164	164	164
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	42	-42	42	-42	42	-42	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	329.0	525.0	329.0	525.0	329.0	525.0	329.0	525.0
Pm+Pl (SUS)	344.2	540.1	344.2	540.1	344.2	540.1	344.2	540.1
Pm+Pl+Q (Total)	344.3	540.1	344.3	540.1	344.3	540.1	344.3	540.1

Nozzle Stress Summation Comparison (kgf/cm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	525.00	1568.83	Passed
Pm+Pl (SUS)	540.10	2353.25	Passed
Pm+Pl+Q (TOTAL)	540.09	4706.50	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: NOZZLE N2

From: 40

Pressure for Reinforcement Calculations	P	163.235	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.40	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	68.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		132.92	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

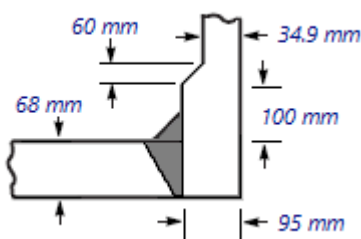
Material		SA-350 LF2	+IT
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	1568.83	kgf/cm ²
Allowable Stress At Ambient	Sna	1682.44	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Hub Height of Integral Nozzle	h	100.0000	mm
Height of Beveled Transition	L'	60.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		95.0000	mm
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	732.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	66.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: NOZZLE N2

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.250	in.
Actual Thickness Used in Calculation	1.375	in.



Nozzle input data check completed without errors.



Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 390.35 \cdot (\exp(163.235/(1568.835 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 21.4018 + 1.5 + 0.0 = 22.9018 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1568.835 \cdot 1.0 \cdot \ln((2 \cdot 33.425 + 390.35) / 390.35) - 0.0 \\ &= 247.997 - 0.0 = 247.9969 \text{ kgf/cm}^2 \end{aligned}$$

This nozzle or its diameter limit intersects weld seam # [2] on this element. The distance between the nozzle and seam is, 220.46 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 488.67 mm

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 9.840 mm
Min. required shell thickness, internal p + c : 64.087 mm

Nozzle neck minimum required thickness : 9.840 mm

Minimum nozzle neck thickness 30.556 mm , is greater than required thickness 9.840 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 493.1,	Allowable	: 1568.8 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3571.0 kgf/cm ²	Passed
Occasional	: 510.2,	Allowable	: 2086.6 kgf/cm ²	Passed
Shear	: 43.9,	Allowable	: 1098.2 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Please check Code paragraph 4.5.17.1 regarding openings in compression. A Part 5 analysis may be required.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1568.8/1650.4, 1] \\ &= 0.951 \end{aligned}$$

Thickness of Nozzle at Shell [tn]:

$$\begin{aligned} &= \text{hub thickness} - \text{corrosion allowance} \\ &= 95.0 - 1.5 \\ &= 93.500 \text{ mm} \end{aligned}$$

Thickness of Nozzle at Top [tn2]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 34.925 - 1.5 \\ &= 33.425 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 1203.0/66.5 \\ &= 18.090 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 1200.0/2 + 1.5 \\ &= 601.500 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{601.5 \cdot 66.5}, 2 \cdot 195.175) \end{aligned}$$



= 199.999 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 66.5, 0.0] + \sqrt{(195.175 \cdot 93.5)} \\ &= 135.088 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 732.0 \\ &= 732.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(66.5 + 0.0) \\ &= 532.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] + t \quad \text{for set-in nozzles} \\ &= \min[135.088, 732.0, 532.0] + 66.5 \\ &= 201.588 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / LR) \\ &= 66.5 + ((0.0 \cdot 1.0) / 200.0) \\ &= 66.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 195.18 + 93.5) / (\sqrt{(1203.0 + 66.5) \cdot 66.5})) \\ &= 1.665 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 66.5 \cdot 200.0 \cdot \max((1.665/5)^{0.85}, 1) \\ &= 133.000 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= A2A + A2B \\ &= 155.678 + 26.644 \\ &= 182.321 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 133.0 + 0.951(182.321 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 306.810 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 93.5 / \ln[1 + 93.5/195.175] \\ &= 238.883 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 66.5 / \ln[1 + 66.5/601.5] \\ &= 634.169 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.405 \cdot 1.0 \\ &= 2475.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn \cdot LH \\ &= 163.235 \cdot 238.883 \cdot 201.588 \\ &= 78607.5 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Force acting on the Shell [fS]:} \\ &= P \cdot Rxs(LR + tn) \\ &= 163.235 \cdot 634.169(200.0 + 93.5) \\ &= 303826.3 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Discontinuity Force from Internal Pressure [fY]:} \\ &= P \cdot Rxs \cdot Rnc \\ &= 163.235 \cdot 634.169 \cdot 195.175 \\ &= 202042.3 \text{ kgf} \end{aligned}$$

$$\begin{aligned} &\text{Area Resisting Internal Pressure [Ap]:} \\ &= (fS + fY + fN)/P \\ &= (303826.3 + 202042.3 + 78607.5)/163.235 \\ &= 3580.6 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\ &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2475.607 / (2 \cdot 3580.582/306.81 - 634.169/66.5) \\ &= 179.3 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\ &= S[t/Rxs] \\ &= 1650.405[66.5/634.169] \\ &= 173.1 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Allowable Working Pressure [Pmax]:} \\ &= \min(Pmax1, Pmax2) \\ &= \min(179.335, 173.064) \\ &= 173.064 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Average Primary Membrane Stress [\sigmaAvg]:} \\ &= (fN + fS + fY) / AT \\ &= (78607.484 + 303826.281 + 202042.312)/306.81 \\ &= 1905.013 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{General Primary Membrane Stress [\sigmaCirc]:} \\ &= P \cdot Rxs / teff \\ &= 163.235 \cdot 634.169/66.5 \\ &= 1556.7 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} &\text{Maximum Local Primary Membrane Stress [PL]:} \\ &= \max(2 \cdot \sigmaAvg - \sigmaCirc, \sigmaCirc) \\ &= \max(2 \cdot 1905.013 - 1556.672, 1556.672) \\ &= 2253.4 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2475.61	kgf/cm ²
Local Primary Membrane Stress	PL	2253.35	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	173.06	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned} &\text{Discontinuity Force Factor [ky]:} \\ &= (Rnc + tn) / Rnc \\ &= (195.175 + 93.5)/195.175 \\ &= 1.479 \text{ For set-in Nozzles} \end{aligned}$$

$$\begin{aligned} &\text{Weld Length of Nozzle to Shell Weld [Lt]:} \\ &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (195.175 + 93.5) \\ &= 453.450 \text{ mm} \end{aligned}$$

$$\begin{aligned} &\text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\ &= 7.071, 0.000, 0.000, \text{ mm} \end{aligned}$$

Required Weld Throat Dimensions :



tc, L41T = 7.071, must be $\geq \min(0.7 * 93.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(202042 \cdot 1.48, 1.5 \cdot 1568.8(182.321 + 0.0), \pi/4 \cdot 163.2 \cdot 195.18^2 \cdot 1.48^2)$$

$$= 106836.938 \text{ kgf}$$

Weld Stress Value [τ]:

$$= \text{fwelds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 106836.938 / (453.45(0.49 \cdot 7.071 + 0.6 \cdot 66.0 + 0.49 \cdot 0.0))$$

$$= 547.105 < \text{or} = \text{to } 1650.405 \text{ Weld Stress Passed}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 74.0088 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 874.0089 mm

Input Echo, WRC297 Item 1, Description: NOZZLE N2 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1200.000	mm	
Shell Thickness	68.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.41	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.40	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	1.5000	mm	
Nozzle Diameter	387.350	mm	
Nozzle Thickness	95.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-350 LF2		
Nozzle Cold S.I. Allowable	SNmc	1682.44	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1568.83	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.*

Design Internal Pressure	Dp	163.24	kgf/cm ²
Radial Load	P	-6653.97	kgf
Circumferential Shear	Vc	5506.74	kgf
Longitudinal Shear	Vl	5506.74	kgf
Circumferential Moment	Mc	7.84	kgf-m.
Longitudinal Moment	Ml	7.84	kgf-m.
Torsional Moment	Mt	10.32	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (19.647) is < 20
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (6.077) is < 10
 check the Limitations of WRC bulletin 297.



Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 577.35 + 2 \cdot 1.65 \cdot \sqrt{634.75(68.0 - 1.5)}$$

$$= 1255.344 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1269.500 mm
Nozzle Outside Diameter	(d) =	577.350 mm
Vessel Thickness used	(T) =	66.500 mm
Nozzle Thickness used	(t) =	93.500 mm
T / t	=	0.711
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.987$$

Nr/P	=	0.076
Mr/P	=	0.132
M0/P	=	0.040
N0/P	=	0.101
MrD/Mc	=	0.269
NrDL/Mc	=	0.118
M0d/Mc	=	0.088
N0DL/Mc	=	0.063
MrD/M1	=	0.113
NrDL/M1	=	0.068
M0D/M1	=	0.039
N0DL/M1	=	0.126

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1399	1562	1399	1562
Outplane Stress Summary	1450	1541	1450	1541

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	119	-119	119	-119
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	699	699	699	699
Inplane Stress Summary	830	591	830	591

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0



Shear Stress Summary 9 9 -9 -9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1450	1541	1450	1541

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	15	15	15	15
Outplane Bending (P)	36	-36	36	-36
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	699	699	699	699

Outplane Stress Summary 750 678 750 678

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	11	11	11	11
Inplane Bending (P)	119	-119	119	-119
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1399	1562	1399	1562

Inplane Stress Summary 1529 1455 1530 1454

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary -9 -9 9 9

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	1529	1455	1530	1454

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525



Hoop Stress Summary 344 540 344 540

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164

Axial Stress Summary 212 126 212 126

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	6	6	-6	-6
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary 6 6 -6 -6

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress 345 540 345 540

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	15	15	15	15
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	329	525	329	525

Hoop Stress Summary 344 540 344 540

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	4	4	4	4
Axial Bending (P)	42	-42	42	-42
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	164	164	164	164

Axial Stress Summary 212 126 212 126

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-6	-6	6	6
Shear due to Torsion	0	0	0	0

Shear Stress Summary -6 -6 6 6

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	345	540	345	540

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1399	1562	1399	1562	1399	1562	1399	1562
Circ. Pl (SUS)	15	15	15	15	11	11	11	11
Circ. Q (SUS)	36	-36	36	-36	119	-119	119	-119
Long. Pm (SUS)	699	699	699	699	699	699	699	699
Long. Pl (SUS)	11	11	11	11	15	15	15	15
Long. Q (SUS)	119	-119	119	-119	36	-36	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1399	1562	1399	1562	1399	1562	1399	1562
Pm+Pl (SUS)	1414	1577	1414	1577	1410	1573	1410	1573
Pm+Pl+Q (Total)	1450	1541	1450	1541	1529	1454	1529	1454

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1562.00	1650.40	Passed
Pm+Pl (SUS)	1577.09	2475.61	Passed
Pm+Pl+Q (TOTAL)	1541.09	4951.21	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	329	525	329	525	329	525	329	525
Circ. Pl (SUS)	15	15	15	15	15	15	15	15
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	164	164	164	164	164	164	164	164
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	42	-42	42	-42	42	-42	42	-42
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	6	6	-6	-6	-6	-6	6	6
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	329.0	525.0	329.0	525.0	329.0	525.0	329.0	525.0
Pm+Pl (SUS)	344.2	540.1	344.2	540.1	344.2	540.1	344.2	540.1



Pm+Pl+Q (Total)	344.3	540.1	344.3	540.1	344.3	540.1	344.3	540.1
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Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	525.00	1568.83	Passed
Pm+Pl (SUS)	540.10	2353.25	Passed
Pm+Pl+Q (TOTAL)	540.09	4706.50	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: Noz N9

Parent: NOZZLE N2

Pressure for Reinforcement Calculations	P	163.258	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-350 LF2	+N+IT
Parent Allowable Stress at Temperature	Sv	1568.83	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1682.44	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	387.35	mm
Shell Finished (Minimum) Thickness	t	34.9250	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

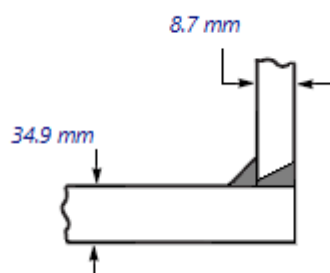
Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N9

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1



Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 45.85 \cdot (\exp(163.258/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 2.5954 + 1.5 + 0.0 = 4.0954 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 7.238 + 45.85) / 45.85) - 0.0 \\ &= 417.66 - 0.0 = 417.6602 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm

Min. required shell thickness, internal p + c : 22.925 mm

Nozzle neck minimum required thickness : 4.920 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 4.920 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1522.2/1568.8, 1] \\ &= 0.970 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 8.738 - 1.5 \\ &= 7.238 \text{ mm} \end{aligned}$$

Shell Diameter to Thickness ratio [D/t]:

$$\begin{aligned} &= D_i/t \\ &= 390.35/33.425 \\ &= 11.678 \text{ must be less than 400.} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= D_i/2 + \text{corrosion allowance} \\ &= 387.35/2 + 1.5 \\ &= 195.175 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{195.175 \cdot 33.425}, 2 \cdot 22.925) \\ &= 45.850 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 53.087 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n} \\ &= \min[1.5 \cdot 33.425, 0.0] + \sqrt{22.925 \cdot 7.238} \\ &= 12.881 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(33.425 + 0.0) \\ &= 267.400 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[12.881, 200.0, 267.4] \\ &= 12.881 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:



$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - tn)) \\ &= 33.425 + ((0.0 \cdot 1.0) / (53.087 - 7.238)) \\ &= 33.425 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot Rn + tn) / (\sqrt{(Di + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 22.92 + 7.238) / (\sqrt{(390.35 + 33.425) \cdot 33.425})) \\ &= 0.446 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 33.425 \cdot 53.087 \cdot \max((0.446/5)^{0.85}, 1) \\ &= 17.744 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= tn \cdot LH \\ &= 7.238 \cdot 12.881 \\ &= 0.932 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 17.744 + 0.97(0.932 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 19.149 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= tn / \ln[1 + tn/Rn] \\ &= 7.238 / \ln[1 + 7.238/22.925] \\ &= 26.378 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 33.425 / \ln[1 + 33.425/195.175] \\ &= 211.447 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1568.835 \cdot 1.0 \\ &= 2353.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 163.258 \cdot 26.378(12.881 + 33.425) \\ &= 1994.2 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 163.258 \cdot 211.447 \cdot 53.087 \\ &= 18326.0 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 163.258 \cdot 211.447 \cdot 22.925 \\ &= 7913.8 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (18326.0 + 7913.8 + 1994.2) / 163.258 \\ &= 172.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2353.252 / (2 \cdot 172.941/19.149 - 211.447/33.425) \\ &= 200.5 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1568.835[33.425/211.447] \\ &= 248.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(200.506, 247.997) \\ &= 200.506 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (1994.157 + 18325.961 + 7913.758) / 19.149 \\ &= 1474.430 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 163.258 \cdot 211.447 / 33.425 \\ &= 1032.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1474.43 - 1032.772, 1032.772) \\ &= 1916.1 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2353.25	kgf/cm ²
Local Primary Membrane Stress	PL	1916.09	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	200.51	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (22.925 + 7.238) \\ &= 47.379 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 7.238, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(7914 \cdot 1.0, 1.5 \cdot 1522.2(0.932 + 0.0), \pi/4 \cdot 163.3 \cdot 22.92^2 \cdot 1.0^2) \\ &= 673.872 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (Lt(0.49 \cdot L41T + 0.6 \cdot tw1 + 0.49 \cdot L43T)) \\ &= 673.872 / (47.379(0.49 \cdot 7.071 + 0.6 \cdot 8.738 + 0.49 \cdot 0.0)) \\ &= 163.345 < \text{ or } = \text{ to } 1568.835 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

This element is Post Weld Heat Treated PWHT

Computed Minimum Design Metal Temperature	:	-52 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	A
From Figure 3.8M- with PWHT, the MDMT is	:	1 °C
Temperature reduction per Graph 3.13	:	53 °C
The computed stress ratio is	:	0.390

The computed thickness ratio [sr]:

$$= (tr \cdot E*) / (tnom - ci - cext)$$



$$= (2.397 \cdot 1.0) / (7.645 - 1.5 - 0.0)$$
$$= 0.390$$

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -52 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -38 °C

The Drop for this Nozzle is : 2.3631 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 237.2881 mm

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Input, Nozzle Desc: Noz N8

From: 50

Pressure for Reinforcement Calculations	P	23.531	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1784.41	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.41	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	1200.00	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		241.42	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

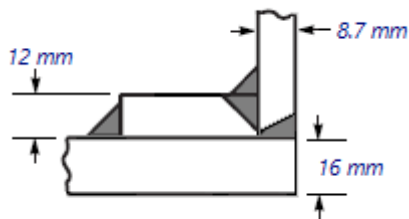
Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.5000	mm
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	8.7376	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	160.3250	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N8

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	1.687	in.
Actual Thickness Used in Calculation	0.344	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 45.85 \cdot (\exp(23.531/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.3571 + 1.5 + 0.0 = 1.8571 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 7.238 + 45.85) / 45.85) - 0.0$$

$$= 417.66 - 0.0 = 417.6602 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 4.920 mm
 Min. required shell thickness, internal p + c : 9.485 mm

Nozzle neck minimum required thickness : 4.920 mm

Minimum nozzle neck thickness 7.645 mm , is greater than required thickness 4.920 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1784.4, 1]$$

$$= 0.853$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1784.4, 1]$$

$$= 0.925$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 8.738 - 1.5$$

$$= 7.238 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 1203.0/14.5$$

$$= 82.966 \text{ must be less than } 400.$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 1200.0/2 + 1.5$$

$$= 601.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, L_r is $\min[LR1, LR2, LR3]$:

Limit [LR1]:

$$= \sqrt{(Reff \cdot t) + W}$$

$$= \sqrt{(601.5 \cdot 14.5) + 50.0}$$

$$= 143.390 \text{ mm}$$



Limit [LR2]:

$$\begin{aligned} &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\ &= \sqrt{((601.5 + 14.5) \cdot (14.5 + 12.0))} \\ &= 127.765 \text{ mm} \end{aligned}$$

Limit [LR3]:

$$\begin{aligned} &= 2 \cdot R_n \\ &= 2 \cdot 22.925 \\ &= 45.850 \text{ mm} \end{aligned}$$

Final value of [LR]:

$$\begin{aligned} &= \min(LR1, LR2, LR3) \\ &= \min(143.39, 127.765, 45.85) \\ &= 45.850 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 53.087 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 14.5, 12.0] + \sqrt{(22.925 \cdot 7.238)} \\ &= 24.881 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 150.0 \\ &= 150.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(14.5 + 12.0) \\ &= 212.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[24.881, 150.0, 212.0] \\ &= 24.881 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 14.5 + ((5.502 \cdot 0.92) / (53.087 - 7.238)) \\ &= 25.599 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, ((2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}}))) \\ &= \min(12, (2 \cdot 22.92 + 7.238) / (\sqrt{((1203.0 + 25.599) \cdot 25.599)})) \\ &= 0.299 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 14.5 \cdot 53.087 \cdot \max((0.299/5)^{0.85}, 1) \\ &= 7.698 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 7.238 \cdot 24.881 \\ &= 1.801 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}_{42}^2 - \text{Area Cut by Reinforcement Limit} \\ &= 0.5 \cdot 12.0^2 - 0.397 \\ &= 0.323 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}_{41}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$



Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e , (LR - t_n) \cdot t_e) \\ &= \min(50.0 \cdot 12.0, (53.087 - 7.238) \cdot 12.0) \\ &= 5.502 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 7.698 + 0.853(1.801 + 0.0) + 0.5 + 0.323 + 0.0 + 0.925(5.502) \\ &= 15.145 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 7.238 / \ln[1 + 7.238/22.925] \\ &= 26.378 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 25.599 / \ln[1 + 25.599/601.5] \\ &= 614.210 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1784.411 \cdot 1.0 \\ &= 2676.6 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.531 \cdot 26.378(24.881 + 14.5) \\ &= 244.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.531 \cdot 614.21 \cdot 53.087 \\ &= 7672.6 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.531 \cdot 614.21 \cdot 22.925 \\ &= 3313.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (7672.6 + 3313.3 + 244.4) / 23.531 \\ &= 477.3 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2676.616 / (2 \cdot 477.264 / 15.145 - 614.21 / 25.599) \\ &= 68.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1784.411 [14.5 / 614.21] \\ &= 42.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(68.578, 42.126) \\ &= 42.126 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (244.44 + 7672.633 + 3313.297) / 15.145 \\ &= 741.501 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 23.531 \cdot 614.21 / 25.599 \\ &= 564.6 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 741.501 - 564.591, 564.591)$$

$$= 918.4 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2676.62	kgf/cm ²
Local Primary Membrane Stress	PL	918.41	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	42.13	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$= \pi/2 \cdot (R_n + t_n)$$

$$= \pi/2 \cdot (22.925 + 7.238)$$

$$= 47.379 \text{ mm}$$

Weld Length of Pad to Shell Weld [LTp]:

$$= \pi/2 \cdot (R_n + t_n + W)$$

$$= \pi/2 \cdot (22.925 + 7.238 + 50.0)$$

$$= 125.919 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$t_c, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 7.238, 6.000) \text{ Passed}$$

$$t_f1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 14.500) \text{ Passed}$$

Weld Load Value [fwelds]:

$$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$$

$$= \min(3313 \cdot 1.0, 1.5 \cdot 1522.2(1.801 + 0.0), \pi/4 \cdot 23.5 \cdot 22.92^2 \cdot 1.0^2)$$

$$= 97.127 \text{ kgf}$$

Discontinuity Force [fws]:

$$= \text{fwelds} \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$$

$$= 97.1 \cdot 14.5 \cdot 1784 / (14.5 \cdot 1784 + 12.0 \cdot 1650)$$

$$= 55.016 \text{ kgf}$$

Discontinuity Force [fwp]:

$$= \text{fwelds} \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$$

$$= 97.1 \cdot 12.0 \cdot 1650 / (14.5 \cdot 1784 + 12.0 \cdot 1650)$$

$$= 42.111 \text{ kgf}$$

Shear Stress [τ1]:

$$= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T}))$$

$$= 55.016 / (47.379 \cdot (0.6 \cdot 8.738 + 0.49 \cdot 0.0))$$

$$= 22.149 \text{ kgf/cm}^2$$

Shear Stress [τ2]:

$$= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T}))$$

$$= 42.111 / (47.379 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$$

$$= 8.334 \text{ kgf/cm}^2$$

Shear Stress [τ3]:

$$= fwp / (L_{Tp} \cdot (0.49 \cdot L_{42T}))$$

$$= 42.111 / (125.919 \cdot (0.49 \cdot 8.485))$$

$$= 8.044 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$

$$= \max(22.149, 8.334, 8.044)$$

$$= 22.1 \text{ must be less than or equal to } 1784.4 \text{ kgf/cm}^2$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :



Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 0.7586 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 166.7586 mm

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Input, Nozzle Desc: Noz N4

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C
Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²
Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		507.42	cm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

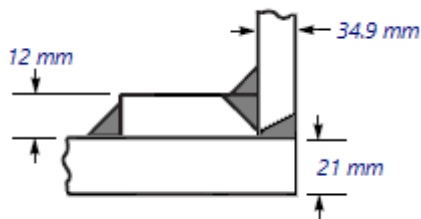
Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		120	
Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.2748	mm
Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation	15.250	in.
Actual Thickness Used in Calculation	1.375	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 393.35 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 3.0538 + 3.0 + 0.0 = 6.0538 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 31.925 + 393.35) / 393.35) - 0.0$$

$$= 228.974 - 0.0 = 228.9739 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.340 mm
Min. required shell thickness, internal p + c	17.977 mm
Min. required shell thickness, external p + c	14.028 mm
Nozzle neck minimum required thickness	11.340 mm

Minimum nozzle neck thickness 30.556 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 74.8,	Allowable	: 1522.2 kgf/cm ²	Passed
Expansion	: 0.0,	Allowable	: 3880.0 kgf/cm ²	Passed
Occasional	: 78.2,	Allowable	: 2024.6 kgf/cm ²	Passed
Shear	: 15.3,	Allowable	: 1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.9, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 34.925 - 3.0$$

$$= 31.925 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$



= 2303.0/19.5
 = 118.103 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300.0/2 + 1.5
 = 1151.500 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(1151.5 \cdot 19.5) + 100.0}$$

$$= 249.847 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))}$$

$$= 192.059 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 196.675$$

$$= 393.350 \text{ mm}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(249.847, 192.059, 393.35)$$

$$= 192.059 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 223.984 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(196.675 \cdot 31.925)}$$

$$= 91.239 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 270.0$$

$$= 270.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(19.5 + 12.0)$$

$$= 252.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[\text{LH1}, \text{LH2}, \text{LH3}]$$

$$= \min[91.239, 270.0, 252.0]$$

$$= 91.239 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot \text{frp}) / (LR - t_n))$$

$$= 19.5 + ((12.0 \cdot 1.) / (223.984 - 31.925))$$

$$= 25.746 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((Di + \text{teff}) \cdot \text{teff})}))$$

$$= \min(12, (2 \cdot 196.68 + 31.925) / (\sqrt{((2303.0 + 25.746) \cdot 25.746)}))$$

$$= 1.737$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 19.5 \cdot 223.984 \cdot \max((1.737/5)^{0.85}, 1)$$



$$= 43.677 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 31.925 \cdot 91.239 \\ &= 29.128 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}42^2 \\ &= 0.5 \cdot 12.0^2 \\ &= 0.720 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, (L_R - t_n) \cdot t_e) \\ &= \min(100.0 \cdot 12.0, (223.984 - 31.925) \cdot 12.0) \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 43.677 + 0.922(29.128 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (12.0) \\ &= 83.752 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 31.925 / \ln[1 + 31.925/196.675] \\ &= 212.238 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 25.746 / \ln[1 + 25.746/1151.5] \\ &= 1164.324 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 23.454 \cdot 212.238(91.239 + 19.5) \\ &= 5512.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.454 \cdot 1164.324 \cdot 223.984 \\ &= 61165.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.454 \cdot 1164.324 \cdot 196.675 \\ &= 53708.1 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (61165.5 + 53708.1 + 5512.4)/23.454 \\ &= 5132.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff}) \\ &= 2476.295 / (2 \cdot 5132.857/83.752 - 1164.324/25.746) \\ &= 32.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1650.863[19.5/1164.324] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:
= min(Pmax1, Pmax2)
= min(32.014, 27.649)
= 27.649 kgf/cm²

Average Primary Membrane Stress [σ Avg]:
= (fN + fS + fY) / AT
= (5512.394 + 61165.508 + 53708.066)/83.752
= 1437.416 kgf/cm²

General Primary Membrane Stress [σ Circ]:
= P • Rxs / teff
= 23.454 • 1164.324/25.746
= 1060.7 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ Avg - σ Circ, σ Circ)
= max(2 • 1437.416 - 1060.658, 1060.658)
= 1814.2 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1814.17	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.65	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld [LT]:
= $\pi/2 \cdot (Rn + tn)$
= $\pi/2 \cdot (196.675 + 31.925)$
= 359.084 mm

Weld Length of Pad to Shell Weld [LTp]:
= $\pi/2 \cdot (Rn + tn + W)$
= $\pi/2 \cdot (196.675 + 31.925 + 100.0)$
= 516.164 mm

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:
= 7.071, 8.485, 0.000, mm

Required Weld Throat Dimensions :
tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 31.925, 6.000)$ Passed
tf1, L42T = 8.485, must be $\geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500)$ Passed

Weld Load Value [fwelds]:
= min(fy • ky, 1.5 • Sn(A2 + A3), $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
= min(53708•1.0, 1.5•1522.2(29.128+0.0), $\pi/4 \cdot 23.5 \cdot 196.68^2 \cdot 1.0^2$)
= 7125.329 kgf

Discontinuity Force [fws]:
= fwelds • t • S / (t • S + te • Sp)
= 7125.3•19.5•1651/(19.5•1651+12.0•1650)
= 4411.384 kgf

Discontinuity Force [fwp]:
= fwelds • te • Sp / (t • S + te • Sp)
= 7125.3•12.0•1650/(19.5•1651+12.0•1650)
= 2713.944 kgf

Shear Stress [τ 1]:
= fws / (LT • (0.6 • tw1 + 0.49 • L43T))
= 4411.384 / (359.084 • (0.6 • 14.275 + 0.49 • 0.0))
= 143.436 kgf/cm²

Shear Stress [τ 2]:
= fwp / (LT • (0.6 • tw2 + 0.49 • L41T))



$$= 2713.944 / (359.084 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$$
$$= 70.868 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (LTP \cdot (0.49 \cdot L42T))$$
$$= 2713.944 / (516.164 \cdot (0.49 \cdot 8.485))$$
$$= 126.461 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$
$$= \max(143.436, 70.868, 126.461)$$
$$= 143.4 \text{ must be less than or equal to } 1650.9 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 31.925 / \ln[1 + 31.925/196.675]$$
$$= 212.238 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 25.746 / \ln[1 + 25.746/1151.5]$$
$$= 1164.324 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.863$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 212.238 (91.239 + 19.5)$$
$$= 247.9 \text{ kgf}$$

Force acting on the Shell [f_S]:

$$= 1.055 \cdot 1164.324 \cdot 223.984$$
$$= 2750.3 \text{ kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 1.055 \cdot 1164.324 \cdot 196.675$$
$$= 2415.0 \text{ kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (2750.3 + 2415.0 + 247.9) / 1.055$$
$$= 5132.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 2476.295 / (2 \cdot 5132.857 / 83.752 - 1164.324 / 25.746)$$
$$= 32.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$
$$= 1650.863 [19.5 / 1164.324]$$
$$= 27.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(32.014, 27.649)$$
$$= 27.649 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (247.864 + 2750.294 + 2414.972) / 83.752$$
$$= 64.633 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:



$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.055 \cdot 1164.324 / 25.746 \\ &= 47.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 64.633 - 47.692, 47.692) \\ &= 81.6 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	81.57	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.65	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 22.9499 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 313.9498 mm

Input Echo, WRC297 Item 1, Description: Noz N4 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.44 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.86 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	387.350	mm



Nozzle Thickness		34.9250	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-1274.71	kgf
Circumferential Shear	Vc	1835.58	kgf
Longitudinal Shear	Vl	1835.58	kgf
Circumferential Moment	Mc	2.48	kgf-m.
Longitudinal Moment	Ml	3.30	kgf-m.
Torsional Moment	Mt	4.13	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$
 $= 955.067 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	31.925	mm
T / t	=	0.987	
d / t	=	14.321	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.686$$

Nr/P	=	0.082
Mr/P	=	0.134
M0/P	=	0.046
N0/P	=	0.112
MrD/Mc	=	0.259
NrDL/Mc	=	0.125
M0d/Mc	=	0.094
N0DL/Mc	=	0.070
MrD/Ml	=	0.135
NrDL/Ml	=	0.076
M0D/Ml	=	0.048
N0DL/Ml	=	0.151

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	845	869	845	869
Outplane Stress Summary	895	848	895	848

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	103	-103	103	-103
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	422	422	422	422
Inplane Stress Summary	536	330	537	329

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	895	848	895	848

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	422	422	422	422
Outplane Stress Summary	472	402	472	401

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	103	-103	103	-103
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	845	869	845	869
Inplane Stress Summary	958	777	960	775

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	958	777	960	775

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188
Hoop Stress Summary	174	202	174	202

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80
Axial Stress Summary	152	13	153	13

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	177	203	177	203

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188

Hoop Stress Summary	174	202	174	202
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80

Axial Stress Summary	152	13	153	12
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-7	-7	7	7
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-7	-7	7	7
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	177	203	177	203
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$
 $= 953.786 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2322.500 mm
Nozzle Outside Diameter (d)	=	657.200 mm
Vessel Thickness used (T)	=	19.500 mm
Nozzle Thickness used (t)	=	131.925 mm
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.088$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.060
Mr/P	=	0.104
M0/P	=	0.031
N0/P	=	0.051
MrD/Mc	=	0.266
NrDL/Mc	=	0.101
M0d/Mc	=	0.080
N0DL/Mc	=	0.074
MrD/M1	=	0.079
NrDL/M1	=	0.072
M0D/M1	=	0.023



NØDL/M1 = 0.062

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	61	-61	61	-61
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1373	1396	1373	1396
Outplane Stress Summary	1451	1352	1452	1351

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	209	-209	209	-209
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	686	686	686	686
Inplane Stress Summary	915	497	916	496

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	9	9	-9	-9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1451	1352	1452	1351

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	61	-61	61	-61
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	686	686	686	686
Outplane Stress Summary	764	642	765	641

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	209	-209	209	-209
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1373	1396	1373	1396

Inplane Stress Summary	1601	1208	1604	1205
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-9	-9	9	9
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Two * Max Shear Stress	1601	1208	1604	1205
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	845	869	845	869	845	869	845	869
Circ. Pl (SUS)	14	14	14	14	10	10	10	10
Circ. Q (SUS)	35	-35	35	-35	103	-103	103	-103
Long. Pm (SUS)	422	422	422	422	422	422	422	422
Long. Pl (SUS)	10	10	10	10	14	14	14	14
Long. Q (SUS)	103	-103	103	-103	35	-35	35	-35
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	845.0	869.0	845.0	869.0	845.0	869.0	845.0	869.0
Pm+Pl (SUS)	859.1	883.1	859.1	883.1	855.2	879.1	855.2	879.1
Pm+Pl+Q (Total)	894.2	848.1	894.2	848.1	958.1	776.2	958.1	776.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	869.00	1650.86	Passed
Pm+Pl (SUS)	883.14	2476.29	Passed
Pm+Pl+Q (TOTAL)	958.13	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Circ. Pl (SUS)	16	16	16	16	19	19	19	19
Circ. Q (SUS)	61	-61	61	-61	208	-208	210	-210
Long. Pm (SUS)	686	686	686	686	686	686	686	686
Long. Pl (SUS)	19	19	19	19	16	16	16	16
Long. Q (SUS)	209	-209	209	-209	61	-61	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Pm+Pl (SUS)	1389	1412	1389	1412	1392	1415	1392	1415
Pm+Pl+Q (Total)	1450	1351	1450	1351	1600	1207	1602	1205

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1396.00	1650.86	Passed
Pm+Pl (SUS)	1415.11	2476.29	Passed
Pm+Pl+Q (TOTAL)	1602.10	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	160	188	160	188	160	188	160	188
Circ. Pl (SUS)	14	14	14	14	14	14	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	80	80	80	80	80	80	80	80
Long. Pl (SUS)	2	2	2	2	2	2	2	2
Long. Q (SUS)	69	-69	69	-69	69	-69	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	160.0	188.0	160.0	188.0	160.0	188.0	160.0	188.0
Pm+Pl (SUS)	174.5	202.4	174.5	202.4	174.5	202.4	174.5	202.4
Pm+Pl+Q (Total)	176.0	202.3	176.0	202.3	176.0	202.3	176.0	202.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	188.00	1522.22	Passed
Pm+Pl (SUS)	202.41	2283.33	Passed
Pm+Pl+Q (TOTAL)	202.26	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: Noz N5

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		767.42	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		18.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

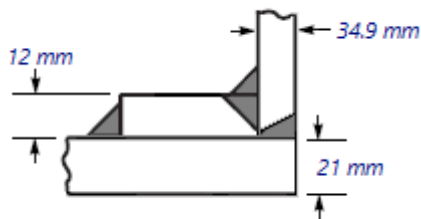
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		15.250	in.
Actual Thickness Used in Calculation		1.375	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 393.35 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 3.0538 + 3.0 + 0.0 = 6.0538 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 31.925 + 393.35) / 393.35) - 0.0$$

$$= 228.974 - 0.0 = 228.9739 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	11.340 mm
Min. required shell thickness, internal p + c	17.977 mm
Min. required shell thickness, external p + c	14.028 mm
Nozzle neck minimum required thickness	11.340 mm

Minimum nozzle neck thickness 30.556 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 74.9	Allowable	: 1522.2 kgf/cm ²	Passed
Expansion	: 0.0	Allowable	: 3879.9 kgf/cm ²	Passed
Occasional	: 78.2	Allowable	: 2024.6 kgf/cm ²	Passed
Shear	: 15.3	Allowable	: 1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.9, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 34.925 - 3.0$$

$$= 31.925 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$



= 2303.0/19.5
 = 118.103 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300.0/2 + 1.5
 = 1151.500 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(1151.5 \cdot 19.5) + 100.0}$$

$$= 249.847 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))}$$

$$= 192.059 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 196.675$$

$$= 393.350 \text{ mm}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(249.847, 192.059, 393.35)$$

$$= 192.059 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 223.984 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(196.675 \cdot 31.925)}$$

$$= 91.239 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 270.0$$

$$= 270.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(19.5 + 12.0)$$

$$= 252.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[\text{LH1}, \text{LH2}, \text{LH3}]$$

$$= \min[91.239, 270.0, 252.0]$$

$$= 91.239 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot \text{frp}) / (LR - t_n))$$

$$= 19.5 + ((12.0 \cdot 1.) / (223.984 - 31.925))$$

$$= 25.746 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((Di + \text{teff}) \cdot \text{teff})}))$$

$$= \min(12, (2 \cdot 196.68 + 31.925) / (\sqrt{((2303.0 + 25.746) \cdot 25.746)}))$$

$$= 1.737$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 19.5 \cdot 223.984 \cdot \max((1.737/5)^{0.85}, 1)$$



$$= 43.677 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 31.925 \cdot 91.239 \\ &= 29.128 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}42^2 \\ &= 0.5 \cdot 12.0^2 \\ &= 0.720 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}41^2 \\ &= 0.5 \cdot 12.0^2 \\ &= 0.720 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e , (L_R - t_n) \cdot t_e) \\ &= \min(100.0 \cdot 12.0, (223.984 - 31.925) \cdot 12.0) \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 43.677 + 0.922(29.128 + 0.0) + 0.72 + 0.72 + 0.0 + 1 \cdot (12.0) \\ &= 83.972 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 31.925 / \ln[1 + 31.925/196.675] \\ &= 212.238 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 25.746 / \ln[1 + 25.746/1151.5] \\ &= 1164.324 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 23.454 \cdot 212.238(91.239 + 19.5) \\ &= 5512.4 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.454 \cdot 1164.324 \cdot 223.984 \\ &= 61165.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.454 \cdot 1164.324 \cdot 196.675 \\ &= 53708.1 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (61165.5 + 53708.1 + 5512.4)/23.454 \\ &= 5132.9 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff}) \\ &= 2476.295 / (2 \cdot 5132.857/83.972 - 1164.324/25.746) \\ &= 32.1 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1650.863[19.5/1164.324] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure [Pmax]:
= min(Pmax1, Pmax2)
= min(32.148, 27.649)
= 27.649 kgf/cm²

Average Primary Membrane Stress [σ Avg]:
= (fN + fS + fY) / AT
= (5512.394 + 61165.508 + 53708.066) / 83.972
= 1433.650 kgf/cm²

General Primary Membrane Stress [σ Circ]:
= P • Rxs / teff
= 23.454 • 1164.324 / 25.746
= 1060.7 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ Avg - σ Circ, σ Circ)
= max(2 • 1433.65 - 1060.658, 1060.658)
= 1806.6 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1806.64	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.65	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:
= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld [LT]:
= $\pi/2 \cdot (Rn + tn)$
= $\pi/2 \cdot (196.675 + 31.925)$
= 359.084 mm

Weld Length of Pad to Shell Weld [LTP]:
= $\pi/2 \cdot (Rn + tn + W)$
= $\pi/2 \cdot (196.675 + 31.925 + 100.0)$
= 516.164 mm

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:
= 8.485, 8.485, 0.000, mm

Required Weld Throat Dimensions :
tc, L41T = 8.485, must be $\geq \min(0.7 \cdot 31.925, 6.000)$ Passed
tf1, L42T = 8.485, must be $\geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500)$ Passed

Weld Load Value [fwelds]:
= min(fy • ky, $1.5 \cdot Sn(A2 + A3)$, $\pi/4 \cdot P \cdot Rn^2 \cdot ky^2$)
= min(53708 • 1.0, $1.5 \cdot 1522.2(29.128 + 0.0)$, $\pi/4 \cdot 23.5 \cdot 196.68^2 \cdot 1.0^2$)
= 7125.329 kgf

Discontinuity Force [fws]:
= fwelds • t • S / (t • S + te • Sp)
= 7125.3 • 19.5 • 1651 / (19.5 • 1651 + 12.0 • 1650)
= 4411.384 kgf

Discontinuity Force [fwp]:
= fwelds • te • Sp / (t • S + te • Sp)
= 7125.3 • 12.0 • 1650 / (19.5 • 1651 + 12.0 • 1650)
= 2713.944 kgf

Shear Stress [τ 1]:
= fws / (LT • (0.6 • tw1 + 0.49 • L43T))
= 4411.384 / (359.084 • (0.6 • 12.0 + 0.49 • 0.0))
= 170.627 kgf/cm²

Shear Stress [τ 2]:
= fwp / (LT • (0.6 • tw2 + 0.49 • L41T))



$$= 2713.944 / (359.084 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 8.485))$$
$$= 66.545 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (LTP \cdot (0.49 \cdot L42T))$$
$$= 2713.944 / (516.164 \cdot (0.49 \cdot 8.485))$$
$$= 126.461 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$
$$= \max(170.627, 66.545, 126.461)$$
$$= 170.6 \text{ must be less than or equal to } 1650.9 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 31.925 / \ln[1 + 31.925/196.675]$$
$$= 212.238 \text{ mm}$$

Shell Radius for Force Calculation [R_{xs}]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 25.746 / \ln[1 + 25.746/1151.5]$$
$$= 1164.324 \text{ mm}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.863$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [f_N]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 212.238 (91.239 + 19.5)$$
$$= 247.9 \text{ kgf}$$

Force acting on the Shell [f_S]:

$$= 1.055 \cdot 1164.324 \cdot 223.984$$
$$= 2750.3 \text{ kgf}$$

Discontinuity Force from External Pressure [f_Y]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 1.055 \cdot 1164.324 \cdot 196.675$$
$$= 2415.0 \text{ kgf}$$

Area Resisting External Pressure [A_p]:

$$= (f_S + f_Y + f_N) / P$$
$$= (2750.3 + 2415.0 + 247.9) / 1.055$$
$$= 5132.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 2476.295 / (2 \cdot 5132.857 / 83.972 - 1164.324 / 25.746)$$
$$= 32.1 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$= S / t / R_{xs}$$
$$= 1650.863 [19.5 / 1164.324]$$
$$= 27.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [P_{max}]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(32.148, 27.649)$$
$$= 27.649 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (f_N + f_S + f_Y) / AT$$
$$= (247.864 + 2750.294 + 2414.972) / 83.972$$
$$= 64.464 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:



$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 1.055 \cdot 1164.324 / 25.746 \\ &= 47.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 64.464 - 47.692, 47.692) \\ &= 81.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	81.24	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.65	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle	:	-46 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 22.9499 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 313.9498 mm

Input Echo, WRC297 Item 1, Description: Noz N5 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.44 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.86 kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	387.350	mm



Nozzle Thickness		34.9250	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	23.45	kgf/cm ²
Radial Load	P	-1233.92	kgf
Circumferential Shear	Vc	1835.58	kgf
Longitudinal Shear	Vl	1835.58	kgf
Circumferential Moment	Mc	2.48	kgf-m.
Longitudinal Moment	Ml	3.30	kgf-m.
Torsional Moment	Mt	4.13	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$
 $= 955.067 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	31.925	mm
T / t	=	0.987	
d / t	=	14.321	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.686$$

Nr/P	=	0.082
Mr/P	=	0.134
M0/P	=	0.046
N0/P	=	0.112
MrD/Mc	=	0.259
NrDL/Mc	=	0.125
M0d/Mc	=	0.094
N0DL/Mc	=	0.070
MrD/Ml	=	0.135
NrDL/Ml	=	0.076
M0D/Ml	=	0.048
N0DL/Ml	=	0.151

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0



Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	845	869	845	869
Outplane Stress Summary	893	849	893	849

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	99	-99	99	-99
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	422	422	422	422
Inplane Stress Summary	532	333	533	332

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	893	849	894	849

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	13	13	13	13
Outplane Bending (P)	33	-33	33	-33
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	422	422	422	422
Outplane Stress Summary	470	403	471	402

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	99	-99	99	-99
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	845	869	845	869
Inplane Stress Summary	955	780	956	778

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	955	780	956	778

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188
Hoop Stress Summary	174	202	174	202

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	67	-67	67	-67
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80
Axial Stress Summary	150	15	150	15

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	176	202	176	202

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	13	13	13	13
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188

Hoop Stress Summary	174	202	174	202
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	67	-67	67	-67
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80

Axial Stress Summary	149	16	151	15
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-7	-7	7	7
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-7	-7	7	7
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Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	176	202	176	202
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1161.25(21.0 - 1.5)}$
 $= 953.786 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	2322.500 mm
Nozzle Outside Diameter (d)	=	657.200 mm
Vessel Thickness used (T)	=	19.500 mm
Nozzle Thickness used (t)	=	131.925 mm
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.088$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.060
Mr/P	=	0.104
M0/P	=	0.031
N0/P	=	0.051
MrD/Mc	=	0.266
NrDL/Mc	=	0.101
M0d/Mc	=	0.080
N0DL/Mc	=	0.074
MrD/M1	=	0.079
NrDL/M1	=	0.072
M0D/M1	=	0.023



NØDL/M1 = 0.062

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	59	-59	59	-59
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1373	1396	1373	1396
Outplane Stress Summary	1449	1353	1449	1353

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	203	-203	203	-203
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	686	686	686	686
Inplane Stress Summary	908	503	909	502

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	9	9	-9	-9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1449	1353	1449	1353

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	59	-59	59	-59
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	686	686	686	686
Outplane Stress Summary	762	643	763	642

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	203	-203	203	-203
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1373	1396	1373	1396

Inplane Stress Summary	1594	1214	1597	1211
------------------------	------	------	------	------

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-9	-9	9	9
----------------------	----	----	---	---

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1594	1214	1597	1211

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	845	869	845	869	845	869	845	869
Circ. Pl (SUS)	13	13	13	13	10	10	10	10
Circ. Q (SUS)	33	-33	33	-33	99	-99	99	-99
Long. Pm (SUS)	422	422	422	422	422	422	422	422
Long. Pl (SUS)	10	10	10	10	13	13	13	13
Long. Q (SUS)	99	-99	99	-99	33	-33	33	-33
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	845.0	869.0	845.0	869.0	845.0	869.0	845.0	869.0
Pm+Pl (SUS)	858.2	882.1	858.2	882.1	855.2	879.1	855.2	879.1
Pm+Pl+Q (Total)	891.2	849.1	891.2	849.1	954.1	780.2	954.1	780.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	869.00	1650.86	Passed
Pm+Pl (SUS)	882.14	2476.29	Passed
Pm+Pl+Q (TOTAL)	954.13	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Circ. Pl (SUS)	16	16	16	16	19	19	19	19
Circ. Q (SUS)	59	-59	59	-59	202	-202	204	-204
Long. Pm (SUS)	686	686	686	686	686	686	686	686
Long. Pl (SUS)	19	19	19	19	16	16	16	16
Long. Q (SUS)	203	-203	203	-203	59	-59	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1373	1396	1373	1396	1373	1396	1373	1396
Pm+Pl (SUS)	1389	1412	1389	1412	1392	1415	1392	1415
Pm+Pl+Q (Total)	1448	1353	1448	1353	1594	1213	1596	1211

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1396.00	1650.86	Passed
Pm+Pl (SUS)	1415.11	2476.29	Passed
Pm+Pl+Q (TOTAL)	1596.10	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	160	188	160	188	160	188	160	188
Circ. Pl (SUS)	13	13	13	13	13	13	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	80	80	80	80	80	80	80	80
Long. Pl (SUS)	2	2	2	2	2	2	2	2
Long. Q (SUS)	67	-67	67	-67	67	-67	67	-67
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	160.0	188.0	160.0	188.0	160.0	188.0	160.0	188.0
Pm+Pl (SUS)	173.5	201.4	173.5	201.4	173.5	201.4	173.5	201.4
Pm+Pl+Q (Total)	174.9	201.3	174.9	201.3	174.9	201.3	174.9	201.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	188.00	1522.22	Passed
Pm+Pl (SUS)	201.41	2283.33	Passed
Pm+Pl+Q (TOTAL)	201.26	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 Licensee: SL User
FileName : 410-E-31008
Nozzle Calcs.: Noz N5

Noz1: 40 5:55pm May 21,2026

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Input, Nozzle Desc: Noz N3

From: 70

Pressure for Reinforcement Calculations	P	23.490	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		657.42	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.82	deg
Diameter		18.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.2748	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	657.2000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm

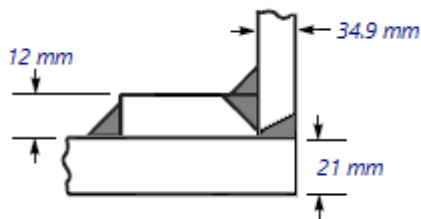
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		15.250	in.
Actual Thickness Used in Calculation		1.375	in.



Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 393.35 \cdot (\exp(23.49/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 3.0585 + 3.0 + 0.0 = 6.0585 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 31.925 + 393.35) / 393.35) - 0.0$$

$$= 228.974 - 0.0 = 228.9739 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c : 11.340 mm
 Min. required shell thickness, internal p + c : 18.002 mm
 Min. required shell thickness, external p + c : 14.028 mm

Nozzle neck minimum required thickness : 11.340 mm

Minimum nozzle neck thickness 30.556 mm , is greater than required thickness 11.340 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 74.9	Allowable	: 1522.2 kgf/cm ²	Passed
Expansion	: 0.0	Allowable	: 3879.9 kgf/cm ²	Passed
Occasional	: 78.3	Allowable	: 2024.6 kgf/cm ²	Passed
Shear	: 15.3	Allowable	: 1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.9, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 34.925 - 3.0$$

$$= 31.925 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D/t$$



= 2303.0/19.5
 = 118.103 must be less than 400.

Effective Pressure Radius [Reff]:
 = Di/2 + corrosion allowance
 = 2300.0/2 + 1.5
 = 1151.500 mm

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

Limit [LR1]:

$$= \sqrt{(\text{Reff} \cdot t) + W}$$

$$= \sqrt{(1151.5 \cdot 19.5) + 100.0}$$

$$= 249.847 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((\text{Reff} + t) \cdot (t + t_e))}$$

$$= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))}$$

$$= 192.059 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 196.675$$

$$= 393.350 \text{ mm}$$

Final value of [LR]:

$$= \min(\text{LR1}, \text{LR2}, \text{LR3})$$

$$= \min(249.847, 192.059, 393.35)$$

$$= 192.059 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 223.984 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)}$$

$$= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(196.675 \cdot 31.925)}$$

$$= 91.239 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$= L_{pr1}$$

$$= 270.0$$

$$= 270.000 \text{ mm}$$

Thickness Limit Candidate [LH3]:

$$= 8(t + t_e)$$

$$= 8(19.5 + 12.0)$$

$$= 252.000 \text{ mm}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min[\text{LH1}, \text{LH2}, \text{LH3}]$$

$$= \min[91.239, 270.0, 252.0]$$

$$= 91.239 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A_5 \cdot \text{frp}) / (LR - t_n))$$

$$= 19.5 + ((12.0 \cdot 1.) / (223.984 - 31.925))$$

$$= 25.746 \text{ mm}$$

Parameter [λ]:

$$= \min(12, (2 \cdot R_n + t_n) / (\sqrt{((Di + \text{teff}) \cdot \text{teff})}))$$

$$= \min(12, (2 \cdot 196.68 + 31.925) / (\sqrt{((2303.0 + 25.746) \cdot 25.746)}))$$

$$= 1.737$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1)$$

$$= 19.5 \cdot 223.984 \cdot \max((1.737/5)^{0.85}, 1)$$



$$= 43.677 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot L_H \\ &= 31.925 \cdot 91.239 \\ &= 29.128 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Pad Fillet Weld [A42]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}42^2 \\ &= 0.5 \cdot 12.0^2 \\ &= 0.720 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}41^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Reinforcing Pad [A5]:

$$\begin{aligned} &= \min(W \cdot t_e, (L_R - t_n) \cdot t_e) \\ &= \min(100.0 \cdot 12.0, (223.984 - 31.925) \cdot 12.0) \\ &= 12.000 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A5 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5) \\ &= 43.677 + 0.922(29.128 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (12.0) \\ &= 83.752 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x/\text{Reff}] \\ &= \cos^{-1}[600.0/1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n)/\text{Reff}] \\ &= \cos^{-1}[(600.0 + 196.675)/1151.5] \\ &= 0.807 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 31.925 / \ln[1 + 31.925/196.675] \\ &= 212.238 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{\text{eff}} / \ln[1 + t_{\text{eff}}/\text{Reff}] \\ &= 25.746 / \ln[1 + 25.746/1151.5] \\ &= 1164.324 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 23.49 \cdot 212.238 (91.239 + 19.5 / \sin[90.0]) \\ &= 5520.8 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.49 \cdot 1164.324 \cdot 223.984 \\ &= 61259.1 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.49 \cdot 1164.324 \cdot 196.675 \\ &= 53790.2 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (61259.1 + 53790.2 + 5520.8) / 23.49 \\ &= 5132.9 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2476.295 / (2 \cdot 5132.857/83.752 - 1164.324/25.746) \\ &= 32.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.863[19.5/1164.324] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(32.014, 27.649) \\ &= 27.649 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ Avg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (5520.826 + 61259.066 + 53790.219) / 83.752 \\ &= 1439.615 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ Circ]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.49 \cdot 1164.324/25.746 \\ &= 1062.3 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1439.615 - 1062.28, 1062.28) \\ &= 1816.9 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1816.95	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.65	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((Rnc+Tn)^2 + (Rn+Tn)^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((196.675+ 31.925)^2 + (196.675+ 31.925)^2)/2)} \\ &= 359.084 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld for non-radial Nozzles [LTP]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((Rnc+Tn+W)^2 + (Rn+Tn+W)^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((196.675+31.925+100.0)^2 + (196.675+31.925+100.0)^2)/2)} \\ &= 516.164 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 31.925, 6.000) \text{ Passed} \\ &tf1, L42T = 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(fy \cdot ky, 1.5 \cdot Sn(A2 + A3), \pi/4 \cdot P \cdot Rn^2 \cdot ky^2) \\ &= \min(53790 \cdot 1.0, 1.5 \cdot 1522.2(29.128+0.0), \pi/4 \cdot 23.5 \cdot 196.68^2 \cdot 1.0^2) \\ &= 7136.228 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + te \cdot Sp) \\ &= 7136.2 \cdot 19.5 \cdot 1651 / (19.5 \cdot 1651 + 12.0 \cdot 1650) \\ &= 4418.132 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$= fwelds \cdot te \cdot Sp / (t \cdot S + te \cdot Sp)$$



$$= 7136.2 \cdot 12.0 \cdot 1650 / (19.5 \cdot 1651 + 12.0 \cdot 1650)$$
$$= 2718.095 \text{ kgf}$$

Shear Stress [τ_1]:

$$= fws / (L_T \cdot (0.6 \cdot tw1 + 0.49 \cdot L_{43T}))$$
$$= 4418.132 / (359.084 \cdot (0.6 \cdot 14.275 + 0.49 \cdot 0.0))$$
$$= 143.655 \text{ kgf/cm}^2$$

Shear Stress [τ_2]:

$$= fwp / (L_T \cdot (0.6 \cdot tw2 + 0.49 \cdot L_{41T}))$$
$$= 2718.095 / (359.084 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$$
$$= 70.977 \text{ kgf/cm}^2$$

Shear Stress [τ_3]:

$$= fwp / (L_{TP} \cdot (0.49 \cdot L_{42T}))$$
$$= 2718.095 / (516.164 \cdot (0.49 \cdot 8.485))$$
$$= 126.654 \text{ kgf/cm}^2$$

Maximum Shear Stress in the Welds:

$$= \max(\tau_1, \tau_2, \tau_3)$$
$$= \max(143.655, 70.977, 126.654)$$
$$= 143.7 \text{ must be less than or equal to } 1650.9 \text{ kgf/cm}^2$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$= \cos^{-1}[Dx/Reff]$$
$$= \cos^{-1}[600.0/1151.5]$$
$$= 1.023$$

Angle [Theta2]:

$$= \cos^{-1}[(Dx+Rn)/Reff]$$
$$= \cos^{-1}[(600.0 + 196.675)/1151.5]$$
$$= 0.807$$

Nozzle Radius for Force Calculation [Rxn]:

$$= tn / \ln[1 + tn/Rn]$$
$$= 31.925 / \ln[1 + 31.925/196.675]$$
$$= 212.238 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= teff / \ln[1 + teff/Reff]$$
$$= 25.746 / \ln[1 + 25.746/1151.5]$$
$$= 1164.324 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.863$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn (LH + t / \sin[\theta]) \text{ (eqn. 4.5.73)}$$
$$= 1.055 \cdot 212.238 (91.239 + 19.5 / \sin[90.0])$$
$$= 247.9 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 1.055 \cdot 1164.324 \cdot 223.984$$
$$= 2750.3 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 1.055 \cdot 1164.324 \cdot 196.675$$
$$= 2415.0 \text{ kgf}$$

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (2750.3 + 2415.0 + 247.9) / 1.055$$
$$= 5132.9 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:



$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff}) \\ &= 2476.295 / (2 \cdot 5132.857/83.752 - 1164.324/25.746) \\ &= 32.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/\text{Rxs}] \\ &= 1650.863[19.5/1164.324] \\ &= 27.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(32.014, 27.649) \\ &= 27.649 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / \text{AT} \\ &= (247.864 + 2750.294 + 2414.972) / 83.752 \\ &= 64.633 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot \text{Rxs} / \text{teff} \\ &= 1.055 \cdot 1164.324/25.746 \\ &= 47.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 64.633 - 47.692, 47.692) \\ &= 81.6 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	81.57	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.65	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:
This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C
Gov. MDMT of the Reinforcement Pad : -29 °C

Flange MDMT including temperature reduction per 3.11:



Flange MDMT with Temp reduction

-85 °C

The Drop for this Nozzle is : 183.6236 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 478.1576 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N3 :

Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.5000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	2300.000	mm	
Shell Thickness	21.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-516 70		
Vessel Cold S.I. Allowable	Smc	1784.44	kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.86	kgf/cm ²
Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	387.350	mm	
Nozzle Thickness	34.9250	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-333 6		
Nozzle Cold S.I. Allowable	SNmc	1641.66	kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh	1522.22	kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	657.200	mm
Thickness of Reinforcing Pad	12.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	23.49	kgf/cm ²
Radial Load	P	-1274.71	kgf
Circumferential Shear	Vc	1835.58	kgf
Longitudinal Shear	Vl	1835.58	kgf
Circumferential Moment	Mc	2.48	kgf-m.
Longitudinal Moment	Ml	3.30	kgf-m.
Torsional Moment	Mt	4.13	kgf-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$$

$$= 955.067 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.500	mm
Nozzle Thickness used	(t) =	31.925	mm
T / t	=	0.987	
d / t	=	14.321	



$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.686$$

Nr/P	=	0.082
Mr/P	=	0.134
M0/P	=	0.046
N0/P	=	0.112
MrD/Mc	=	0.259
NrDL/Mc	=	0.125
M0d/Mc	=	0.094
N0DL/Mc	=	0.070
MrD/M1	=	0.135
NrDL/M1	=	0.076
M0D/M1	=	0.048
N0DL/M1	=	0.151

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	847	870	847	870
Outplane Stress Summary	896	849	896	849

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	103	-103	103	-103
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	423	423	423	423
Inplane Stress Summary	536	331	537	330

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	896	849	896	849

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	35	-35	35	-35



Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	423	423	423	423

Outplane Stress Summary	472	403	473	402
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	103	-103	103	-103
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	847	870	847	870

Inplane Stress Summary	959	778	961	777
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-8	-8	8	8
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	960	778	961	777
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188

Hoop Stress Summary	174	203	174	203
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80

Axial Stress Summary	152	13	153	13
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Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	7	7	-7	-7
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	177	203	177	203

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	14	14	14	14
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	160	188	160	188
Hoop Stress Summary	174	203	174	203

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80
Axial Stress Summary	152	14	153	13

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-7	-7	7	7
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	177	203	177	203

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 457.2 + 2 \cdot 1.65 \cdot \sqrt{1161.25(21.0 - 1.5)}$$

$$= 953.786 \text{ mm}$$



WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 2322.500 mm
 Nozzle Outside Diameter (d) = 657.200 mm
 Vessel Thickness used (T) = 19.500 mm
 Nozzle Thickness used (t) = 131.925 mm
 T / t = 0.148
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.088$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.060
 Mr/P = 0.104
 M0/P = 0.031
 N0/P = 0.051
 MrD/Mc = 0.266
 NrDL/Mc = 0.101
 M0d/Mc = 0.080
 N0DL/Mc = 0.074
 MrD/Ml = 0.079
 NrDL/Ml = 0.072
 M0D/Ml = 0.023
 N0DL/Ml = 0.062

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	61	-61	61	-61
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1375	1398	1375	1398
Outplane Stress Summary	1453	1354	1454	1353

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	209	-209	209	-209
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	687	687	687	687
Inplane Stress Summary	916	498	917	497

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	9	9	-9	-9
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	9	9	-9	-9

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	1453	1354	1454	1354
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	16	16	16	16
Outplane Bending (P)	61	-61	61	-61
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	687	687	687	687

Outplane Stress Summary	765	643	766	642
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	19	19	19	19
Inplane Bending (P)	209	-209	209	-209
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-1	1	1	-1
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1375	1398	1375	1398

Inplane Stress Summary	1603	1210	1606	1207
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-9	-9	9	9
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-9	-9	9	9
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	1603	1210	1606	1207
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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	847	870	847	870	847	870	847	870
Circ. Pl (SUS)	14	14	14	14	10	10	10	10
Circ. Q (SUS)	35	-35	35	-35	103	-103	103	-103
Long. Pm (SUS)	423	423	423	423	423	423	423	423
Long. Pl (SUS)	10	10	10	10	14	14	14	14
Long. Q (SUS)	103	-103	103	-103	35	-35	35	-35

Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	847.0	870.0	847.0	870.0	847.0	870.0	847.0	870.0
Pm+Pl (SUS)	861.1	884.1	861.1	884.1	857.2	880.1	857.2	880.1
Pm+Pl+Q (Total)	896.2	849.1	896.2	849.1	960.1	777.2	960.1	777.2

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	870.00	1650.86	Passed
Pm+Pl (SUS)	884.14	2476.29	Passed
Pm+Pl+Q (TOTAL)	960.13	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1375	1398	1375	1398	1375	1398	1375	1398
Circ. Pl (SUS)	16	16	16	16	19	19	19	19
Circ. Q (SUS)	61	-61	61	-61	208	-208	210	-210
Long. Pm (SUS)	687	687	687	687	687	687	687	687
Long. Pl (SUS)	19	19	19	19	16	16	16	16
Long. Q (SUS)	209	-209	209	-209	61	-61	61	-61
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1375	1398	1375	1398	1375	1398	1375	1398
Pm+Pl (SUS)	1391	1414	1391	1414	1394	1417	1394	1417
Pm+Pl+Q (Total)	1452	1353	1452	1353	1602	1209	1604	1207

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1398.00	1650.86	Passed
Pm+Pl (SUS)	1417.11	2476.29	Passed
Pm+Pl+Q (TOTAL)	1604.10	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	160	188	160	188	160	188	160	188
Circ. Pl (SUS)	14	14	14	14	14	14	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0

Long. Pm (SUS)	80	80	80	80	80	80	80	80
Long. Pl (SUS)	2	2	2	2	2	2	2	2
Long. Q (SUS)	69	-69	69	-69	69	-69	69	-69
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	160.0	188.0	160.0	188.0	160.0	188.0	160.0	188.0
Pm+Pl (SUS)	174.5	202.4	174.5	202.4	174.5	202.4	174.5	202.4
Pm+Pl+Q (Total)	176.0	202.3	176.0	202.3	176.0	202.3	176.0	202.3

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	188.00	1522.22	Passed
Pm+Pl (SUS)	202.41	2283.33	Passed
Pm+Pl+Q (TOTAL)	202.26	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz N6

From: 70

Pressure for Reinforcement Calculations	P	23.601	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		1085.42	cm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-90.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		120	

Flange Material		SA-350 LF1	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
---------------------	-----	--------	----

Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	12.7000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	319.0750	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		50.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

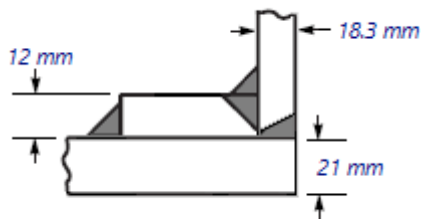
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N6

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		7.187	in.
Actual Thickness Used in Calculation		0.719	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 188.55 \cdot (\exp(23.601/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 1.4731 + 3.0 + 0.0 = 4.4731 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.263 + 188.55) / 188.55) - 0.0$$

$$= 228.412 - 0.0 = 228.4119 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	10.160 mm
Min. required shell thickness, internal p + c :	18.081 mm
Min. required shell thickness, external p + c :	14.028 mm

Nozzle neck minimum required thickness : 10.160 mm

Minimum nozzle neck thickness 15.977 mm , is greater than required thickness 10.160 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	73.7,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3881.1 kgf/cm ²	Passed
Occasional :	80.2,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	30.1,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.9, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 18.263 - 3.0$$

$$= 15.263 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:



$$\begin{aligned}
 &= Di/2 + \text{corrosion allowance} \\
 &= 2300.0/2 + 1.5 \\
 &= 1151.500 \text{ mm}
 \end{aligned}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, Lr is min[LR1, LR2, LR3]:

$$\begin{aligned}
 \text{Limit [LR1]:} \\
 &= \sqrt{(R_{eff} \cdot t) + W} \\
 &= \sqrt{(1151.5 \cdot 19.5) + 50.0} \\
 &= 199.847 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR2]:} \\
 &= \sqrt{((R_{eff} + t) \cdot (t + t_e))} \\
 &= \sqrt{((1151.5 + 19.5) \cdot (19.5 + 12.0))} \\
 &= 192.059 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Limit [LR3]:} \\
 &= 2 \cdot R_n \\
 &= 2 \cdot 94.275 \\
 &= 188.550 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Final value of [LR]:} \\
 &= \min(LR1, LR2, LR3) \\
 &= \min(199.847, 192.059, 188.55) \\
 &= 188.550 \text{ mm}
 \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:
 = 203.812 mm

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH1]:} \\
 &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\
 &= \min[1.5 \cdot 19.5, 12.0] + \sqrt{(94.275 \cdot 15.263)} \\
 &= 49.933 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH2]:} \\
 &= L_{pr1} \\
 &= 270.0 \\
 &= 270.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Thickness Limit Candidate [LH3]:} \\
 &= 8(t + t_e) \\
 &= 8(19.5 + 12.0) \\
 &= 252.000 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Nozzle Wall Length Outside the Vessel [LH]:} \\
 &= \min[LH1, LH2, LH3] \\
 &= \min[49.933, 270.0, 252.0] \\
 &= 49.933 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Effective Vessel Thickness [teff]:} \\
 &= t + ((A5 \cdot f_{rp}) / (LR - t_n)) \\
 &= 19.5 + ((6.0 \cdot 1.) / (203.812 - 15.263)) \\
 &= 22.681 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Parameter } [\lambda]: \\
 &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(Di + t_{eff}) \cdot t_{eff}})) \\
 &= \min(12, (2 \cdot 94.27 + 15.263) / (\sqrt{(2303.0 + 22.681) \cdot 22.681})) \\
 &= 0.887
 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

$$\begin{aligned}
 \text{Area Contributed by the Vessel Wall [A1]:} \\
 &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\
 &= 19.5 \cdot 203.812 \cdot \max((0.887/5)^{0.85}, 1) \\
 &= 39.743 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Area Contributed by the Nozzle Outside the Vessel Wall [A2]:} \\
 &= t_n \cdot LH
 \end{aligned}$$



$$= 15.263 \cdot 49.933$$
$$= 7.621 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}42^2$$
$$= 0.5 \cdot 12.0^2$$
$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}41^2$$
$$= 0.5 \cdot 10.0^2$$
$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$
$$= \min(50.0 \cdot 12.0, (203.812 - 15.263) \cdot 12.0)$$
$$= 6.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$
$$= 39.743 + 0.922(7.621 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (6.0)$$
$$= 53.989 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$
$$= 15.263 / \ln[1 + 15.263/94.275]$$
$$= 101.715 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 22.681 / \ln[1 + 22.681/1151.5]$$
$$= 1162.803 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.863 \cdot 1.0$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn}(LH + t)$$
$$= 23.601 \cdot 101.715(49.933 + 19.5)$$
$$= 1666.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.601 \cdot 1162.803 \cdot 203.812$$
$$= 55933.2 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.601 \cdot 1162.803 \cdot 94.275$$
$$= 25872.3 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (55933.2 + 25872.3 + 1666.8)/23.601$$
$$= 3536.8 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p/AT - R_{xs}/t_{eff})$$
$$= 2476.295 / (2 \cdot 3536.792/53.989 - 1162.803/22.681)$$
$$= 31.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/R_{xs}]$$
$$= 1650.863[19.5/1162.803]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(31.05, 27.685)$$



$$= 27.685 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (1666.798 + 55933.172 + 25872.293) / 53.989 \\ &= 1546.103 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_x / t_{eff} \\ &= 23.601 \cdot 1162.803 / 22.681 \\ &= 1210.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1546.103 - 1209.961, 1209.961) \\ &= 1882.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1882.24	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.68	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [LT]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n) \\ &= \pi/2 \cdot (94.275 + 15.263) \\ &= 172.061 \text{ mm} \end{aligned}$$

Weld Length of Pad to Shell Weld [LTp]:

$$\begin{aligned} &= \pi/2 \cdot (R_n + t_n + W) \\ &= \pi/2 \cdot (94.275 + 15.263 + 50.0) \\ &= 250.601 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 8.485, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\begin{aligned} t_c, L41T &= 7.071, \text{ must be } \geq \min(0.7 \cdot 15.263, 6.000) \text{ Passed} \\ t_{f1}, L42T &= 8.485, \text{ must be } \geq \min(0.6 \cdot 12.000, 0.6 \cdot 19.500) \text{ Passed} \end{aligned}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(25872 \cdot 1.0, 1.5 \cdot 1522.2(7.621 + 0.0), \pi/4 \cdot 23.6 \cdot 94.27^2 \cdot 1.0^2) \\ &= 1647.459 \text{ kgf} \end{aligned}$$

Discontinuity Force [fws]:

$$\begin{aligned} &= fwelds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p) \\ &= 1647.5 \cdot 19.5 \cdot 1651 / (19.5 \cdot 1651 + 12.0 \cdot 1650) \\ &= 1019.964 \text{ kgf} \end{aligned}$$

Discontinuity Force [fwp]:

$$\begin{aligned} &= fwelds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p) \\ &= 1647.5 \cdot 12.0 \cdot 1650 / (19.5 \cdot 1651 + 12.0 \cdot 1650) \\ &= 627.496 \text{ kgf} \end{aligned}$$

Shear Stress [τ_1]:

$$\begin{aligned} &= fws / (L_T \cdot (0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 1019.964 / (172.061 \cdot (0.6 \cdot 12.7 + 0.49 \cdot 0.0)) \\ &= 77.794 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_2]:

$$\begin{aligned} &= fwp / (L_T \cdot (0.6 \cdot t_{w2} + 0.49 \cdot L_{41T})) \\ &= 627.496 / (172.061 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071)) \\ &= 34.196 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress [τ_3]:



$$\begin{aligned} &= fwp / (LrP \cdot (0.49 \cdot L42T)) \\ &= 627.496 / (250.601 \cdot (0.49 \cdot 8.485)) \\ &= 60.224 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Shear Stress in the Welds:

$$\begin{aligned} &= \max(\tau_1, \tau_2, \tau_3) \\ &= \max(77.794, 34.196, 60.224) \\ &= 77.8 \text{ must be less than or equal to } 1650.9 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.263 / \ln[1 + 15.263/94.275] \\ &= 101.715 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 22.681 / \ln[1 + 22.681/1151.5] \\ &= 1162.803 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (L_H + t) \\ &= 1.055 \cdot 101.715 (49.933 + 19.5) \\ &= 74.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1162.803 \cdot 203.812 \\ &= 2499.3 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1162.803 \cdot 94.275 \\ &= 1156.1 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (2499.3 + 1156.1 + 74.5) / 1.055 \\ &= 3536.8 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 3536.792 / 53.989 - 1162.803 / 22.681) \\ &= 31.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.863 [19.5 / 1162.803] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(31.05, 27.685) \\ &= 27.685 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (74.48 + 2499.344 + 1156.089) / 53.989 \\ &= 69.087 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1162.803 / 22.681 \\ &= 54.1 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}})$$

$$= \max(2 \cdot 69.087 - 54.066, 54.066)$$

$$= 84.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	84.11	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.68	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -68 °C

The Drop for this Nozzle is : 5.2286 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 296.2285 mm

Input Echo, WRC297 Item 1, Description: Noz N6 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	1784.44 kgf/cm ²
Vessel Hot S.I. Allowable	Smh	1650.86 kgf/cm ²

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	182.550	mm
Nozzle Thickness	18.2626	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	1641.66 kgf/cm ²



Nozzle Hot S.I. Allowable SNmh 1522.22 kgf/cm²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad 319.075 mm
 Thickness of Reinforcing Pad 12.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUS tained loads.*

Design Internal Pressure	Dp	23.60	kgf/cm ²
Radial Load	P	-550.67	kgf
Circumferential Shear	Vc	815.81	kgf
Longitudinal Shear	Vl	815.81	kgf
Circumferential Moment	Mc	0.49	kgf-m.
Longitudinal Moment	Ml	0.65	kgf-m.
Torsional Moment	Mt	0.82	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(1167.25(21.0 - 1.5))}$$

$$= 716.942 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2334.500 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	31.500 mm
Nozzle Thickness used	(t) =	15.263 mm
T / t	=	2.064
d / t	=	14.354

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.808$$

Nr/P	=	0.140
Mr/P	=	0.142
M0/P	=	0.088
N0/P	=	0.168
MrD/Mc	=	0.217
NrDL/Mc	=	0.175
M0d/Mc	=	0.161
N0DL/Mc	=	0.124
MrD/Ml	=	0.180
NrDL/Ml	=	0.150
M0D/Ml	=	0.131
N0DL/Ml	=	0.141

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0
Normal Pressure Stress	851	874	851	874
Outplane Stress Summary	889	854	889	854



Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	425	425	425	425
Inplane Stress Summary	480	386	480	385

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	7	7	-7	-7

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	889	855	889	854

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	9	9	9	9
Outplane Bending (P)	29	-29	29	-29
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	425	425	425	425
Outplane Stress Summary	463	405	464	405

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	851	874	851	874
Inplane Stress Summary	905	835	906	834

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	C1	Du	D1
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-7	-7	7	7
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	905	835	906	834

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	161	190	161	190
Hoop Stress Summary	171	199	171	199

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	5	5	5	5
Axial Bending (P)	102	-102	102	-102
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80
Axial Stress Summary	187	-15	189	-16

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	15	15	-15	-15
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	15	15	-15	-15

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	197	216	198	218

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	9	9	9	9
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	161	190	161	190



Hoop Stress Summary 171 199 171 199

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Axial Membrane (P)	5	5	5	5
Axial Bending (P)	102	-102	102	-102
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	80	80	80	80

Axial Stress Summary 187 -15 189 -16

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-15	-15	15	15
Shear due to Torsion	0	0	0	0

Shear Stress Summary -15 -15 15 15

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	197	216	198	217
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{1161.25(21.0 - 1.5)}$$

$$= 715.661 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	319.075 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	65.263 mm
T / t	=	0.299
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.499$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.083
Mr/P	=	0.176
M0/P	=	0.051
N0/P	=	0.100
MrD/Mc	=	0.299
NrDL/Mc	=	0.087
M0d/Mc	=	0.092
N0DL/Mc	=	0.063
MrD/Ml	=	0.143
NrDL/Ml	=	0.090
M0D/Ml	=	0.045
N0DL/Ml	=	0.118

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE

	Au	Al	Bu	Bl
--	----	----	----	----



(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	44	-44	44	-44
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	1381	1405	1381	1405
Outplane Stress Summary	1440	1375	1441	1375

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	152	-152	152	-152
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	690	690	690	690
Inplane Stress Summary	855	550	856	549

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	8	8	-8	-8
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	8	8	-8	-8

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	1440	1375	1441	1375

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	14	14	14	14
Outplane Bending (P)	44	-44	44	-44
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	690	690	690	690
Outplane Stress Summary	749	661	750	660

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	152	-152	152	-152
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0

Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1381	1405	1381	1405
Inplane Stress Summary	1546	1265	1547	1263

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-8	-8	8	8
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-8	-8	8	8

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	1546	1265	1547	1264

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	851	874	851	874	851	874	851	874
Circ. Pl (SUS)	9	9	9	9	7	7	7	7
Circ. Q (SUS)	29	-29	29	-29	47	-47	47	-47
Long. Pm (SUS)	425	425	425	425	425	425	425	425
Long. Pl (SUS)	7	7	7	7	9	9	9	9
Long. Q (SUS)	47	-47	47	-47	29	-29	29	-29
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	851.0	874.0	851.0	874.0	851.0	874.0	851.0	874.0
Pm+Pl (SUS)	860.1	883.1	860.1	883.1	858.1	881.1	858.1	881.1
Pm+Pl+Q (Total)	889.1	854.1	889.1	854.1	905.1	834.1	905.1	834.1

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	874.00	1650.86	Passed
Pm+Pl (SUS)	883.11	2476.29	Passed
Pm+Pl+Q (TOTAL)	905.11	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	1381	1405	1381	1405	1381	1405	1381	1405
Circ. Pl (SUS)	14	14	14	14	12	12	12	12
Circ. Q (SUS)	44	-44	44	-44	152	-152	152	-152
Long. Pm (SUS)	690	690	690	690	690	690	690	690
Long. Pl (SUS)	12	12	12	12	14	14	14	14
Long. Q (SUS)	152	-152	152	-152	44	-44	44	-44
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	8	8	-8	-8	-8	-8	8	8
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1381	1405	1381	1405	1381	1405	1381	1405
Pm+Pl (SUS)	1395	1419	1395	1419	1393	1417	1393	1417
Pm+Pl+Q (Total)	1439	1375	1439	1375	1545	1265	1545	1265

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1405.00	1650.86	Passed
Pm+Pl (SUS)	1419.09	2476.29	Passed
Pm+Pl+Q (TOTAL)	1545.08	4952.59	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	161	190	161	190	161	190	161	190
Circ. Pl (SUS)	9	9	9	9	9	9	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	80	80	80	80	80	80	80	80
Long. Pl (SUS)	5	5	5	5	5	5	5	5
Long. Q (SUS)	102	-102	102	-102	102	-102	102	-102
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	15	15	-15	-15	-15	-15	15	15
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	161.0	190.0	161.0	190.0	161.0	190.0	161.0	190.0
Pm+Pl (SUS)	172.6	200.9	172.6	200.9	172.6	200.9	172.6	200.9
Pm+Pl+Q (Total)	195.7	218.1	195.7	218.1	195.7	218.1	195.7	218.1

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	190.00	1522.22	Passed
Pm+Pl (SUS)	200.94	2283.33	Passed
Pm+Pl+Q (TOTAL)	218.07	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: Noz N7

From: 70

Pressure for Reinforcement Calculations	P	23.590	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	1000.0000	mm
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Distance from Bottom/Left Tangent		657.42	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-58.65	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	+IT
Flange Type		Long Weld Neck	

Corrosion Allowance	can	3.0000	mm
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Outside Projection	ho	270.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm

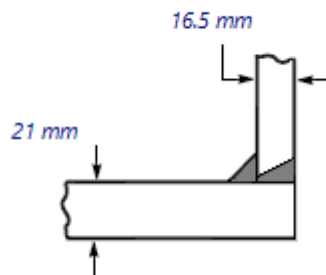
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz N7

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 56.8 \cdot (\exp(23.59/(1522.219 \cdot 1.0)) - 1) + 3.0 + 0.0$$

$$= 0.4436 + 3.0 + 0.0 = 3.4436 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 13.5 + 56.8) / 56.8) - 0.0$$

$$= 591.986 - 0.0 = 591.9862 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	7.800 mm
Min. required shell thickness, internal p + c :	18.073 mm
Min. required shell thickness, external p + c :	14.028 mm

Nozzle neck minimum required thickness : 7.800 mm

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 7.800 mm

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	15.7,	Allowable :	1522.2 kgf/cm ²	Passed
Expansion :	0.0,	Allowable :	3939.1 kgf/cm ²	Passed
Occasional :	20.0,	Allowable :	2024.6 kgf/cm ²	Passed
Shear :	24.0,	Allowable :	1065.6 kgf/cm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.5 - 3.0$$

$$= 13.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:



$$\begin{aligned} &= \min(\sqrt{(\text{Reff} \cdot t)}, 2 \cdot R_n) \\ &= \min(\sqrt{(1151.5 \cdot 19.5)}, 2 \cdot 28.4) \\ &= 56.800 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:
= 70.300 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(28.4 \cdot 13.5)} \\ &= 19.581 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 270.0 \\ &= 270.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[19.581, 270.0, 156.0] \\ &= 19.581 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (70.3 - 13.5)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 28.4 + 13.5) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.330 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 70.3 \cdot \max((0.33/5)^{0.85}, 1) \\ &= 13.708 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 13.5 \cdot 19.581 \\ &= 2.643 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 \\ &= 13.708 + 0.922(2.643 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.646 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / \text{Reff}] \\ &= \cos^{-1}[1000.0 / 1151.5] \\ &= 0.519 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / \text{Reff}] \\ &= \cos^{-1}[(1000.0 + 28.4) / 1151.5] \\ &= 0.467 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:



$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 13.5/\ln[1 + 13.5/28.4] \\ &= 34.714 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 19.5/\ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (LH + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 23.59 \cdot 34.714 (19.581 + 19.5/\sin[90.0]) \\ &= 320.0 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.59 \cdot 1161.225 \cdot 70.3 \\ &= 19257.8 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 23.59 \cdot 1161.225 \cdot 30.025 \\ &= 8224.9 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN) / P \\ &= (19257.8 + 8224.9 + 320.0) / 23.59 \\ &= 1178.6 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 1178.561 / 16.646 - 1161.225 / 19.5) \\ &= 30.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/R_{xs}] \\ &= 1650.863[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(30.179, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (320.034 + 19257.803 + 8224.871) / 16.646 \\ &= 1670.246 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 23.59 \cdot 1161.225 / 19.5 \\ &= 1404.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 1670.246 - 1404.81, 1404.81) \\ &= 1935.7 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1935.68	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [Lt]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{((R_{nc}+T_n)^2 + (R_n+T_n)^2)/2} \\ &= \pi/2 \cdot \sqrt{((30.025+13.5)^2 + (28.4+13.5)^2)/2} \\ &= 67.104 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 0.000, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 13.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(8225 \cdot 1.0, 1.5 \cdot 1522.2(2.643+0.0), \pi/4 \cdot 23.6 \cdot 28.4^2 \cdot 1.0^2) \\ &= 149.438 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds/(L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 149.438/(67.104(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 63.992 < \text{or} = 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x/R_{eff}] \\ &= \cos^{-1}[1000.0/1151.5] \\ &= 0.519 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x+R_n)/R_{eff}] \\ &= \cos^{-1}[(1000.0 + 28.4)/1151.5] \\ &= 0.467 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 13.5 / \ln[1 + 13.5/28.4] \\ &= 34.714 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 19.5 / \ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \text{ (eqn. 4.5.73)} \\ &= 1.055 \cdot 34.714(19.581 + 19.5/\sin[90.0]) \\ &= 14.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 70.3 \\ &= 860.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 30.025 \\ &= 367.7 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N)/P \\ &= (860.9 + 367.7 + 14.3)/1.055 \\ &= 1178.6 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/\text{teff}) \\ &= 2476.295 / (2 \cdot 1178.561 / 16.646 - 1161.225 / 19.5) \\ &= 30.2 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= \text{S}[t/\text{Rxs}] \\ &= 1650.863 [19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(30.179, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / \text{AT} \\ &= (14.307 + 860.917 + 367.691) / 16.646 \\ &= 74.668 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot \text{Rxs} / \text{teff} \\ &= 1.055 \cdot 1161.225 / 19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 74.668 - 62.802, 62.802) \\ &= 86.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	86.53	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

The Drop for this Nozzle is : 81.1214 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 392.5260 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC297 Item 1, Description: Noz N7 :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.5000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2300.000	mm
Shell Thickness	21.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 1784.44	kgf/cm ²
Vessel Hot S.I. Allowable	Smh 1650.86	kgf/cm ²
Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	50.800	mm
Nozzle Thickness	16.5000	mm
Nozzle Stress Concentration Factor	1.000	



Nozzle Material	SA-333 6
Nozzle Cold S.I. Allowable	SNmc 1641.66 kgf/cm ²
Nozzle Hot S.I. Allowable	SNmh 1522.22 kgf/cm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	23.59	kgf/cm ²
Radial Load	P	-132.57	kgf
Circumferential Shear	Vc	203.95	kgf
Longitudinal Shear	Vl	203.95	kgf
Circumferential Moment	Mc	0.03	kgf-m.
Longitudinal Moment	Ml	0.04	kgf-m.
Torsional Moment	Mt	0.05	kgf-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tv, (3.990) is < 5
 check the limitations of WRC bulletin 297.*
*Warning - The ratio, Dn/Tn (5.079) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 83.8 + 2 \cdot 1.65 \cdot \sqrt{(1161.25(21.0 - 1.5))}$
 $= 580.386 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2322.500 mm
Nozzle Outside Diameter	(d) =	83.800 mm
Vessel Thickness used	(T) =	19.500 mm
Nozzle Thickness used	(t) =	13.500 mm
T / t	=	1.444
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.394$$

Nr/P	=	0.207
Mr/P	=	0.271
M0/P	=	0.120
N0/P	=	0.178
MrD/Mc	=	0.264
NrDL/Mc	=	0.168
M0d/Mc	=	0.145
N0DL/Mc	=	-0.112
MrD/Ml	=	0.249
NrDL/Ml	=	0.165
M0D/Ml	=	0.135
N0DL/Ml	=	-0.044

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	0	0	0	0



Normal Pressure Stress	1381	1404	1381	1404
------------------------	------	------	------	------

Outplane Stress Summary	1412	1386	1412	1385
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	690	690	690	690

Inplane Stress Summary	754	641	754	641
------------------------	-----	-----	-----	-----

Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	7	7	-7	-7
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0

Shear Stress Summary	7	7	-7	-7
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Vessel Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	1412	1386	1412	1386
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	25	-25	25	-25
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	690	690	690	690

Outplane Stress Summary	721	671	721	671
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	56	-56	56	-56
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	1381	1404	1381	1404

Inplane Stress Summary	1444	1355	1445	1355
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Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	Dl Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-7	-7	7	7
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-7	-7	7	7

Vessel Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	1445	1355	1445	1355

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	76	48	76
Hoop Stress Summary	54	82	54	82

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	101	-44	101	-44

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	11	11	-11	-11
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	11	11	-11	-11

Nozzle Stresses (kgf/cm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	103	128	104	129

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0

Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	48	76	48	76
Hoop Stress Summary	54	82	54	82

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	4	4	4	4
Axial Bending (P)	72	-72	72	-72
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	24	24	24	24
Axial Stress Summary	101	-44	101	-44

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-11	-11	11	11
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-11	-11	11	11

Nozzle Stresses (kgf/cm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	103	128	104	129

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	1381	1404	1381	1404	1381	1404	1381	1404
Circ. Pl (SUS)	6	6	6	6	7	7	7	7
Circ. Q (SUS)	25	-25	25	-25	56	-56	56	-56
Long. Pm (SUS)	690	690	690	690	690	690	690	690
Long. Pl (SUS)	7	7	7	7	6	6	6	6
Long. Q (SUS)	56	-56	56	-56	25	-25	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	7	7	-7	-7	-7	-7	7	7
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	1381	1404	1381	1404	1381	1404	1381	1404
Pm+Pl (SUS)	1387	1410	1387	1410	1388	1411	1388	1411
Pm+Pl+Q (Total)	1412	1385	1412	1385	1444	1355	1444	1355

Vessel Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	1404.00	1650.86	Passed
Pm+Pl (SUS)	1411.07	2476.29	Passed



Pm+Pl+Q (TOTAL)	1444.07	4952.59	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (kgf/cm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	76	48	76	48	76	48	76
Circ. Pl (SUS)	6	6	6	6	6	6	6	6
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	4	4	4	4	4	4	4	4
Long. Q (SUS)	72	-72	72	-72	72	-72	72	-72
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	11	11	-11	-11	-11	-11	11	11
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	48.0	76.0	48.0	76.0	48.0	76.0	48.0	76.0
Pm+Pl (SUS)	58.0	84.2	58.0	84.2	58.0	84.2	58.0	84.2
Pm+Pl+Q (Total)	102.5	127.9	102.5	127.9	102.5	127.9	102.5	127.9

Nozzle Stress Summation Comparison (kgf/cm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	76.00	1522.22	Passed
Pm+Pl (SUS)	84.15	2283.33	Passed
Pm+Pl+Q (TOTAL)	127.91	4566.66	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Noz K2A

From: 70

Pressure for Reinforcement Calculations	P	23.490	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
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Distance from Bottom/Left Tangent		460.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		30.82	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
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Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm

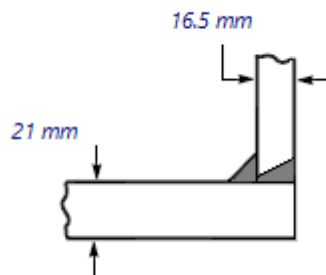
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K2A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 53.8 \cdot (\exp(23.49/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.4183 + 1.5 + 0.0 = 1.9183 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$$

$$= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c	18.002 mm
Min. required shell thickness, external p + c	14.028 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.5 - 1.5$$

$$= 15.000 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$$

$$= 53.800 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 68.800 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 19.5, 0.0] + \sqrt{26.9 \cdot 15.0}$$

$$= 20.087 \text{ mm}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.694 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.863 \cdot 1.0$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t / \sin[\theta]) \quad (\text{eqn. 4.5.73})$$
$$= 23.49 \cdot 33.848 (20.087 + 19.5 / \sin[90.0])$$
$$= 314.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.49 \cdot 1161.225 \cdot 68.8$$
$$= 18766.6 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$
$$= 23.49 \cdot 1161.225 \cdot 26.9$$
$$= 7337.5 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (18766.6 + 7337.5 + 314.8) / 23.49$$
$$= 1124.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$
$$= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5)$$
$$= 32.9 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$
$$= 1650.863 [19.5 / 1161.225]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(32.934, 27.722)$$
$$= 27.722 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (314.751 + 18766.592 + 7337.519) / 16.694$$
$$= 1582.510 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / t_{eff}$$
$$= 23.49 \cdot 1161.225 / 19.5$$
$$= 1398.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1582.51 - 1398.824, 1398.824)$$
$$= 1766.2 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1766.20	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((R_{nc} + T_n)^2 + (R_n + T_n)^2) / 2}$$
$$= \pi/2 \cdot \sqrt{((26.9 + 15.0)^2 + (26.9 + 15.0)^2) / 2}$$
$$= 65.816 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7338 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.498 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.498 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.285 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 33.848(20.087 + 19.5 / \sin[90.0]) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / R_{xs}] \\ &= 1650.863[19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(32.934, 27.722)
= 27.722 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (14.131 + 842.547 + 329.426)/16.694
= 71.049 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 71.049 - 62.802, 62.802)
= 79.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	79.30	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.388 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 26.8883 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.4222 mm

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Input, Nozzle Desc: Noz K2B

From: 70

Pressure for Reinforcement Calculations	P	23.565	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		460.92	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		-30.82	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000	mm

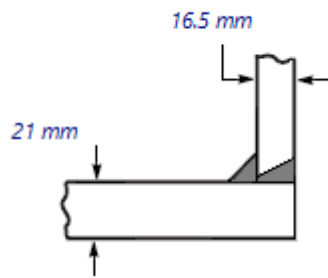
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K2B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.565/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4197 + 1.5 + 0.0 = 1.9197 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c	18.055 mm
Min. required shell thickness, external p + c	14.028 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.9, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than 400.}$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2300.0/2 + 1.5$
 $= 1151.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:
 $= 68.800 \text{ mm}$

Thickness Limit Candidate [LH1]:
 $= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$
 $= \min[1.5 \cdot 19.5, 0.0] + \sqrt{26.9 \cdot 15.0}$
 $= 20.087 \text{ mm}$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.694 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.863 \cdot 1.0$$

$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t / \sin[\theta]) \text{ (eqn. 4.5.73)}$$

$$= 23.565 \cdot 33.848 (20.087 + 19.5 / \sin[90.0])$$

$$= 315.8 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.565 \cdot 1161.225 \cdot 68.8$$

$$= 18826.8 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot R_{xs} \cdot R_{nc}$$

$$= 23.565 \cdot 1161.225 \cdot 26.9$$

$$= 7361.1 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$

$$= (18826.8 + 7361.1 + 315.8) / 23.565$$

$$= 1124.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff})$$

$$= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5)$$

$$= 32.9 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S / (t / R_{xs})$$

$$= 1650.863 [19.5 / 1161.225]$$

$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$

$$= \min(32.934, 27.722)$$

$$= 27.722 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$

$$= (315.762 + 18826.822 + 7361.068) / 16.694$$

$$= 1587.589 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / t_{eff}$$

$$= 23.565 \cdot 1161.225 / 19.5$$

$$= 1403.3 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$

$$= \max(2 \cdot 1587.589 - 1403.313, 1403.313)$$

$$= 1771.9 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1771.87	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi / 2 \cdot \sqrt{((R_{nc} + T_n)^2 + (R_n + T_n)^2) / 2}$$

$$= \pi / 2 \cdot \sqrt{((26.9 + 15.0)^2 + (26.9 + 15.0)^2) / 2}$$

$$= 65.816 \text{ mm}$$

Weld Throat Dimensions, (0.7071 • Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 15.000, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7361 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.6 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.926 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.926 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.472 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 26.9) / 1151.5] \\ &= 0.995 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 33.848(20.087 + 19.5 / \sin[90.0]) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S / [t / R_{xs}] \\ &= 1650.863 [19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(32.934, 27.722)
= 27.722 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (14.131 + 842.547 + 329.426)/16.694
= 71.049 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 71.049 - 62.802, 62.802)
= 79.3 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	79.30	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.389 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 26.8883 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 251.4222 mm

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Input, Nozzle Desc: Noz K3

From: 70

Pressure for Reinforcement Calculations	P	23.490	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Cylinder/Cone Centerline	L1	600.0000	mm
--	----	----------	----

Distance from Bottom/Left Tangent		767.42	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B
Material UNS Number		K03006
Material Specification/Type		Smls. pipe
Allowable Stress at Temperature	Sn	1522.22 kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66 kgf/cm ²

Diameter Basis (for tr calc only)		Inside
Layout Angle		30.82 deg
Diameter		1.5000 in.

Size and Thickness Basis		Actual
Actual Thickness	tn	16.0000 mm

Flange Material		SA-350 LF2
Flange Type		Long Weld Neck

Corrosion Allowance	can	1.5000 mm
---------------------	-----	-----------

Outside Projection	ho	200.0000 mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000 mm
Groove weld depth between Nozzle and Vessel Wgnv		0.0000 mm

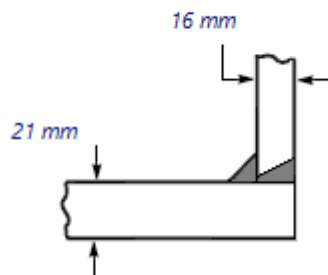
Flange Class		300
Flange Grade		GR 1.1

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K3

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		1.500 in.
Actual Thickness Used in Calculation		0.630 in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 41.1 \cdot (\exp(23.49/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 0.3196 + 1.5 + 0.0 = 1.8196 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 14.5 + 41.1) / 41.1) - 0.0$$

$$= 812.735 - 0.0 = 812.7354 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.020 mm
Min. required shell thickness, internal p + c :	18.002 mm
Min. required shell thickness, external p + c :	14.028 mm

Nozzle neck minimum required thickness	: 6.020 mm
--	------------

Minimum nozzle neck thickness 16.002 mm , is greater than required thickness 6.020 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 16.0 - 1.5$$

$$= 14.500 \text{ mm}$$

Shell Diameter to Thickness ratio [D/t]:

$$= D_i/t$$

$$= 2303.0/19.5$$

$$= 118.103 \text{ must be less than 400.}$$

Effective Pressure Radius [Reff]:

$$= D_i/2 + \text{corrosion allowance}$$

$$= 2300.0/2 + 1.5$$

$$= 1151.500 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

$$= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$$

$$= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 20.55)$$

$$= 41.100 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 55.600 \text{ mm}$$

Thickness Limit Candidate [LH1]:

$$= \min[1.5t, t_e] + \sqrt{R_n \cdot t_n}$$

$$= \min[1.5 \cdot 19.5, 0.0] + \sqrt{20.55 \cdot 14.5}$$

$$= 17.262 \text{ mm}$$



Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[17.262, 200.0, 156.0] \\ &= 17.262 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (55.6 - 14.5)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 20.55 + 14.5) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.261 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 55.6 \cdot \max((0.261/5)^{0.85}, 1) \\ &= 10.842 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 14.5 \cdot 17.262 \\ &= 2.503 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 10.842 + 0.922(2.503 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.650 \text{ cm}^2 \end{aligned}$$

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 20.55) / 1151.5] \\ &= 1.002 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 14.5 / \ln[1 + 14.5 / 20.55] \\ &= 27.158 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$



$$= 1.5 \cdot 1650.863 \cdot 1.0$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot Rxn(LH + t / \sin[\text{theta}]) \text{ (eqn. 4.5.73)}$$
$$= 23.49 \cdot 27.158(17.262 + 19.5/\sin[90.0])$$
$$= 234.5 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= 23.49 \cdot 1161.225 \cdot 55.6$$
$$= 15166.0 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= P \cdot Rxs \cdot Rnc$$
$$= 23.49 \cdot 1161.225 \cdot 20.55$$
$$= 5605.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN)/P$$
$$= (15166.0 + 5605.4 + 234.5)/23.49$$
$$= 894.3 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot Ap/AT - Rxs/teff)$$
$$= 2476.295 / (2 \cdot 894.257/13.65 - 1161.225/19.5)$$
$$= 34.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= S[t/Rxs]$$
$$= 1650.863[19.5/1161.225]$$
$$= 27.7 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(Pmax1, Pmax2)$$
$$= \min(34.645, 27.722)$$
$$= 27.722 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ Avg]:

$$= (fN + fS + fY) / AT$$
$$= (234.517 + 15166.024 + 5605.428)/13.65$$
$$= 1538.908 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ Circ]:

$$= P \cdot Rxs / teff$$
$$= 23.49 \cdot 1161.225/19.5$$
$$= 1398.8 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma\text{Avg} - \sigma\text{Circ}, \sigma\text{Circ})$$
$$= \max(2 \cdot 1538.908 - 1398.824, 1398.824)$$
$$= 1679.0 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1678.99	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$= \pi/2 \cdot \sqrt{((Rnc+Tn)^2 + (Rn+Tn)^2)/2 }$$
$$= \pi/2 \cdot \sqrt{((20.55+ 14.5)^2 + (20.55+ 14.5)^2)/2 }$$
$$= 55.056 \text{ mm}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$



Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 * 14.500, 6.000)$ Passed

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(5605 \cdot 1.0, 1.5 \cdot 1522.2(2.503 + 0.0), \pi/4 \cdot 23.5 \cdot 20.55^2 \cdot 1.0^2) \\ &= 77.910 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= fwelds / (L_t(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 77.91 / (55.056(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 40.663 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Angle [Theta1]:

$$\begin{aligned} &= \cos^{-1}[D_x / R_{eff}] \\ &= \cos^{-1}[600.0 / 1151.5] \\ &= 1.023 \end{aligned}$$

Angle [Theta2]:

$$\begin{aligned} &= \cos^{-1}[(D_x + R_n) / R_{eff}] \\ &= \cos^{-1}[(600.0 + 20.55) / 1151.5] \\ &= 1.002 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 14.5 / \ln[1 + 14.5 / 20.55] \\ &= 27.158 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t / \sin[\theta]) \quad (\text{eqn. 4.5.73}) \\ &= 1.055 \cdot 27.158(17.262 + 19.5 / \sin[90.0]) \\ &= 10.5 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 55.6 \\ &= 680.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 20.55 \\ &= 251.7 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (680.9 + 251.7 + 10.5) / 1.055 \\ &= 894.3 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 894.257 / 13.65 - 1161.225 / 19.5) \\ &= 34.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t / R_{xs}] \\ &= 1650.863[19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:



= min(Pmax1, Pmax2)
= min(34.645, 27.722)
= 27.722 kgf/cm²

Average Primary Membrane Stress [σ_{Avg}]:
= (fN + fS + fY) / AT
= (10.529 + 680.896 + 251.662)/13.65
= 69.091 kgf/cm²

General Primary Membrane Stress [σ_{Circ}]:
= P • Rxs / teff
= 1.055 • 1161.225/19.5
= 62.8 kgf/cm²

Maximum Local Primary Membrane Stress [PL]:
= max(2 • σ_{Avg} - σ_{Circ} , σ_{Circ})
= max(2 • 69.091 - 62.802, 62.802)
= 75.4 kgf/cm²

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	75.38	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated
Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.020

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.296 • 1.0)/(16.0 - 1.5 - 0.0)
= 0.020

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 22.3156 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 246.8496 mm

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Input, Nozzle Desc: Noz K4

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		597.42	cm
-----------------------------------	--	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

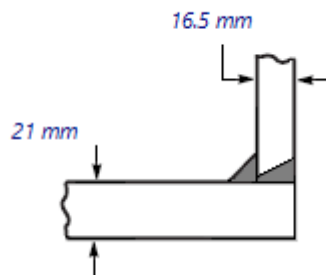
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K4

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4177 + 1.5 + 0.0 = 1.9177 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	17.977 mm
Min. required shell thickness, external p + c :	14.028 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.9, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2300.0/2 + 1.5$
 $= 1151.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:



= 68.800 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(26.9 \cdot 15.0)} \\ &= 20.087 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot frp) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + teff) \cdot teff})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot Leg^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + frn(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.694 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= teff / \ln[1 + teff/Reff] \\ &= 19.5 / \ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 23.454 \cdot 33.848(20.087 + 19.5) \\ &= 314.3 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Force acting on the Shell [fS]:} \\ &= 23.454 \cdot 1161.225 \cdot 68.8 \\ &= 18737.9 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Discontinuity Force from Internal Pressure [fY]:} \\ &= P \cdot Rxs \cdot Rnc \\ &= 23.454 \cdot 1161.225 \cdot 26.9 \\ &= 7326.3 \text{ kgf} \end{aligned}$$

$$\begin{aligned} \text{Area Resisting Internal Pressure [Ap]:} \\ &= (fS + fY + fN)/P \\ &= (18737.9 + 7326.3 + 314.3)/23.454 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Maximum Allowable Working Pressure Candidate [Pmax1]:} \\ &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2476.295 / (2 \cdot 1124.692/16.694 - 1161.225/19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Maximum Allowable Working Pressure Candidate [Pmax2]:} \\ &= S[t/Rxs] \\ &= 1650.863[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Maximum Allowable Working Pressure [Pmax]:} \\ &= \min(Pmax1, Pmax2) \\ &= \min(32.934, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Average Primary Membrane Stress [\sigma Avg]:} \\ &= (fN + fS + fY) / AT \\ &= (314.271 + 18737.93 + 7326.313)/16.694 \\ &= 1580.093 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{General Primary Membrane Stress [\sigma Circ]:} \\ &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1161.225/19.5 \\ &= 1396.7 \text{ kgf/cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Maximum Local Primary Membrane Stress [PL]:} \\ &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1580.093 - 1396.687, 1396.687) \\ &= 1763.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1763.50	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

$$\begin{aligned} \text{Discontinuity Force Factor [ky]:} \\ &= 1.000 \text{ For set-on Nozzles} \end{aligned}$$

$$\begin{aligned} \text{Weld Length of Nozzle to Shell Weld [Lt]:} \\ &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.9 + 15.0) \\ &= 65.816 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Weld Throat Dimensions, (0.7071 \cdot Leg Dimensions) [L41T, L42T, L43T]:} \\ &= 7.071, 0.000, 0.000, \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Required Weld Throat Dimensions :} \\ &tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed} \end{aligned}$$

$$\text{Weld Load Value [fwelds]:}$$



$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7326 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.294 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.294 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.196 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 1.055 \cdot 33.848(20.087 + 19.5) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / A_T - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / (t / R_{xs}) \\ &= 1650.863 [19.5 / 1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(32.934, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / A_T \\ &= (14.131 + 842.547 + 329.426) / 16.694 \\ &= 71.049 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1161.225 / 19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 71.049 - 62.802, 62.802) \\ &= 79.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	79.30	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.026

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.387 \cdot 1.0) / (16.5 - 1.5 - 0.0) \\ &= 0.026 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.7635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.7635 mm

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Input, Nozzle Desc: Noz K5

From: 70

Pressure for Reinforcement Calculations	P	23.454	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Cylindrical Shell	D	2300.00	mm
Design Length of Section	L	6941.6670	mm
Shell Finished (Minimum) Thickness	t	21.0000	mm
Shell Internal Corrosion Allowance	c	1.5000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		677.42	cm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

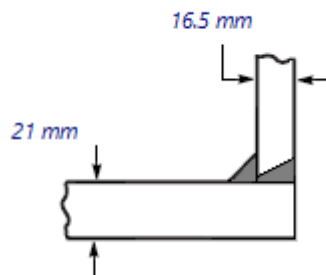
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K5

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.



Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
 $= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
 $= 0.5 \cdot 53.8 \cdot (\exp(23.454/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
 $= 0.4177 + 1.5 + 0.0 = 1.9177 \text{ mm Equation 4.3.1}$

Computed Maximum Allowable Working Pressure [MAWP]:
 $= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
 $= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
 $= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	17.977 mm
Min. required shell thickness, external p + c :	14.028 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
 $= \min[S_n/S, 1]$
 $= \min[1522.2/1650.9, 1]$
 $= 0.922$

Thickness of Nozzle [tn]:
 $= \text{thickness} - \text{corrosion allowance}$
 $= 16.5 - 1.5$
 $= 15.000 \text{ mm}$

Shell Diameter to Thickness ratio [D/t]:
 $= D_i/t$
 $= 2303.0/19.5$
 $= 118.103 \text{ must be less than } 400.$

Effective Pressure Radius [Reff]:
 $= D_i/2 + \text{corrosion allowance}$
 $= 2300.0/2 + 1.5$
 $= 1151.500 \text{ mm}$

Effective Length of Vessel Wall [LR]:
 $= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
 $= \min(\sqrt{1151.5 \cdot 19.5}, 2 \cdot 26.9)$
 $= 53.800 \text{ mm}$

Because the nozzle is set-on, add tn to the result [LR]:



= 68.800 mm

Thickness Limit Candidate [LH1]:

$$\begin{aligned} &= \min[1.5t, t_e] + \sqrt{(R_n \cdot t_n)} \\ &= \min[1.5 \cdot 19.5, 0.0] + \sqrt{(26.9 \cdot 15.0)} \\ &= 20.087 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH2]:

$$\begin{aligned} &= L_{pr1} \\ &= 200.0 \\ &= 200.000 \text{ mm} \end{aligned}$$

Thickness Limit Candidate [LH3]:

$$\begin{aligned} &= 8(t + t_e) \\ &= 8(19.5 + 0.0) \\ &= 156.000 \text{ mm} \end{aligned}$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min[LH1, LH2, LH3] \\ &= \min[20.087, 200.0, 156.0] \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A5 \cdot f_{rp}) / (LR - t_n)) \\ &= 19.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 19.500 \text{ mm} \end{aligned}$$

Parameter [λ]:

$$\begin{aligned} &= \min(12, (2 \cdot R_n + t_n) / (\sqrt{(D_i + t_{eff}) \cdot t_{eff}})) \\ &= \min(12, (2 \cdot 26.9 + 15.0) / (\sqrt{(2303.0 + 19.5) \cdot 19.5})) \\ &= 0.323 \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \cdot \max((\lambda/5)^{0.85}, 1) \\ &= 19.5 \cdot 68.8 \cdot \max((0.323/5)^{0.85}, 1) \\ &= 13.416 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\ &= 13.416 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 16.694 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n/R_n] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff}/R_{eff}] \\ &= 19.5 / \ln[1 + 19.5/1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:



$$\begin{aligned} &= P \cdot Rxn(LH + t) \\ &= 23.454 \cdot 33.848(20.087 + 19.5) \\ &= 314.3 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= 23.454 \cdot 1161.225 \cdot 68.8 \\ &= 18737.9 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= P \cdot Rxs \cdot Rnc \\ &= 23.454 \cdot 1161.225 \cdot 26.9 \\ &= 7326.3 \text{ kgf} \end{aligned}$$

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (18737.9 + 7326.3 + 314.3)/23.454 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= Sallow / (2 \cdot Ap/AT - Rxs/teff) \\ &= 2476.295 / (2 \cdot 1124.692/16.694 - 1161.225/19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= S[t/Rxs] \\ &= 1650.863[19.5/1161.225] \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(Pmax1, Pmax2) \\ &= \min(32.934, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σAvg]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (314.271 + 18737.93 + 7326.313)/16.694 \\ &= 1580.093 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σCirc]:

$$\begin{aligned} &= P \cdot Rxs / teff \\ &= 23.454 \cdot 1161.225/19.5 \\ &= 1396.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma Avg - \sigma Circ, \sigma Circ) \\ &= \max(2 \cdot 1580.093 - 1396.687, 1396.687) \\ &= 1763.5 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1763.50	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	27.72	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld [Lt]:

$$\begin{aligned} &= \pi/2 \cdot (Rn + tn) \\ &= \pi/2 \cdot (26.9 + 15.0) \\ &= 65.816 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$tc, L41T = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:



$$\begin{aligned} &= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2) \\ &= \min(7326 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.294 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= f_{welds} / (L_T(0.49 \cdot L_{41T} + 0.6 \cdot t_{w1} + 0.49 \cdot L_{43T})) \\ &= 133.294 / (65.816(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 58.196 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [R_{xn}]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [R_{xs}]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 19.5 / \ln[1 + 19.5 / 1151.5] \\ &= 1161.225 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [S_{allow}]:

$$\begin{aligned} &= 1.5 \cdot S \\ &= 1.5 \cdot 1650.863 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [f_N]:

$$\begin{aligned} &= P \cdot R_{xn}(L_H + t) \\ &= 1.055 \cdot 33.848(20.087 + 19.5) \\ &= 14.1 \text{ kgf} \end{aligned}$$

Force acting on the Shell [f_S]:

$$\begin{aligned} &= 1.055 \cdot 1161.225 \cdot 68.8 \\ &= 842.5 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [f_Y]:

$$\begin{aligned} &= P \cdot R_{xs} \cdot R_{nc} \\ &= 1.055 \cdot 1161.225 \cdot 26.9 \\ &= 329.4 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [A_p]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (842.5 + 329.4 + 14.1) / 1.055 \\ &= 1124.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max1}]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / t_{eff}) \\ &= 2476.295 / (2 \cdot 1124.692 / 16.694 - 1161.225 / 19.5) \\ &= 32.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [P_{max2}]:

$$\begin{aligned} &= S / t / R_{xs} \\ &= 1650.863 / 19.5 / 1161.225 \\ &= 27.7 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [P_{max}]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(32.934, 27.722) \\ &= 27.722 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (f_N + f_S + f_Y) / AT \\ &= (14.131 + 842.547 + 329.426) / 16.694 \\ &= 71.049 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / t_{eff} \\ &= 1.055 \cdot 1161.225 / 19.5 \\ &= 62.8 \text{ kgf/cm}^2 \end{aligned}$$



Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 71.049 - 62.802, 62.802) \\ &= 79.3 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	79.30	kgf/cm ²
Maximum Allowable External Pressure	Pmax	27.72	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated

Computed Minimum Design Metal Temperature	:	-104 °C
User entered Minimum Design Temperature	:	-29 °C
The Curve for this Material is	:	B

As the Stress Ratio < 0.240, the MDMT is	:	-104 °C
The computed stress ratio is	:	0.026

The computed thickness ratio [sr]:

$$\begin{aligned} &= (tr \cdot E^*) / (tnom - ci - cext) \\ &= (0.387 \cdot 1.0) / (16.5 - 1.5 - 0.0) \\ &= 0.026 \end{aligned}$$

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

The Drop for this Nozzle is : 0.7635 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 221.7635 mm

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Input, Nozzle Desc: Noz K1A

From: 80

Pressure for Reinforcement Calculations	P	23.486	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	650.0000	mm
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User Entered Minimum Design Metal Temperature		-28.89	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
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Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm

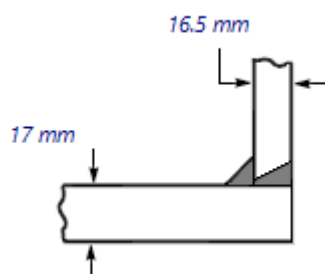
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K1A

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.





Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$
= $0.5 \cdot 53.8 \cdot (\exp(23.486/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$
= $0.4183 + 1.5 + 0.0 = 1.9183$ mm Equation 4.3.1

Computed Maximum Allowable Working Pressure [MAWP]:
= $S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$
= $1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0$
= $674.586 - 0.0 = 674.586$ kgf/cm²

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.592 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness :	6.300 mm
--	----------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:
= $\min[S_n/S, 1]$
= $\min[1522.2/1650.9, 1]$
= 0.922

Thickness of Nozzle [tn]:
= thickness - corrosion allowance
= $16.5 - 1.5$
= 15.000 mm

Effective Pressure Radius [Reff]:
= $0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$
= $0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2]$
= 2069.107 mm

Effective Length of Vessel Wall [LR]:
= $\min(\sqrt{Reff \cdot t}, 2 \cdot R_n)$
= $\min(\sqrt{2069.107 \cdot 15.5}, 2 \cdot 26.9)$
= 53.800 mm

Because the nozzle is set-on, add tn to the result [LR]:
= 68.800 mm

Intermediate value [Cn]:
= $\min(((t + t_e)/t_n)^{0.35}, 1)$
= $\min(((15.5 + 0.0)/15.0)^{0.35}, 1)$
= $\min(1.012, 1)$
= 1.000

Angle for non-radial nozzles in Elliptical Heads [theta]:
= $\text{atan}((h/R) \cdot (DR/(\sqrt{R^2 - DR^2})))$
= $\text{atan}((576.5/1151.5) \cdot (650.0/(\sqrt{1151.5^2 - 650.0^2})))$
= 0.330 radians

Intermediate value [Cp]:
= $\exp((0.35 \cdot d_i - X_o)/(16t))$
= $\exp((0.35 \cdot 2303.0 - 689.641)/(16 \cdot 15.5))$
= 1.599



Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1}) \\ &= \min(0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.0)}, 200.0) \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / (LR - t_n)) \\ &= 15.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 15.500 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 15.5 \cdot 68.8 \\ &= 10.664 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\ &= 10.664 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.942 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 15.5 / \ln[1 + 15.5 / 2069.107] \\ &= 2076.859 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.486 \cdot 33.848(20.087 + 15.5) \\ &= 282.9 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot LR) / 2 \\ &= (23.486 \cdot 2076.859 \cdot 68.8) / 2 \\ &= 16779.3 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (23.486 \cdot 2076.859 \cdot 28.557) / 2 \\ &= 6964.7 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (16779.3 + 6964.7 + 282.9) / 23.486 \\ &= 1023.0 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff})) \\ &= 2476.295 / (2 \cdot 1023.034/13.942 - 2076.859/(2 \cdot 15.5)) \\ &= 31.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot \text{S}/[\text{t}/\text{Rxs}] \\ &= 2 \cdot 1650.863[15.5/2076.859] \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(31.048, 24.641) \\ &= 24.641 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (282.901 + 16779.311 + 6964.741) / 13.942 \\ &= 1723.316 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff}) \\ &= 23.486 \cdot 2076.859 / (2 \cdot 15.5) \\ &= 1573.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1723.316 - 1573.456, 1573.456) \\ &= 1873.2 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1873.18	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.64	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((28.557 + 15.0)^2 + (26.9 + 15.0)^2)/2)} \\ &= 67.131 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(6965 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.5 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.476 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 133.476 / (67.131(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 57.135 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= \text{tn} / \ln[1 + \text{tn}/\text{Rn}] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$



$$= 15.5 / \ln[1 + 15.5 / 2069.107]$$
$$= 2076.859 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.863$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 33.848 (20.087 + 15.5)$$
$$= 12.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 68.8) / 2$$
$$= 753.5 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 28.557) / 2$$
$$= 312.7 \text{ kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (753.5 + 312.7 + 12.7) / 1.055$$
$$= 1023.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$
$$= 2476.295 / (2 \cdot 1023.034 / 13.942 - 2076.859 / (2 \cdot 15.5))$$
$$= 31.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$
$$= 2 \cdot 1650.863 [15.5 / 2076.859]$$
$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(31.048, 24.641)$$
$$= 24.641 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (12.703 + 753.451 + 312.742) / 13.942$$
$$= 77.383 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 1.055 \cdot 2076.859 / (2 \cdot 15.5)$$
$$= 70.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 77.383 - 70.654, 70.654)$$
$$= 84.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	84.11	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.64	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated



Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.388 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 14.2524 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.1376 mm

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Input, Nozzle Desc: Noz K1B

From: 80

Pressure for Reinforcement Calculations	P	23.569	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	650.0000	mm
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User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	16.5000	mm

Flange Material		SA-350 LF2	
Flange Type		Long Weld Neck	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	0.0000	mm

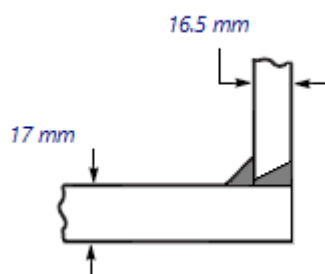
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: Noz K1B

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		2.000	in.
Actual Thickness Used in Calculation		0.650	in.





Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$\begin{aligned} &= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o \\ &= 0.5 \cdot 53.8 \cdot (\exp(23.569/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0 \\ &= 0.4197 + 1.5 + 0.0 = 1.9197 \text{ mm Equation 4.3.1} \end{aligned}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$\begin{aligned} &= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq} \\ &= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 15.0 + 53.8) / 53.8) - 0.0 \\ &= 674.586 - 0.0 = 674.586 \text{ kgf/cm}^2 \end{aligned}$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	6.300 mm
Min. required shell thickness, internal p + c :	21.654 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness	: 6.300 mm
--	------------

Minimum nozzle neck thickness 16.510 mm , is greater than required thickness 6.300 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$\begin{aligned} &= \min[S_n/S, 1] \\ &= \min[1522.2/1650.9, 1] \\ &= 0.922 \end{aligned}$$

Thickness of Nozzle [tn]:

$$\begin{aligned} &= \text{thickness} - \text{corrosion allowance} \\ &= 16.5 - 1.5 \\ &= 15.000 \text{ mm} \end{aligned}$$

Effective Pressure Radius [Reff]:

$$\begin{aligned} &= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2] \\ &= 0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2] \\ &= 2069.107 \text{ mm} \end{aligned}$$

Effective Length of Vessel Wall [LR]:

$$\begin{aligned} &= \min(\sqrt{Reff \cdot t}, 2 \cdot R_n) \\ &= \min(\sqrt{2069.107 \cdot 15.5}, 2 \cdot 26.9) \\ &= 53.800 \text{ mm} \end{aligned}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 68.800 \text{ mm}$$

Intermediate value [Cn]:

$$\begin{aligned} &= \min(((t + t_e)/t_n)^{0.35}, 1) \\ &= \min(((15.5 + 0.0)/15.0)^{0.35}, 1) \\ &= \min(1.012, 1) \\ &= 1.000 \end{aligned}$$

Angle for non-radial nozzles in Elliptical Heads [theta]:

$$\begin{aligned} &= \text{atan}((h/R) \cdot (DR/(\sqrt{R^2 - DR^2}))) \\ &= \text{atan}((576.5/1151.5) \cdot (650.0/(\sqrt{1151.5^2 - 650.0^2}))) \\ &= 0.330 \text{ radians} \end{aligned}$$

Intermediate value [Cp]:

$$\begin{aligned} &= \exp((0.35 \cdot d_i - X_o)/(16t)) \\ &= \exp((0.35 \cdot 2303.0 - 689.641)/(16 \cdot 15.5)) \\ &= 1.599 \end{aligned}$$



Effective Nozzle Wall Length Outside the Vessel [LH]:

$$\begin{aligned} &= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1}) \\ &= \min(0.0 + 1.0 \cdot \sqrt{(26.9 \cdot 15.0)}, 200.0) \\ &= 20.087 \text{ mm} \end{aligned}$$

Effective Vessel Thickness [teff]:

$$\begin{aligned} &= t + ((A_5 \cdot f_{rp}) / (LR - t_n)) \\ &= 15.5 + ((0.0 \cdot 1.0) / (68.8 - 15.0)) \\ &= 15.500 \text{ mm} \end{aligned}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$\begin{aligned} &= t \cdot LR \\ &= 15.5 \cdot 68.8 \\ &= 10.664 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$\begin{aligned} &= t_n \cdot LH \\ &= 15.0 \cdot 20.087 \\ &= 3.013 \text{ cm}^2 \end{aligned}$$

Area Contributed by the Outside Fillet Weld [A41]:

$$\begin{aligned} &= 0.5 \cdot \text{Leg}^2 \\ &= 0.5 \cdot 10.0^2 \\ &= 0.500 \text{ cm}^2 \end{aligned}$$

The total area contributed by A1 through A43 [AT]:

$$\begin{aligned} &= A1 + f_{rn}(A2 + A3) + A41 + A42 + A43 \\ &= 10.664 + 0.922(3.013 + 0.0) + 0.5 + 0.0 + 0.0 \\ &= 13.942 \text{ cm}^2 \end{aligned}$$

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= t_n / \ln[1 + t_n / R_n] \\ &= 15.0 / \ln[1 + 15.0 / 26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$\begin{aligned} &= t_{eff} / \ln[1 + t_{eff} / R_{eff}] \\ &= 15.5 / \ln[1 + 15.5 / 2069.107] \\ &= 2076.859 \text{ mm} \end{aligned}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$\begin{aligned} &= 1.5 \cdot S \cdot E \\ &= 1.5 \cdot 1650.863 \cdot 1.0 \\ &= 2476.3 \text{ kgf/cm}^2 \end{aligned}$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn}(LH + t) \\ &= 23.569 \cdot 33.848(20.087 + 15.5) \\ &= 283.9 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot LR) / 2 \\ &= (23.569 \cdot 2076.859 \cdot 68.8) / 2 \\ &= 16838.7 \text{ kgf} \end{aligned}$$

Discontinuity Force from Internal Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc}) / 2 \\ &= (23.569 \cdot 2076.859 \cdot 28.557) / 2 \\ &= 6989.4 \text{ kgf} \end{aligned}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting Internal Pressure [Ap]:

$$\begin{aligned} &= (f_S + f_Y + f_N) / P \\ &= (16838.7 + 6989.4 + 283.9) / 23.569 \\ &= 1023.0 \text{ cm}^2 \end{aligned}$$



Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= \text{Sallow} / (2 \cdot \text{Ap}/\text{AT} - \text{Rxs}/(2 \cdot \text{teff})) \\ &= 2476.295 / (2 \cdot 1023.034/13.942 - 2076.859/(2 \cdot 15.5)) \\ &= 31.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot \text{S}/[\text{t}/\text{Rxs}] \\ &= 2 \cdot 1650.863[15.5/2076.859] \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(\text{Pmax1}, \text{Pmax2}) \\ &= \min(31.048, 24.641) \\ &= 24.641 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (\text{fN} + \text{fS} + \text{fY}) / \text{AT} \\ &= (283.903 + 16838.727 + 6989.403) / 13.942 \\ &= 1729.418 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= \text{P} \cdot \text{Rxs} / (2 \cdot \text{teff}) \\ &= 23.569 \cdot 2076.859 / (2 \cdot 15.5) \\ &= 1579.0 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{\text{Avg}} - \sigma_{\text{Circ}}, \sigma_{\text{Circ}}) \\ &= \max(2 \cdot 1729.418 - 1579.027, 1579.027) \\ &= 1879.8 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	1879.81	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.64	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14

Discontinuity Force Factor [ky]:

$$= 1.000 \text{ For set-on Nozzles}$$

Weld Length of Nozzle to Shell Weld for non-radial Nozzles [LT]:

$$\begin{aligned} &= \pi/2 \cdot \sqrt{(((\text{Rnc} + \text{Tn})^2 + (\text{Rn} + \text{Tn})^2)/2)} \\ &= \pi/2 \cdot \sqrt{(((28.557 + 15.0)^2 + (26.9 + 15.0)^2)/2)} \\ &= 67.131 \text{ mm} \end{aligned}$$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

$$= 7.071, 0.000, 0.000, \text{ mm}$$

Required Weld Throat Dimensions :

$$\text{tc, L41T} = 7.071, \text{ must be } \geq \min(0.7 \cdot 15.000, 6.000) \text{ Passed}$$

Weld Load Value [fwelds]:

$$\begin{aligned} &= \min(\text{fy} \cdot \text{ky}, 1.5 \cdot \text{Sn}(\text{A2} + \text{A3}), \pi/4 \cdot \text{P} \cdot \text{Rn}^2 \cdot \text{ky}^2) \\ &= \min(6989 \cdot 1.0, 1.5 \cdot 1522.2(3.013 + 0.0), \pi/4 \cdot 23.6 \cdot 26.9^2 \cdot 1.0^2) \\ &= 133.949 \text{ kgf} \end{aligned}$$

Weld Stress Value [τ]:

$$\begin{aligned} &= \text{fwelds} / (\text{LT}(0.49 \cdot \text{L41T} + 0.6 \cdot \text{tw1} + 0.49 \cdot \text{L43T})) \\ &= 133.949 / (67.131(0.49 \cdot 7.071 + 0.6 \cdot 0.025 + 0.49 \cdot 0.0)) \\ &= 57.337 < \text{or} = \text{to } 1650.863 \text{ Weld Stress Passed} \end{aligned}$$

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$$\begin{aligned} &= \text{tn} / \ln[1 + \text{tn}/\text{Rn}] \\ &= 15.0 / \ln[1 + 15.0/26.9] \\ &= 33.848 \text{ mm} \end{aligned}$$

Shell Radius for Force Calculation [Rxs]:

$$= \text{teff} / \ln[1 + \text{teff}/\text{Reff}]$$



$$= 15.5 / \ln[1 + 15.5 / 2069.107]$$
$$= 2076.859 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S$$
$$= 1.5 \cdot 1650.863$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 1.055 \cdot 33.848 (20.087 + 15.5)$$
$$= 12.7 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 68.8) / 2$$
$$= 753.5 \text{ kgf}$$

Discontinuity Force from External Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (1.055 \cdot 2076.859 \cdot 28.557) / 2$$
$$= 312.7 \text{ kgf}$$

Note: Rnc in the above formula is divided by sin(theta).

Area Resisting External Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (753.5 + 312.7 + 12.7) / 1.055$$
$$= 1023.0 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= Sallow / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$
$$= 2476.295 / (2 \cdot 1023.034 / 13.942 - 2076.859 / (2 \cdot 15.5))$$
$$= 31.0 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S[t / R_{xs}]$$
$$= 2 \cdot 1650.863 [15.5 / 2076.859]$$
$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(31.048, 24.641)$$
$$= 24.641 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σAvg]:

$$= (fN + fS + fY) / AT$$
$$= (12.703 + 753.451 + 312.742) / 13.942$$
$$= 77.383 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σCirc]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 1.055 \cdot 2076.859 / (2 \cdot 15.5)$$
$$= 70.7 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 77.383 - 70.654, 70.654)$$
$$= 84.1 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	84.11	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.64	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle MDMT per 3.7, (Nozzle to Shell/Head Weld), Curve: B

This element is Not Post Weld Heat Treated



Computed Minimum Design Metal Temperature : -104 °C
User entered Minimum Design Temperature : -29 °C
The Curve for this Material is : B

As the Stress Ratio < 0.240, the MDMT is : -104 °C
The computed stress ratio is : 0.026

The computed thickness ratio [sr]:
= (tr • E*)/(tnom - ci - cext)
= (0.389 • 1.0)/(16.5 - 1.5 - 0.0)
= 0.026

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including temperature reduction per 3.11:
Flange MDMT with Temp reduction : -48 °C

The Drop for this Nozzle is : 14.2524 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 232.1376 mm

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Input, Nozzle Desc: MH1

From: 80

Pressure for Reinforcement Calculations	P	23.528	kgf/cm ²
Temperature for Internal Pressure	Temp	80	°C
Design External Pressure	Pext	1.05	kgf/cm ²
Temperature for External Pressure	Tempex	37	°C

Parent Material		SA-516 70	+N+IT
Parent Allowable Stress at Temperature	Sv	1650.86	kgf/cm ²
Parent Allowable Stress At Ambient	Sva	1784.44	kgf/cm ²

Inside Diameter of Elliptical Head	D	2300.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	17.0000	mm
Head Internal Corrosion Allowance	c	1.5000	mm
Head External Corrosion Allowance	co	0.0000	mm

Distance from Head Centerline	L1	0.0000	mm
-------------------------------	----	--------	----

User Entered Minimum Design Metal Temperature		-28.89	°C
---	--	--------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6	+IT
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	1522.22	kgf/cm ²
Allowable Stress At Ambient	Sna	1641.66	kgf/cm ²

Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		80	

Flange Material		SA-350 LF2	+IT
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	1.5000	mm
---------------------	-----	--------	----

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	21.0000	mm

Pad Material		SA-516 70	+IT
Pad Allowable Stress at Temperature	Sp	1650.40	kgf/cm ²
Pad Allowable Stress At Ambient	Spa	1784.41	kgf/cm ²
Diameter of Pad along vessel surface	Dp	809.6000	mm
Thickness of Pad	te	12.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm
Reinforcing Pad Width		100.0000	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

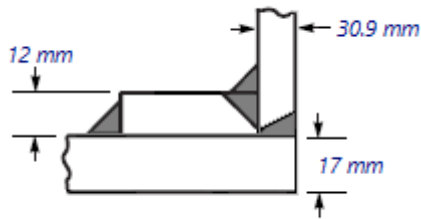
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: MH1

ASME Code, Section VIII, Div. 2, Section 4.5.1 - 4.5.14

Actual Inside Diameter Used in Calculation		21.564	in.
Actual Thickness Used in Calculation		1.218	in.



Nozzle input data check completed without errors.

Cylindrical Shell Calculation - Section 4.3.3.1

Computed Minimum Required Thickness [t]:

$$= 0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + c_i + c_o$$

$$= 0.5 \cdot 550.726 \cdot (\exp(23.528/(1522.219 \cdot 1.0)) - 1) + 1.5 + 0.0$$

$$= 4.2891 + 1.5 + 0.0 = 5.7891 \text{ mm Equation 4.3.1}$$

Computed Maximum Allowable Working Pressure [MAWP]:

$$= S \cdot E \cdot \ln((2 \cdot t + D) / D) - P_{liq}$$

$$= 1522.219 \cdot 1.0 \cdot \ln((2 \cdot 29.437 + 550.726) / 550.726) - 0.0$$

$$= 154.606 - 0.0 = 154.6062 \text{ kgf/cm}^2$$

Nozzle Minimum Thickness per 4.5.4:

Min. required nozzle neck thickness, table 4.5.2 + c :	9.840 mm
Min. required shell thickness, internal p + c :	21.623 mm
Min. required shell thickness, external p + c :	8.416 mm

Nozzle neck minimum required thickness	: 9.840 mm
--	------------

Minimum nozzle neck thickness 27.076 mm , is greater than required thickness 9.840 mm

Nozzle Calculations per Section 4.5: Internal Pressure Case:

Nozzle Material Factor [frn]:

$$= \min[S_n/S, 1]$$

$$= \min[1522.2/1650.9, 1]$$

$$= 0.922$$

Pad Material Factor [frp]:

$$= \min[S_p/S, 1]$$

$$= \min[1650.4/1650.9, 1]$$

$$= 1.000$$

Thickness of Nozzle [tn]:

$$= \text{thickness} - \text{corrosion allowance}$$

$$= 30.937 - 1.5$$

$$= 29.437 \text{ mm}$$

Effective Pressure Radius [Reff]:

$$= 0.9/6 \cdot D_i [2 + (D_i/(2h))^2]$$

$$= 0.9/6 \cdot 2303.0 [2 + (2303.0/(2 \cdot 576.5))^2]$$

$$= 2069.107 \text{ mm}$$

Effective Length of Vessel Wall [LR]:

For Nozzles with reinforcing pads, L_r is $\min[LR1, LR2, LR3]$:

Limit [LR1]:

$$= \sqrt{((Reff \cdot t) + W)}$$

$$= \sqrt{(2069.107 \cdot 15.5) + 100.0}$$

$$= 279.084 \text{ mm}$$

Limit [LR2]:

$$= \sqrt{((Reff + t) \cdot (t + t_e))}$$

$$= \sqrt{((2069.107 + 15.5) \cdot (15.5 + 12.0))}$$



$$= 239.430 \text{ mm}$$

Limit [LR3]:

$$= 2 \cdot R_n$$

$$= 2 \cdot 275.363$$

$$= 550.726 \text{ mm}$$

Final value of [LR]:

$$= \min(LR1, LR2, LR3)$$

$$= \min(279.084, 239.43, 550.726)$$

$$= 239.430 \text{ mm}$$

Because the nozzle is set-on, add tn to the result [LR]:

$$= 268.867 \text{ mm}$$

Intermediate value [Cn]:

$$= \min(((t + t_e)/t_n)^{0.35}, 1)$$

$$= \min(((15.5 + 12.0)/29.437)^{0.35}, 1)$$

$$= \min(0.976, 1)$$

$$= 0.976$$

Intermediate value [Cp]:

$$= \exp((0.35 \cdot d_i - X_o) / (16t))$$

$$= \exp((0.35 \cdot 2303.0 - 304.8) / (16 \cdot 15.5))$$

$$= 7.547$$

Effective Nozzle Wall Length Outside the Vessel [LH]:

$$= \min(t_e + F_p \cdot \sqrt{(R_n \cdot t_n)}, L_{pr1})$$

$$= \min(12.0 + 0.976 \cdot \sqrt{(275.363 \cdot 29.437)}, 300.0)$$

$$= 99.913 \text{ mm}$$

Effective Vessel Thickness [teff]:

$$= t + ((A5 \cdot f_{rp}) / (LR - t_n))$$

$$= 15.5 + ((12.0 \cdot 1) / (268.867 - 29.437))$$

$$= 20.511 \text{ mm}$$

Compute Areas A1-A43 (No Pad) or A1-A5 (With Pad) :

Area Contributed by the Vessel Wall [A1]:

$$= t \cdot LR$$

$$= 15.5 \cdot 268.867$$

$$= 41.674 \text{ cm}^2$$

Area Contributed by the Nozzle Outside the Vessel Wall [A2]:

$$= t_n \cdot LH$$

$$= 29.437 \cdot 99.913$$

$$= 29.412 \text{ cm}^2$$

Area Contributed by the Pad Fillet Weld [A42]:

$$= 0.5 \cdot \text{Leg}_{42}^2$$

$$= 0.5 \cdot 12.0^2$$

$$= 0.720 \text{ cm}^2$$

Area Contributed by the Outside Fillet Weld [A41]:

$$= 0.5 \cdot \text{Leg}_{41}^2$$

$$= 0.5 \cdot 10.0^2$$

$$= 0.500 \text{ cm}^2$$

Area Contributed by the Reinforcing Pad [A5]:

$$= \min(W \cdot t_e, (LR - t_n) \cdot t_e)$$

$$= \min(100.0 \cdot 12.0, (268.867 - 29.437) \cdot 12.0)$$

$$= 12.000 \text{ cm}^2$$

The total area contributed by A1 through A5 [AT]:

$$= A1 + \text{frn}(A2 + A3) + A41 + A42 + A43 + \text{frp}(A5)$$

$$= 41.674 + 0.922(29.412 + 0.0) + 0.5 + 0.72 + 0.0 + 1 \cdot (12.0)$$

$$= 82.011 \text{ cm}^2$$

Nozzle Radius for Force Calculation [Rxn]:

$$= t_n / \ln[1 + t_n/R_n]$$



$$= 29.437 / \ln[1 + 29.437/275.363]$$
$$= 289.832 \text{ mm}$$

Shell Radius for Force Calculation [Rxs]:

$$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$$
$$= 20.511 / \ln[1 + 20.511/2069.107]$$
$$= 2079.347 \text{ mm}$$

Allowable Local Primary Membrane Stress [Sallow]:

$$= 1.5 \cdot S \cdot E$$
$$= 1.5 \cdot 1650.863 \cdot 1.0$$
$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$= P \cdot R_{xn} (LH + t)$$
$$= 23.528 \cdot 289.832 (99.913 + 15.5)$$
$$= 7870.1 \text{ kgf}$$

Force acting on the Shell [fS]:

$$= (P \cdot R_{xs} \cdot LR) / 2$$
$$= (23.528 \cdot 2079.347 \cdot 268.867) / 2$$
$$= 65767.5 \text{ kgf}$$

Discontinuity Force from Internal Pressure [fY]:

$$= (P \cdot R_{xs} \cdot R_{nc}) / 2$$
$$= (23.528 \cdot 2079.347 \cdot 275.363) / 2$$
$$= 67356.4 \text{ kgf}$$

Area Resisting Internal Pressure [Ap]:

$$= (fS + fY + fN) / P$$
$$= (65767.5 + 67356.4 + 7870.1) / 23.528$$
$$= 5992.7 \text{ cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff}))$$
$$= 2476.295 / (2 \cdot 5992.719 / 82.011 - 2079.347 / (2 \cdot 20.511))$$
$$= 25.9 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$= 2 \cdot S / (t / R_{xs})$$
$$= 2 \cdot 1650.863 [15.5 / 2079.347]$$
$$= 24.6 \text{ kgf/cm}^2$$

Maximum Allowable Working Pressure [Pmax]:

$$= \min(P_{max1}, P_{max2})$$
$$= \min(25.942, 24.612)$$
$$= 24.612 \text{ kgf/cm}^2$$

Average Primary Membrane Stress [σ_{Avg}]:

$$= (fN + fS + fY) / AT$$
$$= (7870.074 + 65767.531 + 67356.438) / 82.011$$
$$= 1719.216 \text{ kgf/cm}^2$$

General Primary Membrane Stress [σ_{Circ}]:

$$= P \cdot R_{xs} / (2 \cdot t_{eff})$$
$$= 23.528 \cdot 2079.347 / (2 \cdot 20.511)$$
$$= 1192.6 \text{ kgf/cm}^2$$

Maximum Local Primary Membrane Stress [PL]:

$$= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ})$$
$$= \max(2 \cdot 1719.216 - 1192.609, 1192.609)$$
$$= 2245.8 \text{ kgf/cm}^2$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	2245.82	kgf/cm ²
Maximum Allowable Working Pressure	Pmax	24.61	kgf/cm ²

Strength of Nozzle Attachment Welds per 4.5.14



Discontinuity Force Factor [ky]:

= 1.000 For set-on Nozzles

Weld Length of Nozzle to Shell Weld [Lt]:

$= \pi/2 \cdot (R_n + t_n)$
 $= \pi/2 \cdot (275.363 + 29.437)$
 $= 478.779 \text{ mm}$

Weld Length of Pad to Shell Weld [LtP]:

$= \pi/2 \cdot (R_n + t_n + W)$
 $= \pi/2 \cdot (275.363 + 29.437 + 100.0)$
 $= 635.858 \text{ mm}$

Weld Throat Dimensions, (0.7071•Leg Dimensions) [L41T, L42T, L43T]:

= 7.071, 8.485, 0.000, mm

Required Weld Throat Dimensions :

tc, L41T = 7.071, must be $\geq \min(0.7 \cdot 29.437, 6.000)$ Passed
tf1, L42T = 8.485, must be $\geq \min(0.6 \cdot 12.000, 0.6 \cdot 15.500)$ Passed

Weld Load Value [fwlds]:

$= \min(f_y \cdot k_y, 1.5 \cdot S_n(A_2 + A_3), \pi/4 \cdot P \cdot R_n^2 \cdot k_y^2)$
 $= \min(67356 \cdot 1.0, 1.5 \cdot 1522.2(30.036 + 0.0), \pi/4 \cdot 23.5 \cdot 275.36^2 \cdot 1.0^2)$
 $= 14011.261 \text{ kgf}$

Discontinuity Force [fws]:

$= fwlds \cdot t \cdot S / (t \cdot S + t_e \cdot S_p)$
 $= 14011.3 \cdot 15.5 \cdot 1651 / (15.5 \cdot 1651 + 12.0 \cdot 1650)$
 $= 7898.214 \text{ kgf}$

Discontinuity Force [fwp]:

$= fwlds \cdot t_e \cdot S_p / (t \cdot S + t_e \cdot S_p)$
 $= 14011.3 \cdot 12.0 \cdot 1650 / (15.5 \cdot 1651 + 12.0 \cdot 1650)$
 $= 6113.046 \text{ kgf}$

Shear Stress [τ_1]:

$= fws / (L_t \cdot (0.6 \cdot tw_1 + 0.49 \cdot L_{43T}))$
 $= 7898.214 / (478.779 \cdot (0.6 \cdot 15.5 + 0.49 \cdot 0.0))$
 $= 177.383 \text{ kgf/cm}^2$

Shear Stress [τ_2]:

$= fwp / (L_t \cdot (0.6 \cdot tw_2 + 0.49 \cdot L_{41T}))$
 $= 6113.046 / (478.779 \cdot (0.6 \cdot 12.0 + 0.49 \cdot 7.071))$
 $= 119.721 \text{ kgf/cm}^2$

Shear Stress [τ_3]:

$= fwp / (L_{tP} \cdot (0.49 \cdot L_{42T}))$
 $= 6113.046 / (635.858 \cdot (0.49 \cdot 8.485))$
 $= 231.228 \text{ kgf/cm}^2$

Maximum Shear Stress in the Welds:

$= \max(\tau_1, \tau_2, \tau_3)$
 $= \max(177.383, 119.721, 231.228)$
 $= 231.2$ must be less than or equal to 1650.9 kgf/cm²

Nozzle Calculations per Section 4.5: External Pressure Case:

Nozzle Radius for Force Calculation [Rxn]:

$= t_n / \ln[1 + t_n/R_n]$
 $= 29.437 / \ln[1 + 29.437/275.363]$
 $= 289.832 \text{ mm}$

Shell Radius for Force Calculation [Rxs]:

$= t_{eff} / \ln[1 + t_{eff}/R_{eff}]$
 $= 20.511 / \ln[1 + 20.511/2069.107]$
 $= 2079.347 \text{ mm}$

Allowable Local Primary Membrane Stress [Sallow]:

$= 1.5 \cdot S$
 $= 1.5 \cdot 1650.863$



$$= 2476.3 \text{ kgf/cm}^2$$

Force acting on the Nozzle [fN]:

$$\begin{aligned} &= P \cdot R_{xn} (LH + t) \\ &= 1.055 \cdot 289.832 (99.913 + 15.5) \\ &= 352.8 \text{ kgf} \end{aligned}$$

Force acting on the Shell [fS]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot LR)/2 \\ &= (1.055 \cdot 2079.347 \cdot 268.867)/2 \\ &= 2948.0 \text{ kgf} \end{aligned}$$

Discontinuity Force from External Pressure [fY]:

$$\begin{aligned} &= (P \cdot R_{xs} \cdot R_{nc})/2 \\ &= (1.055 \cdot 2079.347 \cdot 275.363)/2 \\ &= 3019.2 \text{ kgf} \end{aligned}$$

Area Resisting External Pressure [Ap]:

$$\begin{aligned} &= (fS + fY + fN)/P \\ &= (2948.0 + 3019.2 + 352.8)/1.055 \\ &= 5992.7 \text{ cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax1]:

$$\begin{aligned} &= S_{allow} / (2 \cdot A_p / AT - R_{xs} / (2 \cdot t_{eff})) \\ &= 2476.295 / (2 \cdot 5992.719 / 82.011 - 2079.347 / (2 \cdot 20.511)) \\ &= 25.9 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure Candidate [Pmax2]:

$$\begin{aligned} &= 2 \cdot S / t / R_{xs} \\ &= 2 \cdot 1650.863 / (15.5 / 2079.347) \\ &= 24.6 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Allowable Working Pressure [Pmax]:

$$\begin{aligned} &= \min(P_{max1}, P_{max2}) \\ &= \min(25.942, 24.612) \\ &= 24.612 \text{ kgf/cm}^2 \end{aligned}$$

Average Primary Membrane Stress [σ_{Avg}]:

$$\begin{aligned} &= (fN + fS + fY) / AT \\ &= (352.77 + 2947.976 + 3019.197) / 82.011 \\ &= 77.062 \text{ kgf/cm}^2 \end{aligned}$$

General Primary Membrane Stress [σ_{Circ}]:

$$\begin{aligned} &= P \cdot R_{xs} / (2 \cdot t_{eff}) \\ &= 1.055 \cdot 2079.347 / (2 \cdot 20.511) \\ &= 53.5 \text{ kgf/cm}^2 \end{aligned}$$

Maximum Local Primary Membrane Stress [PL]:

$$\begin{aligned} &= \max(2 \cdot \sigma_{Avg} - \sigma_{Circ}, \sigma_{Circ}) \\ &= \max(2 \cdot 77.062 - 53.458, 53.458) \\ &= 100.7 \text{ kgf/cm}^2 \end{aligned}$$

Summary of Nozzle Pressure/Stress Results:

Allowed Local Primary Membrane Stress	Sallow	2476.29	kgf/cm ²
Local Primary Membrane Stress	PL	100.67	kgf/cm ²
Maximum Allowable External Pressure	Pmax	24.61	kgf/cm ²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.



Impact Test Temperature provided per Specification -46 °C

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld at Pad OD for pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle-Shell/Head Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Gov. MDMT of the Nozzle : -46 °C

Gov. MDMT of the Reinforcement Pad : -29 °C

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -85 °C

The Drop for this Nozzle is : 22.4143 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 339.4142 mm

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Input Echo, Tubesheet Number 1, Description: U-TUBE SHEET

Shell Data:

Main Shell Description: SH.1

Shell Maximum Design Pressure	Psd,max	23.45	kgf/cm ²
Shell Maximum Operating Pressure	Psox,max	23.45	kgf/cm ²
Shell Minimum Operating Pressure	Psox,min	0.00	kgf/cm ²
Shell Thickness	ts	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.5000	mm
Shell External Corrosion Allowance	casext	0.0000	mm
Inside Diameter of Shell	Ds	1200.000	mm
Shell Temperature for Internal Pressure	Ts	80.00	°C
Shell Material		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Shell Material UNS Number		K02700	
Shell Allowable Stress at Temperature	Ss	1650.41	kgf/cm ²
Shell Allowable Stress at Ambient		1784.41	kgf/cm ²

Channel Data:

Channel Description: CH. SHELL

Channel Type:		Cylinder	
Channel Maximum Design Pressure	Ptd,max	163.16	kgf/cm ²
Channel Maximum Operating Pressure	Ptox,max	163.16	kgf/cm ²
Channel Minimum Operating Pressure	Ptox,min	0.00	kgf/cm ²
Channel Thickness	tc	68.0000	mm
Channel Internal Corrosion Allowance	cac	1.5000	mm
Channel External Corrosion Allowance	cacext	0.0000	mm
Inside Diameter of Channel	Dc	1200.000	mm
Channel Design Temperature	TEMPC	80.00	°C
Channel Flange Matl		SA-516 70	

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Channel Material UNS Number		K02700	
Channel Allowable Stress at Temperature	Sc	1650.41	kgf/cm ²
Channel Allowable Stress at Ambient		1784.41	kgf/cm ²

Tube Data:

Number of Tube Holes	Nt	1066	
Tube Wall Thickness	et	2.5000	mm
Tube Outside Diameter	D	25.0000	mm
Total Straight Tube Length	Lt	7997.00	mm
Straight Tube Length (bet. inner tubsht faces) L		7687.00	mm
Design Temperature of the Tubes		80.00	°C
Tube Material		SA-334 1	
Tube Material UNS Number		K03008	
Is this a Welded Tube		No	
Tube Material Specification used		Smls. tube	
Tube Allowable Stress at Temperature		1308.09	kgf/cm ²
Tube Allowable Stress At Ambient		1407.13	kgf/cm ²
Tube Yield Stress At design Temperature	Syt	1962.14	kgf/cm ²
Tube Pitch (Center to Center Spacing)	P	32.0000	mm
Tube Layout Pattern		Triangular	
Fillet Weld Leg	af	2.5000	mm
Groove Weld Leg	ag	2.5000	mm
Tube-Tubesheet Joint Weld Type		Full Strength	
Method for Tube-Tubesheet Jt. Allowable		4.21.2	
Is the Tube Inset		No	
Radius to Outermost Tube Hole Center	ro	584.500	mm
Largest Center-to-Center Tube Distance	U1	47.0000	mm
Length of Expanded Portion of Tube	ltx	0.0000	mm
Tube-side pass partition groove depth	hg	8.0000	mm

Tubesheet Data:



Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 80.00 °C
 Tubesheet Material SA-350 LF2

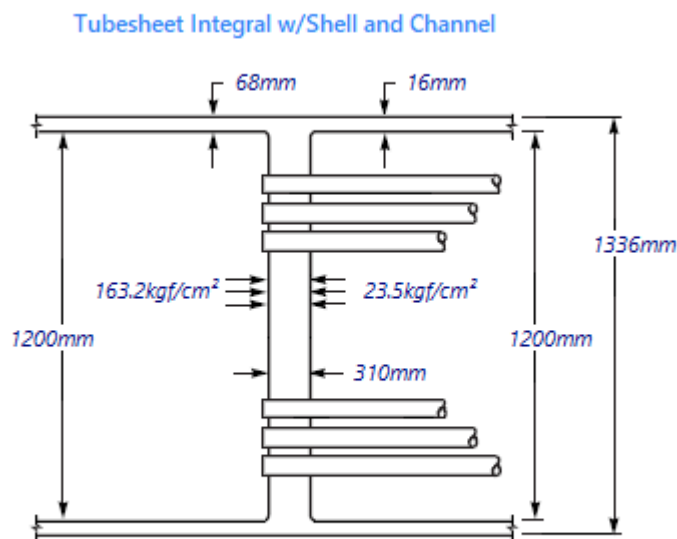
See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Tubesheet Material UNS Number K03011
 Tubesheet Allowable Stress at Temperature S 1568.83 kgf/cm²
 Tubesheet Allowable Stress at Ambient Tt 1682.44 kgf/cm²
 Thickness of Tubesheet h 310.0000 mm
 Tubesheet Corr. Allowance (Shell side) Cats 1.5000 mm
 Tubesheet Corr. Allowance (Channel side) Catc 1.5000 mm
 Tubesheet Outside Diameter A 1336.000 mm

Area of the Untubed Lanes AL 549.4 cm²

Junction Stress Reduction option Analyze as Simply Supported
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No



Configuration a:

ASME Division 2 Tubesheet Results per Part 4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²



TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	80.0 °C	0.20326E+07 kgf/cm ²
Elastic Mod. at Ambient Temperature	21.1 °C	0.20670E+07 kgf/cm ²

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Computed Minimum Required Thickness [t]:
= $0.5 \cdot D \cdot (\exp(P/(S \cdot E)) - 1) + ci$ Equation 4.3.1
= $0.5 \cdot 20.0 \cdot (\exp(163.158/(1308.1 \cdot 1.0)) - 1) + 0$
= 1.3284 + 0
= 1.3284 mm

Tube Required Thickness under External Pressure (Shellside pressure) :

Determine Parameter [Mx]:

= $L / \sqrt{Ro \cdot t}$
= $8271.5 / \sqrt{12.5 \cdot 2.5}$
= 1479.651

Compute the Factor [Ch]:

Since $Mx > 2 \cdot (Do/t)^{0.94}$, $Ch = 0.55 / (Do/t)$
= $0.55 / (25.0/2.5)$
= 0.0550

Elastic Hoop Compressive Membrane Failure Stress [Fhe]:

= $1.6 \cdot Ch \cdot Ey(t/Do)$
= $1.6 \cdot 0.055 \cdot 2032634 (2.5/25.0)$
= 17887.176 kgf/cm²

Ratio of Elastic Hoop Compressive Membrane Failure Stress to Yield [Fr]:

= Fhe / Fy
= $17887.2/1962.1$
= 9.116

Predicted Buckling Stress [Fic]:

Engineering Yield to Tensile Stress ratio [R]:

= $\sigma_{ys} / \sigma_{uts}$
= $1962.1/3866.9$
= 0.507

Ratio [AE]:

= Fhe / E
= $17887.2/2032634$
= 0.008800

Factor [m1]:

= $(\ln(r) + (\epsilon_{psp} - \epsilon_{sys})) / \ln(\ln(1 + \epsilon_{psp}) / \ln(1 + \epsilon_{sys}))$
= $(\ln(0.507) + (0.00002 - 0.002)) / \ln(\ln(1 + 0.00002) / \ln(1 + 0.002))$
= 0.148

Curve Fitting Constant for the Elastic Region of the Stress-Strain Curve [A1]:

= $\sigma_{ys}(1 + \epsilon_{sys}) / \ln(1 + \epsilon_{sys})^{m1}$
= $1962.1 (1 + 0.002) / \ln(1 + 0.002)^{0.147778}$
= 4926.163 kgf/cm²

Curve Fitting Exponent for, Ferritic Steel [m2]:

= $m2f(m2f^2 - R)$
= $0.6 (1.0 - 0.507)$
= 0.296

Curve Fitting Constant [A2]:

= $(\sigma_{uts} \cdot \exp(m2)) / (m2^{m2})$
= $(3866.9 \cdot \exp(0.296)) / (0.296^{0.296})$
= 7450.251 kgf/cm²

Material Parameter for Stress-Strain Curve Model [K]:

= $1.5 \cdot R1.5 - 0.5 \cdot R2.5 - R3.5$



$$= 1.5 \cdot 0.5071.5 - 0.5 \cdot 0.5072.5 - 0.5073.5$$

$$= 0.357$$

Compute the Allowable Stress per 4.4.3

	Fic	H	D1	D2	D3	D4	Tangent Modulus	Iteration Diff
1	1933	-2.084	0.312E-05	0.287E-05	0.912E-05	-0.792E-05	0.130E+06	0.605E-02
2	966	-4.924	0.573E-07	0.573E-07	0.175E-05	-0.175E-05	0.165E+07	0.821E-02
3	1450	-3.504	0.594E-06	0.592E-06	0.459E-05	-0.456E-05	0.586E+06	0.632E-02
4	1691	-2.794	0.145E-05	0.142E-05	0.663E-05	-0.644E-05	0.282E+06	0.279E-02
5	1812	-2.439	0.215E-05	0.207E-05	0.782E-05	-0.733E-05	0.192E+06	0.624E-03
6	1752	-2.617	0.177E-05	0.172E-05	0.721E-05	-0.690E-05	0.233E+06	0.128E-02
7	1782	-2.528	0.195E-05	0.189E-05	0.751E-05	-0.712E-05	0.212E+06	0.382E-03
8	1797	-2.484	0.205E-05	0.198E-05	0.766E-05	-0.723E-05	0.202E+06	0.107E-03
9	1789	-2.506	0.200E-05	0.193E-05	0.759E-05	-0.717E-05	0.207E+06	0.141E-03
10	1793	-2.495	0.203E-05	0.196E-05	0.762E-05	-0.720E-05	0.204E+06	0.181E-04
11	1795	-2.489	0.204E-05	0.197E-05	0.764E-05	-0.721E-05	0.203E+06	0.440E-04
12	1794	-2.492	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.129E-04
13	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.720E-05	0.204E+06	0.264E-05
14	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.512E-05
15	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.721E-05	0.204E+06	0.124E-05
16	1794	-2.493	0.203E-05	0.196E-05	0.763E-05	-0.720E-05	0.204E+06	0.699E-06

Factor of Safety [FS]:

Since $Fic > 0.55 F_y$ and $Fic < F_y$

$$= 2.407 - 0.741 \cdot Fic / F_y$$

$$= 2.407 - 0.741 \cdot 1794.25/1962.14$$

$$= 1.729$$

Allowable Circumferential Compressive Stress [Fha]:

$$= Fic/FS$$

$$= 1794.252/1.729$$

$$= 1037.499 \text{ kgf/cm}^2$$

Allowable External Pressure at the given Thickness [Pa]:

$$= 2 \cdot Fha(t / Do)$$

$$= 2 \cdot 1037.499 (2.5/25.0)$$

$$= 207.500 \text{ kgf/cm}^2$$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	1.3284 mm
Allowable Internal Pressure at Corroded thickness	291.89 kgf/cm ²
Required Internal Design Pressure	163.16 kgf/cm ²
Allowable External Pressure at Corroded thickness	207.50 kgf/cm ²
Required External Design Pressure	23.45 kgf/cm ²
Required Thickness due to Shell Side pressure	0.6257 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

Note:

The Tubesheet stress reduction option was Simply Supported. Performing the 1st pass by including the stiffness of integral shell/channel to ensure that the junction stress is < SPS.

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 584.5 + 25.0$$

$$= 1194.000 \text{ mm}$$



Basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (32.0 - 25.0) / 32.0$$

$$= 0.219$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ_s]:

$$= D_s / D_o \text{ (Configurations a, b, c)}$$

$$= 1200.0 / 1194.0$$

$$= 1.0050$$

Ratio [ρ_c]:

$$= D_c / D_o \text{ (Configurations a, e, f)}$$

$$= 1200.0 / 1194.0$$

$$= 1.0050$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$= l_{tx} / h$$

$$= 0 / 310.0$$

$$= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d^*]:

$$= \max(dt - 2t_t (E_t/E) \cdot (St/S) \cdot (\rho); dt - 2t_t)$$

$$= \max(25.0 - 2 \cdot 2.5 (2032634/2032634) \cdot (1308/1568) \cdot (0); 25.0 - 2 \cdot 2.5)$$

$$= 25.0000 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{(1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2))}$$

$$= 32.0 / \sqrt{(1 - 4 \cdot \min(549.43; 4 \cdot 1194.0 \cdot 32.0) / (\pi \cdot 1194.0^2))}$$

$$= 32.8152 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (32.8152 - 25.0) / 32.8152$$

$$= 0.2382$$

$$h/p = 9.687500; \mu^* = 0.238159; E^* = 410040 \text{ kgf/cm}^2$$

$$E^*/E = 0.201728; \nu^* = 0.409245$$

Note: As h/p (9.688) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / \sqrt{(D_s + t_s) / t_s}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{(1200.0 + 16.0) / 16.0}$$

$$= 0.0130 \text{ 1/mm}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.013 \cdot 2032634 \cdot 16.0^3 / (6(1 - 0.3^2))$$

$$= 19872634.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1200.0 + 16.0) \cdot 19872634 / 310.0^3 \cdot (1 + 310.0 \cdot 0.013 + (310.0 \cdot 0.013)^2 / 2)$$

$$= 64249.5 \text{ kgf/cm}^2$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1200.0^2 / (4 \cdot 2032634 \cdot 16.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 16.0)))$$

$$= 0.94308\text{E-}02 \text{ mm/kgf/cm}^2$$

Parameter [w_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$



$$= 1.005 \cdot 19872634 \cdot 0.013 \cdot 0.009431 (1 + 310.0 \cdot 0.013)$$

$$= 12372.2480 \text{ mm}^2$$

Channel coefficient [β_c]:

$$= ((12(1 - v_c^2))^{1/4}) / \sqrt{((D_c + t_c)t_c)}$$

$$= ((12(1 - 0.3^2))^{1/4}) / \sqrt{((1200.0 + 68.0)68.0)}$$

$$= 0.0062 \text{ 1/mm}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.006 \cdot 2032634 \cdot 68.0^3 / (6(1 - 0.3^2))$$

$$= 724661056.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(1200.0 + 68.0 / \cos(0)) \cdot 0.72466E+09 / 310.0^3 \cdot (1.0 + 310.0 \cdot 0.006 \cdot (1.0 + 1.0) / 2 + (310.0 \cdot 0.006)^2 / 2 \cdot 1.0)$$

$$= 881020.9 \text{ kgf/cm}^2$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 1200.0^2 / (4 \cdot 2032634 \cdot 68.0) \cdot (1 - 1200.0 \cdot 0.3 / (2(1200.0 + 68.0)))$$

$$= 0.22348E-02 \text{ mm/kgf/cm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.005 \cdot 0.72466E+09 [0.002235 (1.0 + 0.006 \cdot 1.0 \cdot 310.0) - 0 (1.0 / 310.0 + 1.0 \cdot 0.006 / 2)]$$

$$= 29413.7227 \text{ mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1336.0 / 1194.0$$

$$= 1.1189$$

Coefficient [F]:

$$= (1 - v_s) / E_s \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.41) / 410040 \cdot (64249.48 + 881020 + 2032634 \cdot \ln(1.12))$$

$$= 1.6910$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2 / 16 \cdot [(\rho_s - 1) \cdot (\rho_s^2 + 1) \cdot P_s - (\rho_c - 1) \cdot (\rho_c^2 + 1) \cdot P_t]$$

$$= 1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 23.454 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 163.158]$$

$$= -125732.8906 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega_c \cdot P_t - \omega_s \cdot P_s$$

$$= -125732.89 + 29413.723 \cdot 163.158 - 12372.248 \cdot 23.454$$

$$= 4383176.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M_s - D_o^2 / 32 \cdot F (P_s - P_t)) / (1 + F)$$

$$= (4383176 - 1194.0^2 / 32 \cdot 1.691 (23.45 - 163.16)) / (1 + 1.69)$$

$$= 5539906.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$



Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= M_p + D_o^2/64 \cdot (3 + \nu) \cdot (P_s - P_t) \\ &= 5539906 + 1194.0^2/64 \cdot (3 + 0.409) \cdot (23.45 - 163.16) \\ &= -5069624.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|M_p|; |M_o|) \\ &= \max(|5539906|; |-5069624.5|) \\ &= 5539906.5000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 5539906 / (0.2382 \cdot (310.0 - 8.0)^2) \\ &= 1530.2869 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1568.83 \\ &= 3137.67 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 5542786 / (0.2382 \cdot (219.1505 - 8.0)^2) \\ &= 3132.0461 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 219.1505 + 0 + 0 \\ &= 219.1505 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(219.1505; 152.224) \\ &= 219.1505 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |23.45 - 163.16| \\ &= 139.704 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot \min(0.8 \cdot 1568.835; 0.533 \cdot 2349.607) \\ &= 284.504 \text{ kgf/cm}^2 \end{aligned}$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1252.341) \cdot |23.45 - 163.16| + 0 \\ &= 152.2240 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 1200.0^2 / (4 \cdot 16.0 (1200.0 + 16.0)) \cdot 23.454 \\ &= 433.9759 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu) / E \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 438116.31 / 16.0^2 \cdot [0.013 \cdot 0.2207 + 6(1 - 0.409) / 410040 \cdot \end{aligned}$$



$$1194.0/310.0^3 \cdot (1 + 310.0 \cdot 0.013/2) \cdot (5539906 + 1194.0^2/32 \cdot (23.45 - 163.16)) \\ = 1009.272 \text{ kgf/cm}^2$$

Shell Membrane + Bending Stress [σ_s]:
= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq$ SPS,s)
= 1443.248 kgf/cm² (Should be $\leq$ 4955.82)

Axial Channel Membrane Stress [$\sigma_{c,m}$]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $1200.0^2 / (4 \cdot 68.0 (1200.0 + 68.0) \cdot \cos(0)) \cdot 163.158$
= 681.213 kgf/cm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 15976032 / 68.0^2 \cdot [(0.002235 \cdot 1.0 \cdot 0.006 - 0 \cdot 1.0) \cdot 163.158 - 6((1 - 0.409) / 410040) \cdot 1194.0 / 310.0^3 \cdot (1.0 + 0.006 \cdot 310.0 / 2 \cdot 1.0) \cdot (5539906 + 1194.0^2 / 32 \cdot (23.454 - 163.158))]$
= 2559.269 kgf/cm²

Channel Membrane + Bending Stress [σ_c]:
= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq$ SPS,c)
= 3240.482 kgf/cm² (Should be $\leq$ 4955.82)

Note: The Cylinder-to-Tubesheet Junction stress is < SPS. So, the Simply Supported option can be used.

Re-running the analysis by ignoring the integral channel/shell stiffness. Only modified portions of the output will be re-printed.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate Shell Side parameters for the Simply Supported case:
 $\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate Channel side parameters for the Simply Supported condition:
 $\beta_c, k_c, \lambda_c, \delta_c$ are all = 0

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Coefficient [F]:
= $(1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
= $(1 - 0.41) / 410040 \cdot (0 + 0 + 2032634 \cdot \ln(1.12))$
= 0.3291

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (P_s, P_t) [Mts]:
= $D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_c - 1) \cdot (p_c^2 + 1) \cdot P_t]$
= $1194.0^2 / 16 \cdot [(1.005 - 1) \cdot (1.005^2 + 1) \cdot 23.454 - (1.005 - 1) \cdot (1.005^2 + 1) \cdot 163.158]$
= -125732.8906 kgf/cm²•mm²

Moment acting on unperforated tubesheet rim [M*]:
= $M_{ts} + \omega_c \cdot P_t - \omega_s \cdot P_s$
= -125732.89 + 0 • 163.158 - 0 • 23.454
= -125732.8906 kgf/cm²•mm²

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on center of tubesheet [Mo]:
= $M_p + D_o^2 / 64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$
= $1446436 + 1194.0^2 / 64 \cdot (3 + 0.409) \cdot (23.45 - 163.16)$



$$= -9163095.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= |M_o| \\ &= |-9163095.0| \\ &= 9163095.0000 \text{ kgf/cm}^2 \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 9163095 / (0.2382 (310.0 - 8.0)^2) \\ &= 2531.1191 \text{ kgf/cm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{II}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 1568.83 \\ &= 3137.67 \text{ kgf/cm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu \cdot (h - h_g')^2) \\ &= 6 \cdot 9163095 / (0.2382 (279.2456 - 8.0)^2) \\ &= 3137.6250 \text{ kgf/cm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + C_{ats} + C_{atc} \\ &= 279.2456 + 0 + 0 \\ &= 279.2456 \text{ mm} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(H_{reqB}; H_{reqS}) \\ &= \max(279.2456; 152.224) \\ &= 279.2456 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned} &= |P_s - P_t| \\ &= |23.45 - 163.16| \\ &= 139.704 \text{ kgf/cm}^2 \end{aligned}$$

Shear Stress check [τ_{limit}]:

$$\begin{aligned} &= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y) \\ &= 4 \cdot 0.219 \cdot 310.0 / 1194.0 \cdot \min(0.8 \cdot 1568.835; 0.533 \cdot 2349.607) \\ &= 284.504 \text{ kgf/cm}^2 \end{aligned}$$

|Ps - Pt| is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$\begin{aligned} &= 1 / (4 \cdot \mu) \cdot (D_o / \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot |P_s - P_t| + C_{ats} + C_{atc} \\ &= 1 / (4 \cdot 0.219) \cdot (1194.0 / 1252.341) \cdot |23.45 - 163.16| + 0 \\ &= 152.2240 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Axial Shell Membrane Stress [σ_s, m]:

$$\begin{aligned} &= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s \\ &= 1200.0^2 / (4 \cdot 16.0 (1200.0 + 16.0)) \cdot 23.454 \\ &= 433.9759 \text{ kgf/cm}^2 \end{aligned}$$

Axial Shell Bending Stress [σ_s, b]:

$$\begin{aligned} &= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E \cdot \\ &\quad D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))] \\ &= 6 \cdot 0 / 16.0^2 \cdot [0 \cdot 0.2207 + 6(1 - 0.409) / 410040 \cdot \\ &\quad 1194.0 / 310.0^3 \cdot (1 + 310.0 \cdot 0 / 2) \cdot (1446436 + 1194.0^2 / 32 \cdot \\ &\quad (23.45 - 163.16))] \\ &= -0.000 \text{ kgf/cm}^2 \end{aligned}$$



Shell Membrane + Bending Stress [σ_s]:
= $|\sigma_{s,m}| + |\sigma_{s,b}|$ (Should be $\leq$ SPS,s)
= 433.976 kgf/cm² (Should be $\leq$ 4955.82)

Axial Channel Membrane Stress [$\sigma_{c,m}$]:
= $D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$
= $1200.0^2 / (4 \cdot 68.0 (1200.0 + 68.0) \cdot \cos(0)) \cdot 163.158$
= 681.213 kgf/cm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:
= $6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$
= $6 \cdot 0 / 68.0^2 \cdot [(0 \cdot 1.0 \cdot 0 - 0 \cdot 1.0) \cdot 163.158 - 6((1 - 0.409) / 410040) \cdot 1194.0 / 310.0^3 \cdot (1.0 + 0 \cdot 310.0 / 2 \cdot 1.0) \cdot (1446436 + 1194.0^2 / 32 \cdot (23.454 - 163.158))]$
= 0.000 kgf/cm²

Channel Membrane + Bending Stress [σ_c]:
= $|\sigma_{c,m}| + |\sigma_{c,b}|$ (Should be $\leq$ SPS,c)
= 681.213 kgf/cm² (Should be $\leq$ 4955.82)

Analysis Finished.

Tube Metal Area [At]:
= $\pi (d_o - t) t$
= $\pi (25.0 - 2.5) 2.5$
= 1.767 cm²

Tube Weld Size Results per VIII-2 4.21.2:

Tube Strength [Ft]:
= $\pi \cdot t (d_o - t) S$
= $\pi \cdot 2.5 (25.0 - 2.5) 1308.09$
= 2311.583 kgf

Groove Weld Strength [Fg]:
= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
= $0.85 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1308.09$
= 2311.5833 kgf

Weld Strength Factor [fw]:
= S / S_w where $S_w = \min(S \text{ and } S_t)$
= $1308.09 / 1308.09$
= 1.000

Design Strength Ratio [fd]:
= 1 for full strength welds

Fillet Weld Strength [Ff]:
= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
= $0.55 \cdot \pi \cdot 2.5 (25.0 + 0.67 \cdot 2.5) 1308.09$
= 1507.2808 kgf

Minimum Length of Weld [ar]:
= $2(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o)$
= $2(\sqrt{(0.75 \cdot 25.0)^2 + 1.07 \cdot 2.5 \cdot (25.0 - 2.5) 1.0 \cdot 1.0} - 0.75 \cdot 25.0)$
= 3.083 mm

Minimum Required Fillet Weld Leg [afr]: 1.5416 mm
Minimum Required Groove Weld Leg [agr]: 1.5416 mm

Tube-Tubesheet Jt allowable, 2311.58 is $\geq$ tube strength 2311.58 kgf

Note: This tube-tubesheet joint is a Full Strength joint



Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	2531.1 <=	3137.7 kgf/cm ²	Ok
Tubesheet shear stress	615.0 <=	1252.3 kgf/cm ²	Ok
Stress in Shell at Tubesheet	434.0 <=	4955.8 kgf/cm ²	Ok
Stress in Channel at Tubesheet	681.2 <=	4955.8 kgf/cm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	279.246	310.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.542	2.500	Ok
Tube-Tubesheet Groove Weld Leg :	1.542	2.500	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	301.131 ...	2956	3138	718 1252	Ps+Pt D1	Ok
D2uc	119.148 ...	425	3138	103 1252	Ps+Pt D2	Ok
D3uc	279.246 ...	2531	3138	615 1252	Ps+Pt D3	Ok
D1c	303.392 ...	3001	3138	725 1252	Ps+Pt-c D1	Ok
D2c	120.932 ...	431	3138	104 1252	Ps+Pt-c D2	Ok
D3c	281.449 ...	2570	3138	621 1252	Ps+Pt-c D3	Ok
Max:	303.3917 ... mm	0.956	0.579	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	4956	681	4956	Ok
D2uc		434	4956	...	4956	Ok
D3uc		434	4956	681	4956	Ok
D1c		...	4956	699	4956	Ok
D2c		481	4956	...	4956	Ok
D3c		481	4956	699	4956	Ok
Max		0.097	0.141			

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:



Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	303.392	310.000	Ok
Tube thickness :	1.328	2.500	Ok
Tube-tubesheet fillet weld leg :	1.542	2.500	Ok
Tube-tubesheet groove weld leg :	1.542	2.500	Ok

Given Shell Thickness: 16.0000 mm
 Given Channel Thickness: 68.0000 mm

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	170.578	1.000	170.578	1.000
Tubesheet Shear Stress	281.749	1.000	281.749	1.000
Shell Stress (Axial, Junction)	281.749	1.000	54.922	1.000
Tubesheet-Channel Junction	227.347	1.000	241.813	1.000
Tube Pressure Stress	291.891	1.000	207.499	1.000
Minimum MAWP	170.578		54.922	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	185.723	1.000	185.723	1.000
Tubesheet Shear Stress	305.770	1.000	305.770	1.000
Shell Stress (Axial, Junction)	305.770	1.000	65.490	1.000
Tubesheet-Channel Junction	255.048	1.000	289.311	1.000
Tube Pressure Stress	313.992	1.000	219.790	1.000
Minimum MAPnc	185.723		65.490	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(310.0/4, \min(310.0, 16.0))$$

$$= 77.500 \text{ mm}$$

MDMT reduction ratio:

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(23.45/65.49, 1)$$

$$= 0.358$$

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

Channel Side MDMT calculation:

Governing thickness on the shell side per figure 3.10:

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(310.0/4, \min(310.0, 68.0))$$

$$= 77.500 \text{ mm}$$

MDMT reduction ratio:



= min(pt/Tubeside MAPnc, 1)
= min(163.16/185.72, 1)
= 0.879

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification -46 °C

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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